2D–Hydrogel-Based Flexible Materials for Soft Robotics, Wearable Electronics, and Biomedical Applications: Progress and Challenges

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Abstract

Flexible and mechanically compliant energy-storage systems are essential for the development of next-generation soft mechatronic technologies, yet maintaining reliable operation under deformation, dehydration, and interfacial stress remains challenging. Hydrogels provide hydrated ionic pathways, tunable mechanics, adhesion, and opportunities for self-healing, while two-dimensional (2D) materials offer diverse electronic, electrochemical, and interfacial properties for engineering multifunctional composites. This review critically examines 2D material–hydrogel systems for flexible supercapacitors, microsupercapacitors, batteries, and hybrid energy storage devices. It considers graphene derivatives, MXenes, transition-metal dichalcogenides, layered double hydroxides, and carbon nitride, alongside emerging MBenes, 2D metal–organic frameworks, covalent organic frameworks, and related organic nanosheets. Beginning with hydrogel chemistry and the synthesis, surface functionality, and structural characteristics of 2D materials, the review connects constituent properties to network formation, ion transport, electronic conduction, and charge-storage mechanisms. Particular attention is given to nanosheet dispersion and restacking, polymer–filler interactions, dynamic crosslinking, pore accessibility, and solvent retention across hydrogel, organohydrogel, and related ionogel architectures. Comparative assessment distinguishes demonstrated hydrogel-based devices from adjacent electrode studies and emerging integration concepts, accounting for differences in conductivity, aqueous stability, processability, and evidence maturity. Applications in soft robotics, wearable electronics, smart textiles, and biomedical interfaces are evaluated through device-level requirements for mechanical compliance, electrochemical stability, encapsulation, and operational lifetime. The review further examines performance normalization, coupled mechanical–electrochemical–environmental testing, material-specific degradation, manufacturing safety, and biological and end-of-life considerations. By linking material chemistry and composite architecture to operating conditions and system integration, it identifies priorities for reproducible benchmarking, mechanistic aging studies, scalable fabrication, and application-appropriate degradation. This materials-to-systems perspective establishes design principles for reliable, sustainable, and application-matched energy storage in soft mechatronics.

Keywords: Two-dimensional materials; hydrogel nanocomposites; flexible energy storage; graphene derivatives; MXenes; two-dimensional organic frameworks; interfacial engineering; soft robotics; wearable electronics; bioelectronics.

1. Introduction

Soft mechatronic systems require energy-storage units to function as integral components of deformable structures rather than rigid accessories. Their power sources must accommodate platform mechanics and circuitry while maintaining function. This extends evaluation beyond conventional metrics such as capacitance, energy density, and cycling stability. A material with high static capacitance may be unsuitable if electrical contact deteriorates during deformation, dehydration disrupts ion transport, or rigidity restricts actuation. Conversely, hydrogels with moderate electrochemical performance may offer advantages through hydrated ionic pathways, adhesion, flexibility, and compatibility with skin, textiles, and compliant joints [1-3].

Combining hydrogels with two-dimensional (2D) materials offers a route to address these demands. This family includes MXenes, MBenes, graphene derivatives, transition-metal dichalcogenides, inorganic nanosheets, and organic nanosheets such as 2D metal–organic frameworks and covalent organic frameworks. Their structures, surface chemistries, porosities, electronic properties, and redox-active sites create opportunities for charge storage and transport. MXenes are important because their conductivity and surface terminations can support electrochemical activity in hydrated polymer networks; however, oxidation sensitivity and processing requirements make comparison with alternative 2D materials essential [4-6].

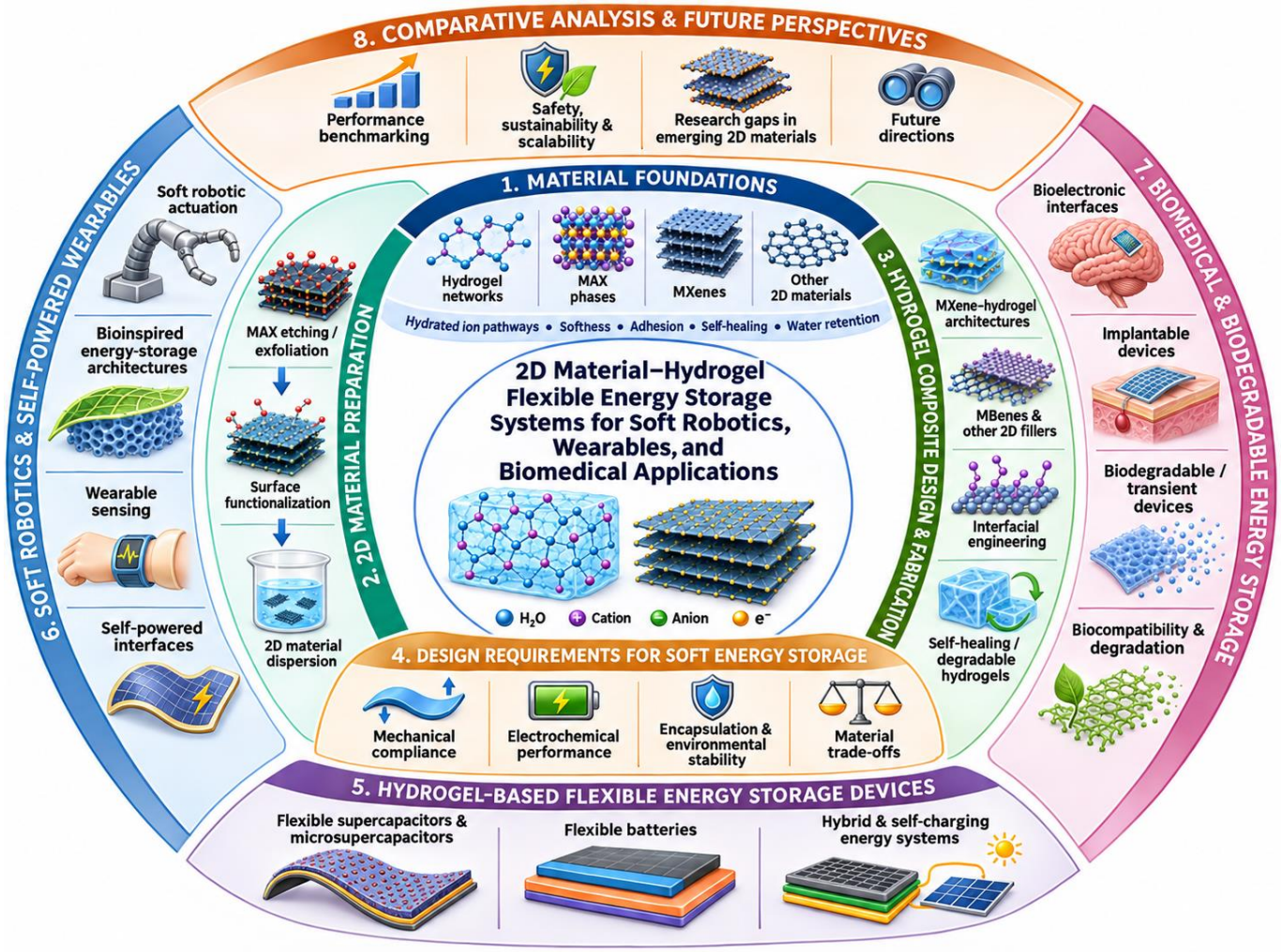


Fig. 1. Integrated material-to-device roadmap of 2D material–hydrogel flexible energy storage systems for soft robotics, wearables, and biomedical applications. The schematic summarizes the overall research framework of the review, covering material foundations (hydrogel networks, MAX phases, MXenes, and other two-dimensional materials), 2D material preparation, hydrogel composite design and fabrication, design requirements for soft energy storage, and hydrogel-based flexible energy storage devices, together with their emerging roles in soft robotics and self-powered wearables and biomedical and biodegradable energy storage systems. It further highlights the links between composition, structure, processing, device configuration, and application-oriented performance, while outlining key themes in comparative analysis and future perspectives. Reproduced with permission from Ref. [7]. Copyright 2022, Elsevier. Reproduced with permission from Ref. [8]. Copyright 2022, Elsevier. Reproduced with permission from Ref. [9]. Copyright 2021, Wiley-VCH. Reproduced with permission from Ref. [10]. Copyright 2022, Elsevier. Reproduced with permission from Ref. [11]. Copyright 2017, Elsevier. Reproduced with permission from Ref. [12]. Copyright 2024, Royal Society of Chemistry.

This review adopts a materials-to-systems framework linking hydrogel chemistry, 2D material structure and functionalization, interfacial interactions, composite-network formation, electrode architecture, and device integration. It considers how dispersion, nanosheet restacking, polymer–filler coupling, ionic transport, and dynamic crosslinking influence mechanical compliance and electrochemical performance. Flexible supercapacitors, microsupercapacitors, batteries, and hybrid energy-storage systems are examined alongside encapsulation and environmental-stability requirements.

Operational reliability is emphasized over isolated peak performance. Wearable devices must withstand sweat exposure, repeated bending, adhesion–reapplication cycles, and dehydration, while soft robots experience bending, torsion, compression, and repetitive actuation. Biomedical and biodegradable platforms additionally require assessment of cytocompatibility, sterilization tolerance, degradation kinetics, and decomposition products [13-15]. Capacitance, resistance, adhesion, hydration, and device geometry evolve together; electrode fracture, electrolyte displacement, or stiff encapsulation can undermine promising materials [2, 5, 16]. Accordingly, this review addresses material foundations, composite design, device performance, soft robotic and wearable applications, biomedical integration, and sustainability challenges. Fig. 1 summarizes links among material synthesis, structure, properties, device architectures, and wearable and biomedical integration.

1.1 Flexible Energy Storage Needs in Soft Mechatronics

Soft mechatronics integrates sensing, actuation, computation, communication, and energy delivery within mechanically deformable structures. Robotic grippers undergo bending, stretching, compression, and torsion, while skin-mounted electronics experience repetitive body movements, external

pressure, perspiration, and temperature fluctuations. Energy storage components must therefore maintain electrode–electrolyte contact, ionic conductivity, electrical safety, and operational stability during complex deformation [2, 16, 17].

The increasing demand for distributed autonomy further highlights the importance of localized energy storage. Soft robots and wearable devices require power for sensing, data acquisition, wireless communication, memory, and intermittent actuation. Flexible microsupercapacitors provide rapid power delivery, hydrogel-based batteries support extended operation, and hybrid systems enable localized energy generation, storage, and utilization [18-20]. However, miniaturization alone cannot eliminate mechanical and interfacial incompatibilities between conventional energy storage components and compliant substrates. Rigid batteries introduce stress concentrations, liquid electrolytes present leakage risks, and dry solid electrolytes may exhibit excessive stiffness or limited ionic conductivity. Hydrogel nanomaterials offer an alternative by retaining electrolytes within flexible, hydrated, and conformable polymer networks [1, 21]. Consequently, flexible energy storage is a fundamental requirement governing autonomy, safety, reliability, and operational lifetime. Soft robots require distributed power for actuation and feedback control, wearable electronics demand lightweight and skin-conformal modules, and biomedical interfaces require mechanical compatibility, biocompatibility, stable ionic exchange, and potentially controlled degradability [2, 13].

These applications impose distinct electrochemical and mechanical requirements. Electronic skin benefits from rapid charging, low interfacial impedance, and prolonged bending endurance; robotic actuators require pulse-power capability and deformation-compatible encapsulation; and transient biomedical sensors require predictable degradation into nontoxic products. Such demands have stimulated interest in hydrogels, MXenes, graphene derivatives, conductive polymers, and hybrid architectures [16, 22]. Performance assessment must therefore reflect the intended application. Areal energy density may be more informative than gravimetric capacitance for epidermal devices, whereas volumetric performance and pulse-power response are particularly relevant to embedded robotic systems. Likewise, biomedical suitability requires evaluating the complete device, including electrodes, current collectors, encapsulation, and degradation products, rather than electrolyte biocompatibility alone [13, 14, 17].

1.2 Hydrogels and Two-Dimensional Nanomaterials as Enabling Materials

Hydrogels offer tunable physicochemical and mechanical properties across molecular, network, and mesoscopic scales. Monomer chemistry influences charge density, hydrogen bonding, hydrophobic interactions, and degradation behavior. Crosslinking regulates structural stability, energy dissipation, and self-healing, whereas porosity, filler distribution, and solvent composition affect ionic transport and fatigue resistance. These characteristics enable hydrogels to function as electrolytes, adhesive interfaces, substrates, encapsulants, and sensing components [23-25].

MXenes complement hydrogel networks through their two-dimensional conductive structures and chemically active surfaces. Their general formula, $M_{n+1}X_nT_x$, represents transition-metal carbides, nitrides, or carbonitrides with surface terminations such as –O, –OH, –F, and –Cl. These groups facilitate dispersion in polar media and interactions with polymer chains, water molecules, and ions. MXenes contribute to electrochemical energy storage through electric double-layer capacitance, pseudocapacitive reactions, ion intercalation, and efficient electronic transport [26-29].

The primary design challenge involves regulating interactions between hydrated polymer networks and conductive MXene nanosheets. Insufficient interfacial interactions promote aggregation and restacking, while excessively strong interactions may restrict interlayer accessibility and ionic transport. Similarly, excessive water may intensify dehydration-related instability and limit electrochemical voltage stability, whereas insufficient hydration reduces ion mobility. Effective composites therefore require coordinated optimization of water content, MXene loading, surface terminations, crosslinking density, electrolyte concentration, and encapsulation [4, 30, 31]. This interfacial balance governs simultaneous ionic and electronic conduction. Polymer networks control swelling, hydration, and mechanical energy dissipation, while MXene nanosheets establish conductive pathways and electrochemically active interfaces. Reversible interactions can preserve structural integrity during deformation, whereas uncontrolled aggregation limits ion-accessible conductive regions. Accordingly, MXene concentration, surface chemistry, polymer composition, and solvent characteristics must be optimized as interconnected design parameters [4, 30, 32].

1.3 Why MXenes? Opportunities and Limitations of Other 2D Materials

MXenes combine conducting sheets, ion-accessible galleries, and polar polymer interfaces. Recent $V_4C_3T_x$/PVA electrolytes [33] and MXene-cross-linked polypyrrole electrodes [34] demonstrate different functions: electronic percolation supports an electrode but can short an electrolyte separator. Alternatives include MXene/rGO hydrogel electrodes [35], MoS_2 gel membranes [36], boride anodes [37], and MBene-containing colloidal electrolytes [38]. Selection must therefore consider electrode chemistry, hydration, processing, and deformation together. A constituent's conductivity does not establish superiority in a packaged hydrogel device, and a hydrated colloid does not demonstrate a persistent polymer network.

1.4 Scope and Organization of the Review

This review integrates MXene synthesis, hydrogel chemistry, flexible electrochemistry, soft robotics, and wearable and biomedical electronics within a unified framework. The central objective is to evaluate materials and devices according to their combined mechanical, electrochemical, and application-specific performance rather than isolated material properties [2, 16, 39].

The discussion encompasses three interconnected levels. At the constituent-material level, hydrogel composition, MXene surface chemistry, and additional two-dimensional fillers determine fundamental transport and mechanical properties. At the composite-architecture level, nanosheet orientation, porosity, dynamic bonding, solvent composition, and interfacial compatibility govern mixed ionic–electronic transport. At the device-system level, electrodes, separators, electrolytes, current collectors, encapsulation, interconnects, energy harvesters, sensors, and circuits must operate as an integrated compliant platform [5, 20, 40]. Different material strategies address specific limitations: organohydrogels improve dehydration and freezing resistance; MXene/GO aerogels mitigate restacking and enhance porosity; dynamic crosslinks improve fatigue resistance and self-healing; printable MXene inks facilitate miniaturized device integration; and biodegradable composites support transient and sustainable applications [15, 31, 39, 41, 42].

The review therefore provides an application-oriented assessment of MXene–hydrogel flexible energy storage, encompassing MAX-to-MXene conversion, hydrogel network engineering, ion transport, charge-storage mechanisms, flexible supercapacitors, batteries, hybrid energy systems, soft robotics, wearable sensing, human–machine interfaces, smart textiles, and biomedical electronics. Section 2 introduces material foundations; Section 3 examines composite design and fabrication; Section 4 addresses mechanical, electrochemical, and integration requirements; Section 5 discusses flexible supercapacitors, microsupercapacitors, batteries, and hybrid systems; Section 6 explores soft robotics and self-powered wearable technologies; Section 7 evaluates biocompatible, biodegradable, and biomedical applications; and Section 8 critically compares reported systems, identifies unresolved challenges, and outlines future research directions.

2. Material Foundations of Hydrogels and Two-Dimensional Materials

Hydrogel–MXene energy storage systems rely on interconnected electronic, ionic, and mechanical networks. Electronic conduction occurs through MXene sheets, conductive polymers, graphene derivatives, carbon nanotubes, and current collectors, while ionic transport is facilitated by water, salts, ionic liquids, polymer-bound charges, and interconnected pores. Mechanical integrity arises from polymer chains, crosslinking interactions, filler–polymer interfaces, and hierarchical architectures. Sustained connectivity among these networks during deformation and environmental exposure is essential for device stability [4, 21, 30]. Processing-induced structural variations significantly influence material performance. Vacuum-filtered MXene films offer high conductivity but limited stretchability, whereas freeze-cast MXene/graphene aerogels enhance ionic accessibility at the expense of packing density. Polymer-rich hydrogels improve flexibility and adhesion but may increase electronic resistance, while redox-active gels enhance capacitance yet promote self-discharge. Consequently, material optimization requires balancing competing electrochemical and mechanical properties [41, 43, 44]. Processing parameters, including sonication duration, washing procedures, centrifugation speed, freeze-casting orientation, printing shear rate, and post-gelation solvent exchange, influence flake dimensions, interlayer spacing, pore connectivity, and filler distribution. Therefore, reproducible performance evaluation requires comprehensive reporting of processing conditions, dispersion characteristics, filler loading, solvent content, and mechanical conditioning before electrochemical testing [41, 43, 45].

2.1 Hydrogel Chemistry, Synthesis, Ionic Transport, and Mechanics

Hydrogels for energy storage are commonly fabricated through in situ polymerization, physical gelation, freeze–thaw processing, or post-gelation solvent/ion exchange. In situ polymerization enables filler incorporation and structural control but requires removal of residual monomers and initiators for biomedical applications. Physical gelation provides reversible network formation, whereas freeze–thaw processing generates crystalline domains, particularly in poly(vinyl alcohol) (PVA), enhancing toughness without extensive chemical cross-linking [21, 23, 46].

Hydrogel ionic conductivity depends on electrolyte concentration, water distribution, polymer-chain mobility, transport tortuosity, ion pairing, and filler–polymer interactions. MXene incorporation can enhance ionic transport through hydrophilic interfaces and interlayer water, although excessive loading may increase tortuosity and restrict water mobility [3, 47]. Free water facilitates ion diffusion but promotes evaporation and potentially MXene oxidation, whereas bound water improves retention at the expense of ion mobility. Intermediate water states, hydrogen-bond networks, and nanoconfined water further influence ionic transport. Consequently, conductivity evaluation should consider hydration, swelling, temperature, humidity, and mechanical deformation, including repeated bending, compression, dehydration, rehydration, and substrate attachment [3, 47, 48]. Mechanical compliance must satisfy application-specific requirements, including conformability for epidermal sensors, fatigue resistance for soft robotics, abrasion tolerance for textiles, and tissue-compatible stiffness for implantable devices. Combining covalent cross-links with reversible physical interactions, ionic coordination, and hydrogen bonding enables structural integrity, energy dissipation, and self-healing [2, 24, 25].

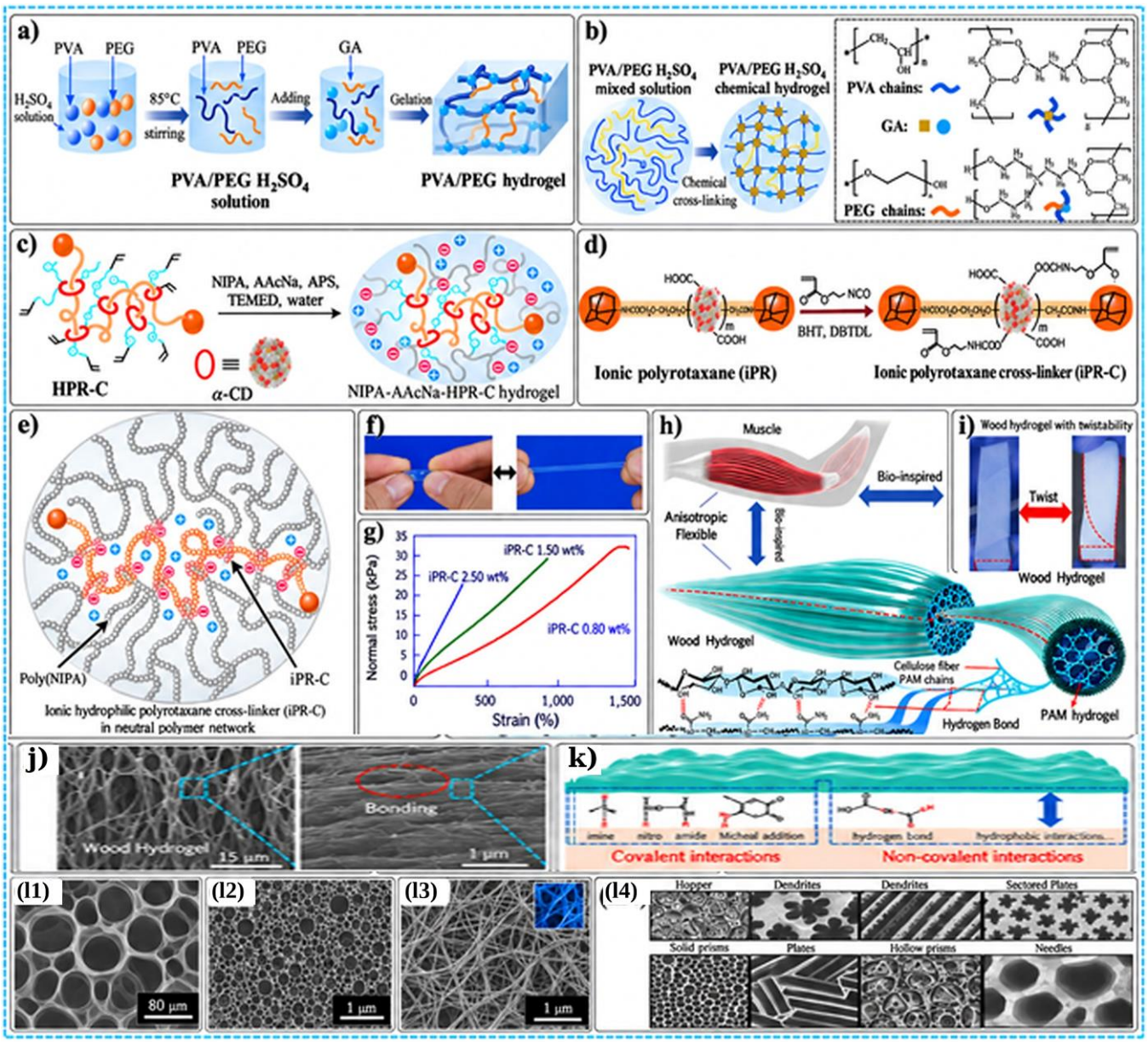


Fig. 2. Preparation, cross-linking mechanisms, structural characteristics, mechanical properties, and morphology of hydrogels. (a) Preparation of the PVA/PEG hydrogel electrolyte; (b) cross-linking reaction involving PVA, PEG, and glutaraldehyde. Reproduced with permission from Ref. [49]. Copyright 2020, Elsevier . (c) Synthesis of HPR-C from HPR and 2-acryloyloxyethyl isocyanate using DBTDL as a catalyst and BHT as a polymerization inhibitor in DMSO; (d) preparation of the NIPA–AAcNa–HPR-C hydrogel using HPR-C as a cross-linker, NIPA as the primary monomer, AAcNa as a comonomer, APS as an initiator, and TEMED as an accelerator in water; (e) synthesis of iPR-C from iPR and 2-acryloyloxyethyl isocyanate using DBTDL as a catalyst and BHT as a polymerization inhibitor in DMSO; (f) schematic illustration of swollen iPR-C within the NIPA–iPR-C hydrogel; (g) elongated configuration of the NIPA–iPR-C hydrogel. Reproduced from Ref. [50]. Copyright 2014, Nature Publishing Group under a Creative Commons license. (h) Schematic illustration and microscopic network structure of the wood hydrogel; (i) photographs of a 7-cm-long wood hydrogel specimen twisted by 180°; schematic representation of hydrogen bonding and covalent cross-linking between PAM chains in the wood hydrogel; (j) cross-sectional SEM image revealing the PAM network and wood channels, with magnified views illustrating hydrogen-bond-mediated cross-linking between CNFs and PAM chains. Reproduced with permission from Ref. [51]. Copyright 2018, Wiley-VCH. (k) Schematic illustration of hydrogel self-adhesion mechanisms. Reproduced with permission from Ref. [52]. Copyright 2021, Royal Society of Chemistry. (l1) SEM image of the OP hydrogel; (l2) SEM image of the OPP hydrogel; (l3) SEM image of the OPPC hydrogel. (l4) Representative pore morphology of a three-dimensional porous hydrogel formed through controlled ice crystal growth. Reproduced with permission from Ref. [53] for panel (m4); Copyright 2021, Elsevier.

Ionic conductivity is experimentally expressed as:

$$\sigma = \frac{L}{RA} \quad (1)$$

where σ represents ionic conductivity, L is electrolyte thickness, R is measured resistance, and A is electrode area. Reliable comparisons require consistent measurement geometry, hydration, temperature, and electrode contact.

The simplified Nernst–Einstein relationship is:

$$\sigma = \frac{\sum_i z_i^2 F^2 D_i C_i}{R_g T} \quad (2)$$

where z_i is ionic valence, F is the Faraday constant, D_i is ionic diffusivity, C_i is mobile-ion concentration, R_g is the gas constant, and T is absolute temperature.

Fig. 2 highlights the interplay among cross-linking chemistry, solvent retention, ionic transport, and mechanical compliance. Chemical networks maintain dimensional stability, reversible

interactions dissipate energy, and hydrated pores and filler–polymer interfaces support transport and structural continuity under deformation [2, 21, 23-25, 47]. The PVA/polyethylene glycol (PEG) electrolyte was prepared by dissolving 1.5 g PVA and PEG in 15 mL of 0.5 mol L^{-1} H_2SO_4 at 85 °C. After cooling, 3.5 mL glutaraldehyde (GA) solution was added, followed by casting into a Petri dish. PEG-to-PVA mass ratios of 5%, 10%, 15%, 20%, and 25% were investigated [49]. Fig. 2(a) and (b) illustrate preparation and acid-catalyzed cross-linking, respectively, involving PVA hydroxyl groups, PEG terminal hydroxyl groups, and GA aldehyde groups to form acetal or hemiacetal linkages. The resulting flexible films could be removed and shaped as required.

Polyrotaxane (PR)-based cross-linkers provide another approach to improving hydrogel extensibility. Functionalization of PR with 2-acryloyloxyethyl isocyanate produces PR-C, whose isocyanate groups react with α-cyclodextrin (α-CD) hydroxyl groups to form carbamate linkages. However, N-isopropylacrylamide (NIPA) hydrogels containing PR-C exhibited only modest mechanical improvements over conventional N,N′-methylenebisacrylamide (BIS)-cross-linked networks. Solvent exchange from dimethyl sulfoxide (DMSO) to water also caused slight opacity, suggesting PR-C contraction due to limited solubility and α-CD aggregation, which restricts sliding-ring mobility [50]. Although alkaline conditions reduce aggregation, their practical applicability is limited. Hydroxypropylated PR cross-linkers (HPR-C), synthesized as illustrated in Fig. 2(c), improved transparency without substantially enhancing extensibility. Fig. 2(d) further illustrates NIPA–AAcNa–HPR-C hydrogel preparation using ionic comonomers. Thermoresponsive poly(NIPA) networks exhibit a volume transition near 32 °C [50]. Randomly incorporated ionic monomers generally increase the transition temperature, while excessive ionic content suppresses volume changes through strong hydration. Grafted ionic chains, however, largely preserve the poly(NIPA) transition. Accordingly, ionic PR-derived cross-linkers (iPR-C), illustrated in Fig. 2(e), were developed to enhance water compatibility and mechanical performance. Carboxyl-functionalized α-CD units improve cross-linker solubility, maintaining extended iPR-C configurations within neutral poly(NIPA) networks, as depicted in Fig. 2(f) and (g), thereby facilitating sliding interactions and mechanical extensibility.

Wood-derived hydrogels demonstrate hierarchical mechanical reinforcement through anisotropic architectures. Fig. 2(h) and (i) illustrate their aligned structure and reversible twisting deformation [51]. Following delignification, preserved cellulose nanofiber (CNF) frameworks are infiltrated with acrylamide (AM), ammonium persulfate (APS), and N,N′-methylenebisacrylamide (MBA). Polymerization at 60 °C generates a polyacrylamide (PAM) network reinforced by CNF–PAM hydrogen bonding, as illustrated in Fig. 2 (j). Tensile strength and elastic modulus reach 36 MPa and 310 MPa longitudinally, compared with 0.54 MPa and 0.135 MPa transversely, respectively. Unmodified PAM exhibits corresponding values of only 0.072 MPa and 0.01 MPa. This reinforcement also provides optical anisotropy, transparency, and nanofluidic ionic transport, supporting multifunctional biomedical applications [51].

Self-adhesion is particularly important for epidermal and organ-mounted sensing devices [52]. Fig. 2(k) illustrates adhesion through covalent interactions involving hydroxyl, ether, amino, carboxyl, and catechol functionalities, including imine and amide bond formation. Noncovalent mechanisms include hydrogen bonding, hydrophobic interactions, cation–π interactions, and mechanical interlocking. Hydrogel morphology further determines mechanical and transport characteristics. Fig. 2(l1)–(l3) present scanning electron microscopy (SEM) images of OP, OPP, and OPPC hydrogels. OP exhibits an average pore size of 140 ± 80 μm, decreasing to 0.4 ± 0.4 μm following PVA/borax incorporation in OPP through increased dual-network cross-linking [53]. Additional carbon nanotube (CNT) incorporation in OPPC slightly reduces pore dimensions, potentially through hydrogen bonding with O-CMCS or PVA chains. Pore architecture also depends on polymer concentration and freezing conditions. Higher polymer concentrations restrict ice-crystal growth, while lower freezing temperatures generally produce finer pores through accelerated crystallization. Unlike conventional copolymerization, which produces irregular pore distributions, directional freezing controls pore dimensions and orientation through regulated ice-crystal growth, as illustrated in Fig. 2(l4) [53]. These relationships establish a structural basis for optimizing hydrogel networks toward integrated ionic conductivity, mechanical compliance, and functional stability.

2.2 MXenes: Synthesis, Surface Functionalization, and Key Properties

MAX phases are layered, nanolaminated materials with the general formula $M_{n+1}AX_n$, where M represents an early transition metal, A denotes an A-group element, and X typically represents carbon and/or nitrogen. The parameter n generally ranges from 1 to 3, corresponding to the 211, 312, and 413 structural families, while higher-order 514 phases have also been reported. As illustrated in Fig. 3(a), high-throughput computational screening provides an efficient strategy for identifying novel MAX phases and potential precursors for two-dimensional (2D) MXenes. Using density functional theory (DFT), the study investigated 1,122 candidate compositions involving 11 transition metals, 17 A-site elements, two X elements (C and N), and three structural types. Thermodynamic formability was evaluated by comparing candidate formation energies with those of competing phases, while phonon calculations assessed local dynamical stability. Of the candidates investigated, 466 were predicted to have a high probability of experimental realization, and 136 were identified as potentially exfoliable, corresponding to 26 single-transition-metal MXene compositions. Exfoliation feasibility was further assessed by considering exfoliation energy and the relative strengths of M–A and M–X bonds, helping identify candidates that may undergo selective A-layer removal while retaining their 2D frameworks [54].

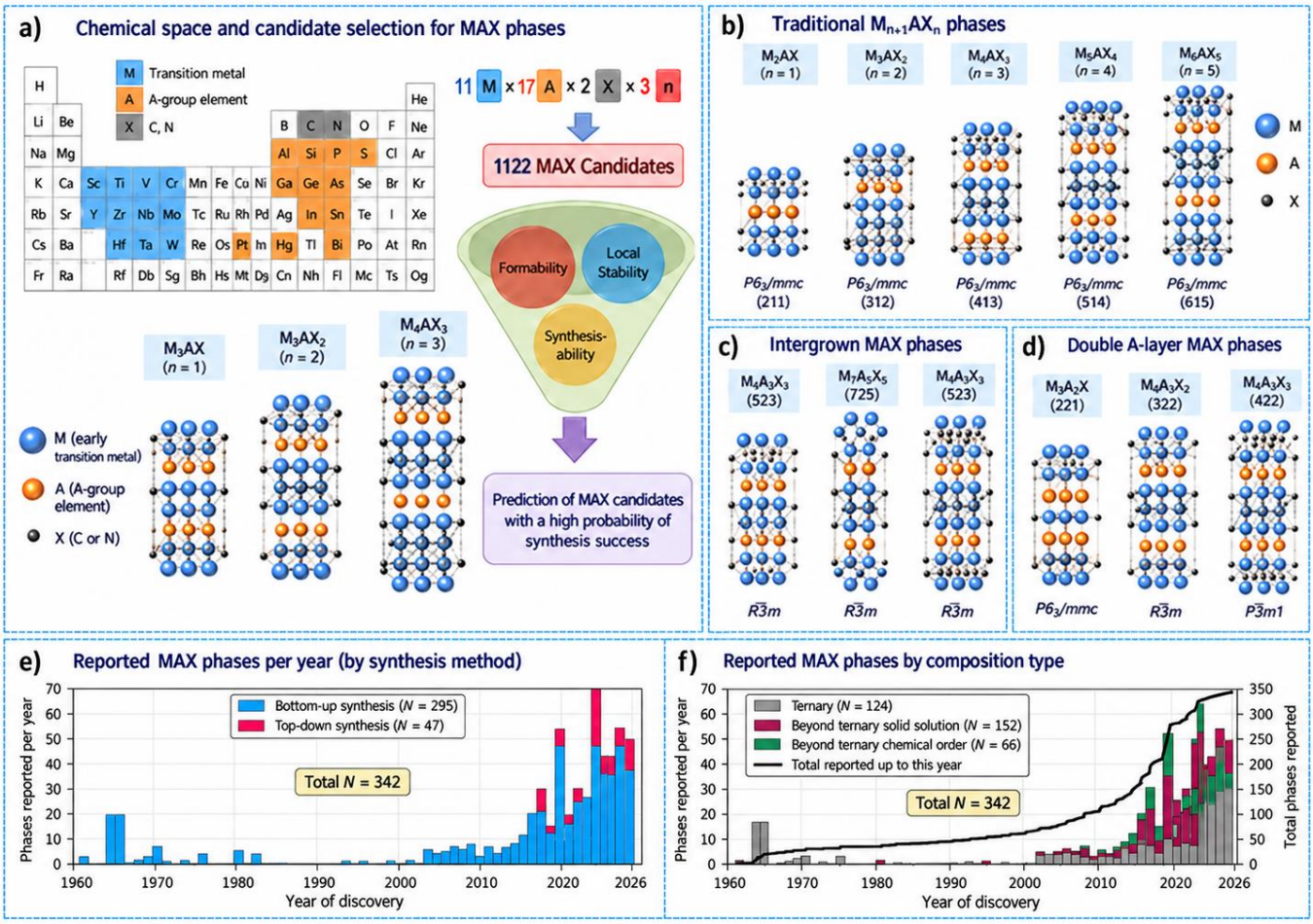


Fig. 3. discovery, crystal structures, and historical development of MAX phases. a) Schematic illustration of the MAX/MXene chemical search space, showing combinations of M, A, and X elements across three structural types and the assessment of synthesizability based on formability and local stability [54]. b–d) Crystal structures of b) conventional $M_{n+1}AX_n$ phases, c) hybrid MAX phases with alternating $M_{n+1}X_n$ layer thicknesses, and d) double-A-layer MAX phases. M, A, and X atoms are represented in blue, orange, and dark gray, respectively; black rectangles indicate unit cells, and phase abbreviations are provided in parentheses [55]. e) Historical distribution of synthesized MAX phases according to synthesis route (bottom-up or top-down) and compositional complexity; f) cumulative number of reported MAX phases, indicated by the gray line and right-hand axis [55]. Panel a) reproduced with permission from Ref. [54]. Copyright 2021, The Royal Society of Chemistry. Panels b–f) reproduced from Ref. [55] under the Creative Commons Attribution 4.0 International License (CC BY 4.0). © 2023 , published by Elsevier Ltd.

Panels (b–d) illustrate the structural diversity of MAX phases. Conventional MAX phases generally crystallize in the hexagonal $P6_3/mmc$ space group, whereas hybrid phases containing alternating $M_{n+1}X_n$ slab thicknesses may adopt the R-3m structure. Double-A-layer MAX phases exhibit additional structural variations and may crystallize in $P6_3/mmc$, R-3m, or P-3m1 symmetry. In these representations, M atoms are shown in blue, A atoms in orange, and X atoms in dark gray, while black rectangles delineate the unit cells [55]. Panels (e–f) summarize the historical development of MAX-phase research by distinguishing bottom-up synthesis, including powder metallurgy and thin-film deposition, from top-down approaches that modify existing phases through post-synthesis elemental-exchange reactions. The historical distribution also distinguishes conventional three-element compositions from phases containing additional elements. The cumulative trend highlights the rapid expansion of the MAX-phase family, with the review reporting at least 342 phases, including 69 reported in 2023 [55].

The generalized selective etching process for MAX-to-MXene conversion can be summarized as:

$$M_{n+1}AX_n + xHF \rightarrow M_{n+1}X_nT_x + \text{soluble byproducts} \quad (3)$$

where M denotes an early transition metal, A is the selectively removed element, X is carbon or nitrogen, and T_x represents surface terminations introduced during etching and washing.

For Al-containing MAX phases [54, 55], a simplified HF-based Al-removal reaction is:

$$M_{n+1}AlX_n + 3HF \rightarrow AlF_3 + \frac{3}{2}H_2 + M_{n+1}X_n \quad (4)$$

This idealized reaction does not represent the full aqueous chemistry, but it clarifies why etchant selection affects surface chemistry and safety.

Hydroxyl termination can be represented in simplified form as:

$$M_{n+1}X_n + 2H_2O \rightarrow M_{n+1}X_n(OH)_2 + H_2 \quad (5)$$

Termination, intercalation, washing, and delamination occur concurrently; therefore, the final MXene generally contains a distribution of -O, -OH, -F, -Cl, and adsorbed species determined by the processing route.

The conversion of MAX phases into MXenes generally involves selective A-layer etching, followed by intercalation and delamination. Conventional hydrofluoric acid (HF) etching has motivated the development of alternative approaches, including LiF/HCl mixtures, ammonium bifluoride, molten-salt etching, Lewis-acidic etching, supercritical-fluid methods, and chemical vapor deposition. These techniques differ in etching efficiency, defect formation, surface termination chemistry, scalability, and dispersion stability [56, 57]. The resulting MXene surfaces commonly contain oxygen-, hydroxyl-, and halogen-containing terminations, such as –O, –OH, –F, and, depending on the processing route, –Cl. These surface groups influence hydrophilicity, interlayer spacing, electrical conductivity, oxidation stability, and interactions with polymer matrices [28, 30, 54, 55]. MXene flake dimensions, precursor quality, and etching conditions also affect hydrogel integration: larger flakes can enhance conductivity and mechanical reinforcement but may sediment or align during gelation, whereas smaller flakes can improve dispersion but may be more susceptible to oxidation. Surface modification, interlayer engineering, and hybridization with graphene, conductive polymers, metal oxides, layered double hydroxides, and carbon nanotubes can further improve structural stability and electrochemical performance [30, 41, 42, 56]. These relationships provide a foundation for designing conductive, mechanically robust, and electrochemically active 2D material–hydrogel composites for energy storage and related applications.

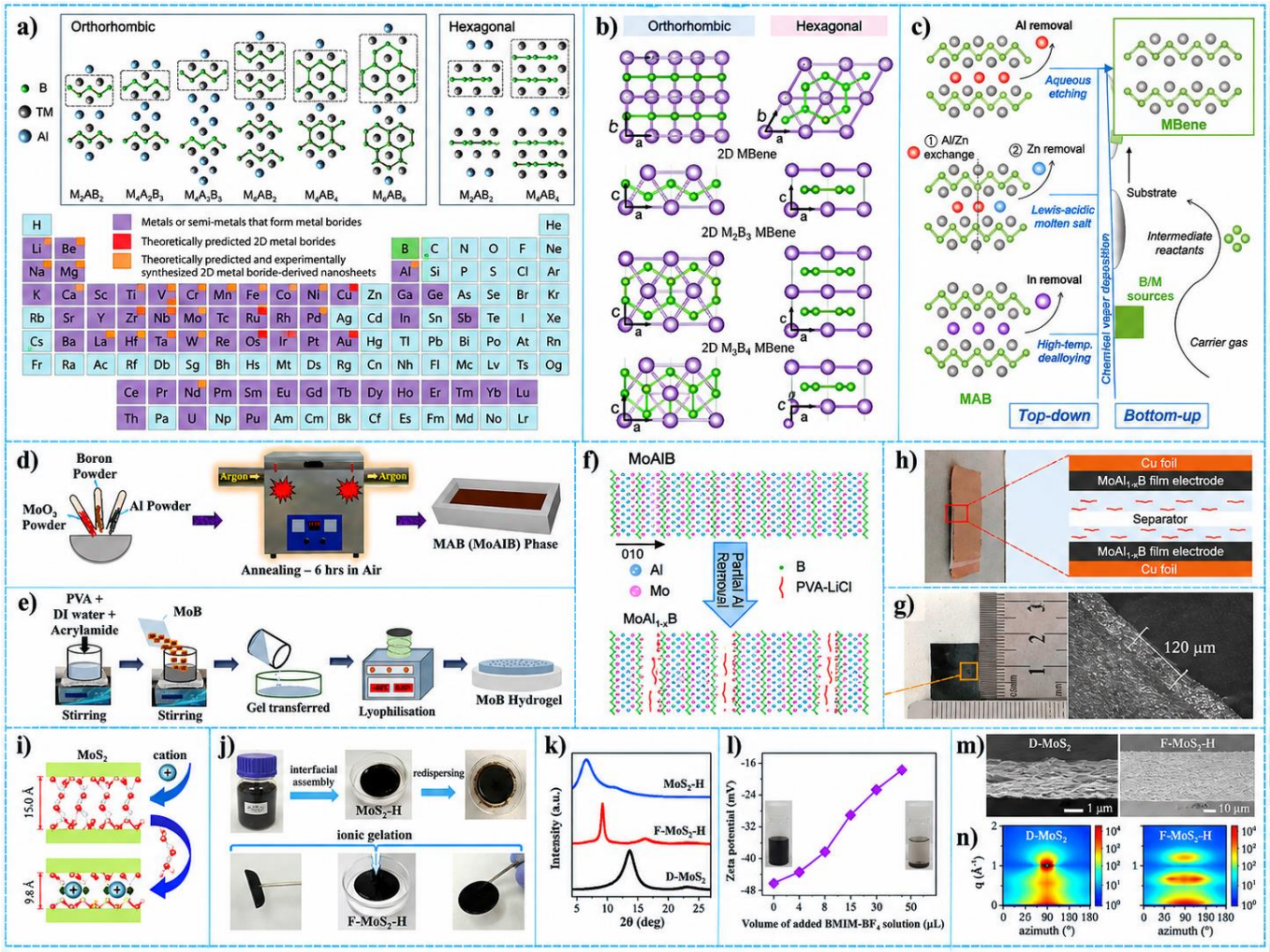


Fig. 4. Crystal structures, synthesis routes, and hydrogel-related electrochemical architectures of 2D borides and MoS_2. (a) Representative MAB structures and elemental diversity; (b) orthorhombic and hexagonal MBene-related motifs; and (c) top–down and bottom–up fabrication strategies. Reproduced with permission from Ref. [29]. Copyright 2026, Elsevier. (d) Stepwise synthesis of layered MoB and (e) MoB-derived hydrogel processing and lyophilization. Reproduced with permission from Ref. [56]. Copyright 2025, Wiley. (f) Al deintercalation to form $MoAl_{1-x}B$, (g) flexible electrode morphology, and (h) all-solid-state supercapacitor assembly using PVA–LiCl electrolyte. Reproduced with permission from Ref. [58]. Copyright 2025, American Chemical Society; licensed under CC BY 4.0. (i) Cation-mediated MoS_2 assembly and (j) formation of an ion-stabilized hydrogel membrane. Reproduced with permission from Ref. [36]. Copyright 2026, Elsevier.

2.3 MBenes: An Emerging Class of Two-Dimensional Transition-Metal Borides

MBenes are an emerging family of two-dimensional (2D) transition-metal borides attracting attention for their compositionally tunable electronic structures and potential applications in electrochemical energy storage. Their development parallels the production of MXenes from layered MAX phases, $M_{n+1}AX_n$ (n = 1–3), although MBenes originate principally from metal–aluminum–boron (MAB) precursors rather than carbide or nitride MAX phases. Research on MAX phases and MXene nanocomposites provides a useful structural comparison, particularly regarding precursor composition, selective etching, surface termination, and conductive electrode design [26, 27, 59, 60]. However, this analogy does not establish equivalent synthetic accessibility, chemical stability, or hydrogel compatibility. Unlike $Ti_3C_2T_x$ MXene, whose surface groups are conventionally represented by T_x, MBene compositions and terminal groups must be established experimentally for each treatment. Consequently, a layered appearance or increased electrode capacitance cannot independently establish the formation of a chemically distinct, fully delaminated MBene phase.

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MBene synthesis can involve top–down wet-chemical etching, molten-salt processing, and high-temperature dealloying, whereas physical or chemical vapor deposition represents a possible bottom–up approach (Fig. 4(c)) [29]. In one reported MoB-related route, MoO_3 (2 g), Al (1 g), and B (3.5 g) were ground for 3 h, pelletized, and annealed under Ar at 1150 °C for 6 h using a heating rate of 180 °C h^{-1}. Subsequent ultrasonication in 15, 20, or 25 wt% NaOH for 6 h preceded hydrothermal treatment at 180 °C for 48 h. The products were repeatedly washed, centrifuged at 6000 rpm for 10 min per cycle, and dried at 80 °C overnight, yielding samples designated MoB-15, MoB-20, and MoB-25. Figure 4(d,e) presents this processing sequence alongside the preparation of a MoB-derived hydrogel architecture (MoBH) and its lyophilization [56]. These procedures illustrate how precursor processing and removal of Al-containing species influence the resulting material, but do not alone verify complete exfoliation.

Electrochemical studies further demonstrate the importance of composition and device architecture. Fluoride-free $MoAl_{1-x}B$ supercapacitor electrodes [58], layered $MoAlB/Mo_2AlB_2$ lithium-storage electrodes [61], microwave-prepared MoB sodium-ion anodes [62], and boride-based sulfur hosts [63] illustrate distinct charge-storage functions. In the system shown in Fig. 4(f–h), treating MoAlB with 1 wt% NaOH for 24 h at room temperature achieved approximately 25% Al deintercalation, producing hydroxyl-containing $MoAl_{1-x}B$ surfaces. The resulting 1/24-$MoAl_{1-x}B$ film electrode exhibited an areal capacitance of 2006.60 mF cm^{-2}, with both surface-controlled and diffusion-controlled charge-storage contributions [58]. Its all-solid-state supercapacitor incorporated a separate poly(vinyl alcohol)–LiCl (PVA–LiCl) gel electrolyte. Similarly, the boride electrodes reported by Aghamohammadi et al. employed a PVA/H_2SO_4 gel electrolyte [64]. Neither configuration, by itself, demonstrates a uniformly MBene-filled polymer hydrogel.

Recent MoO_x/MoB anodes [37] and hydrated $Mo_4/_3B_2T_2$ colloidal zinc electrolytes [38] further distinguish borides used as electrochemically active electrodes from those employed to regulate ionic transport. For comparison, Fig. 4(i,j) illustrates cation-mediated gelation of negatively charged metallic-phase MoS_2 sheets, producing an ion-stabilized membrane, F-MoS_2-H, containing approximately 85 wt% liquid. Its interconnected transport pathways support electrochemical operation at mass loadings reaching 11.2 mg cm^{-2} [36]. Although this MoS_2 system demonstrates a hydrated 2D network, its mechanism cannot be assumed to apply directly to MBenes. Future MBene–hydrogel studies should therefore establish nanosheet retention, oxidation resistance, electronic percolation, swelling behavior, and long-term mechanical stability. Performance comparisons must distinguish active boride electrodes, separate gel electrolytes, hydrated colloids, and genuinely integrated boride–polymer

networks rather than attributing device-level improvements indiscriminately to hydrogel incorporation.

2.4 Other Two-Dimensional Materials for Hydrogel-Based Energy Storage

Beyond MBenes, graphene-based frameworks [35], LDH/carbon-hydrogel electrode pairs [65], MoS_2 hydrogel membranes [36], MoS_2-assisted polymer networks [66], and carbon-nitride/MXene heterostructures [67] offer complementary design approaches. Their functions and electrochemical metrics depend on whether the 2D material constitutes an electrode, a conductive scaffold, or a component of the hydrated polymer network. Accordingly, evidence from energy storage must be distinguished from related sensing and interface studies. This distinction also prevents misleading comparisons when cycling stability, specific capacity, or flexibility arise from different measurement protocols and cell configurations.

2.4.1 Graphene and Graphene Oxide

Graphene-derived materials are attractive for hydrogel-based devices because they can establish conductive networks and reinforce polymer frameworks. Oxygen-containing groups promote graphene oxide (GO) dispersion and polymer interactions, whereas reduction increases the fraction of conductive carbon domains while altering available binding sites. Consequently, sheet dimensions, defect density, reduction degree, and electrolyte accessibility jointly influence performance. Network continuity can enhance electron collection, but excessive sheet stacking may restrict accessible surface area and impede ion diffusion, offsetting gains in bulk conductivity. Graphene-quantum-dot/graphene assemblies [68] and nitrogen-doped MXene/reduced graphene oxide (rGO) hydrogels [35] illustrate interfacial and scaffold engineering. $ZnMoO_4$/rGO photo-rechargeable electrodes [69] and NiCo-LDH/carbon-hydrogel electrode pairs [65] demonstrate alternative distributions of electroactive and conductive components.

Importantly, sodium-polyacrylate gel electrolytes coupled with rGO electrodes [70] differ structurally from nanosheet-filled polymer hydrogels. MXene@polypyrrole/rGO hydrogels employ conducting-polymer connections [71], while GO-assisted systems have been investigated for electrolyte self-healing [72]. Likewise, rGO/polyacrylamide hydrogels demonstrate low-temperature and sensing functionality [73], but these results cannot independently establish superior storage performance. Meaningful comparisons must account for polymer content, network porosity, electrolyte retention, and device configuration rather than attributing capacitance changes solely to conductivity.

2.4.2 Transition-Metal Dichalcogenides (TMDs)

Transition-metal dichalcogenides, particularly MoS_2, exhibit phase- and layer-dependent electrochemical behavior. Ionic gelation has enabled thick MoS_2 hydrogel membranes for capacitive storage [36], whereas GO-gel-coated MoS_2/MoO_2 composites have served as sodium-ion anodes [74]. Because their charge-storage mechanisms and reported metrics differ, direct capacitance–capacity rankings are inappropriate. Mechanically activated MoS_2-assisted polymerization [66] additionally demonstrates polymer-network modification rather than equivalent electrochemical performance.

Ion-coordinated PVA–MoS_2 photodetectors [75], printable alginate/MoS_2/silver-nanowire sensors [76], and multifunctional bioelectrical hydrogels [77] provide insights into processing and mixed transport. Nevertheless, their sensing performance should not be extrapolated to energy-storage devices. Future investigations should assess phase stability, scaffold contributions, and coupled mechanical–electrochemical durability under comparable conditions. Phase changes, layer restacking, and electrolyte-induced degradation also warrant examination because they may obscure the contribution of the original MoS_2 structure during prolonged cycling. Evidence for flexible TMD–hydrogel storage remains heterogeneous, and MoS_2 findings cannot be assumed to apply to WS_2.

2.4.3 Other Emerging 2D Nanosheets

Layered double hydroxides (LDHs) offer composition-tunable redox sites but generally benefit from conductive scaffolds. NiCo-LDH/rGO paired with carbon-hydrogel electrodes [65], GO/MnO_2/Co–Ni LDH hybrids [78], and cerium-containing LDH arrays [79] exemplify different architectures. Carbon nitride provides nitrogen-rich interfaces, as illustrated by sulfur-doped carbon-nitride/MXene composites [67], photo-rechargeable interfaces [80], fluorine-doped porous electrodes [81], and oxide/LDH hybrids [82]. Although these studies demonstrate promising heterostructure strategies, they do not necessarily validate polymer-hydrogel integration. Hydrogel-specific assessments of nanosheet retention, component release, mechanical fatigue, chemical compatibility, and long-term electrochemical stability remain necessary before these materials can be considered established load-bearing or multifunctional hydrogel fillers. Suitable controls should isolate each nanosheet's contribution from that of the polymer matrix and conductive additives, including mechanical testing before and after cycling.

2.5 Two-Dimensional MOFs, COFs, and Related Organic Nanosheets for Hydrogel-Based Energy Storage

Two-dimensional organic frameworks extend hydrogel composite design beyond inorganic sheets by allowing coordination environments, covalent linkages, and molecular recognition sites to be selected independently. Their attraction is chemically addressable porosity rather than an assumed combination of high conductivity and aqueous stability. A layered crystal is not necessarily an isolated nanosheet: thickness distributions, lateral dimensions, and stacking must establish the dimensionality actually incorporated into a device. Likewise, a hydrated electrode scaffold, a polymer hydrogel electrolyte, and a dried framework aerogel represent different material states. This distinction is essential when transferring results from framework electrochemistry to flexible hydrogel systems [83, 84]. Fig. 5 highlights studies linking molecular architecture to integration opportunities while critically outlining the experimental limitations and scope of each approach.

2.5.1 2D MOFs: Coordination Networks, Pore Architecture, and Redox/Electronic Transport

In layered MOFs, metal–ligand connectivity defines the in-plane network, while interlayer interactions control stacking

and access to channels. Conjugated ligands coordinated to suitable metal centres can support electronic delocalization, whereas many carboxylate and imidazolate frameworks remain poorly conducting. Fig. 5(a) reproduces the layered Ni3(HITP)2 structure, illustrating coordination-linked sheets and accessible pore space [85]. Its source investigates oxygen reduction, so the structural illustration supplies no hydrogel-storage performance evidence. Additive-free Ni3(HITP)2 supercapacitors nevertheless establish that an intact conductive framework can serve directly as a charge-storage electrode [86]. Such results should be separated from pyrolysed MOF-derived carbons, whose coordination chemistry and electronic structure have changed.

Solvothermal growth, interfacial assembly, substrate-directed synthesis, and exfoliation offer different compromises between crystallinity, thickness, yield, and electrical contact. Direct growth can reduce binder resistance, but substrate coverage and adhesion may limit scale. Exfoliation exposes sites and shortens diffusion distances while introducing defects, fragmentation, and restacking during drying. The conductivity of a pressed powder also includes grain-boundary and contact contributions; it cannot simply be assigned to a dispersed sheet in a swollen gel. A systematic Ni3(HITP)2 comparison showed that morphology can govern capacitance more strongly than bulk conductivity [87]. Hierarchical ion pathways and interfacial contact therefore require optimization alongside intrinsic electronic transport.

Redox-active metals and ligands may provide Faradaic storage, but a metal-containing framework does not automatically behave pseudocapacitively. In Ni3(HITP)2 with aqueous NaTFSI, operando scattering supports persistent anion anchoring and predominantly sodium-ion redistribution, rather than ion intercalation [88]. This electrolyte-specific mechanism underlies Fig. 5(g). In alkaline systems, coordination networks can also reconstruct into hydroxides or oxyhydroxides; post-cycling diffraction alone may miss a thin active surface layer. Operando spectroscopy, metal-release measurements, and appropriate precursor controls are needed before assigning capacity to the original framework. These distinctions become especially consequential when hydrogel ligands compete for metal sites or change the local pH.

2.5.2 2D COFs: Linkage Chemistry, Conjugation, and Electroactive Pores

COFs replace coordination bonds with extended covalent connectivity, enabling pore geometry and electroactive functionality to be programmed through monomer selection. Boronate linkages offer structural order but can be hydrolytically vulnerable; imine stability depends on substitution and electrolyte conditions. β-Ketoenamine, imide, and carbon-linked architectures offer alternative stability profiles rather than universal aqueous resistance. Exfoliation, interfacial polymerization, and confined growth can produce thin layers, but thickness and crystallinity require independent verification [83]. A low-angle diffraction peak alone neither proves monolayer formation nor establishes continuous through-plane ion channels in the hydrated composite.

Electroactive carbonyls, diimides, quinones, and nitrogen-containing heterocycles introduce ion-coupled redox centres. Fig. 5(b) shows the redox-active diimide walls and channel architecture of DTP–ANDI–COF, whose integration with carbon nanotubes illustrates the need to connect insulating or weakly conducting framework domains electronically [89]. The source demonstrates lithium-battery electrodes, not hydrogel devices. Conjugation and ordered stacking can assist transport, but neither guarantees useful macroscopic conductivity. Polymer infiltration may improve wetting while obstructing precisely the pores intended to expose redox sites. Consequently, pore accessibility should be measured after composite formation rather than inferred from nitrogen adsorption on the isolated powder.

The COF/rGO aerogel route in Fig. 5(d) shows another useful division of labour: graphene provides a connected scaffold, while the anthraquinone-containing framework contributes redox functionality and modifies assembly [90]. Its hydrogel precursor is freeze-dried before electrode evaluation, making it processing evidence rather than proof of an operating polymer-hydrogel composite. Triazine frameworks broaden the chemistry through nitrogen-rich pores and robust linking motifs, although high-temperature synthesis can generate disorder that complicates structural assignments [91]. Across these systems, more redox groups increase theoretical capacity only if electrons and compensating ions can reach them at the relevant loading and charging rate.

2.5.3 Other Organic 2D Nanosheets

2.5.3.1 Conjugated Polymer Nanosheets and Organic Electronic Networks

Conjugated polymer sheets span covalently connected two-dimensional networks and thin assemblies of conventional polymer chains. These categories should remain distinct. Solid-state coupling of preorganized phenazine precursors produces a layered conjugated aromatic polymer that can be exfoliated, as shown by the synthesis scheme and microscopy in Fig. 5(c) [92]. By contrast, ice templating organizes PEDOT:PSS into sheets through confinement and phase segregation, without creating a crystallographically defined two-dimensional covalent lattice [93]. Both approaches offer accessible interfaces, but their transport anisotropy, defect populations, and processing histories differ.

Within a hydrogel, conducting polymers can bridge poorly connected framework particles or supply their own doping-related charge storage. Their conductivity depends on oxidation state, counterions, hydration, and pH, so dry-film values provide an incomplete benchmark. Repeated ion uptake can cause swelling, dedoping, and mechanical fatigue. Carbon or metal-oxide reinforcement may improve continuity but dilutes the organic fraction and introduces additional electrochemical contributions. Thus, framework-free polymer controls and equal-loading comparisons are essential. The flexible devices reported for ice-templated polymer composites demonstrate processability and device fabrication; they do not establish that every conjugated sheet remains conductive inside an independently crosslinked hydrogel [93].

2.5.3.2 Hydrogen-Bonded Organic Frameworks and Molecular Assemblies

Hydrogen-bonded frameworks exploit directional noncovalent interactions, frequently combined with π-stacking, to organize

electroactive molecules. Reversibility can facilitate assembly but also allows competitive water binding, dissolution, and reorganization. The BBQPH framework combines carbonyl and nitrogen-containing sites with intermolecular locking; its aqueous zinc-battery study supports coupled Zn2+/H+ storage [94]. Fig. 5(e) presents the hydrogen-bonded structure and discharge profiles with ion-storage assignments specific to the investigated electrolytes. The reported capacity of 498.6 mAh g−1 at 0.2 A g−1 is normalized to cathode active mass, not the packaged cell. Its liquid-electrolyte architecture establishes molecular-storage feasibility, while hydrogel incorporation remains a separate validation step. Stable cycling of one hydrogen-bonded assembly cannot establish wet stability for the entire material class.

2.5.4 Hydrogel Integration: Interfacial Coupling, Ion Transport, and Flexible Device Roles

Hydrogel integration comprises several experimentally distinguishable architectures. Framework particles or sheets may reside within the electrolyte as ion-regulating fillers; a framework-rich electrode may contact a separate gel; or a hydrated, electronically connected framework composite may itself function as an electrode. These configurations distribute electronic conduction, ionic transport, and mechanical support differently. Framework-containing electrolyte films should remain electronically insulating across their thickness, whereas electrodes require continuous electron pathways. Adding a conductive sheet to an electrolyte therefore demands checks for leakage current and internal short circuits, alongside any increase in ionic conductivity.

Direct integration is exemplified by a sulfonic-acid-functionalized COF/PVA gel electrolyte developed for zinc-ion batteries [95]. The proposed function involves Zn2+ interactions and deposition regulation, rather than using the COF as a high-conductivity electrode. This is stronger evidence of a framework-containing hydrogel device than a powder electrode tested in liquid electrolyte. However, gel incorporation does not by itself establish exfoliated nanosheet morphology or resolve the individual contributions of sulfonation, polymer hydration, and framework porosity. Matched PVA, nonfunctionalized-COF/PVA, and sulfonated molecular-additive controls would make that attribution more convincing. Temperature-dependent durability should additionally distinguish electrolyte water retention from intrinsic framework heat resistance.

Hydrogen bonding between polymer hydroxyl or amide groups and framework surfaces can improve dispersion and stress transfer. Coordination of polymer carboxylates with exposed metal sites offers stronger anchoring but may compete with the original linker environment. Electrostatic association and π interactions can connect charged gels or conjugated scaffolds to organic sheets. Stronger coupling is not invariably beneficial: extensive adsorption may occlude redox sites, suppress swelling, or immobilize ions. Conversely, weak attachment can permit aggregation and filler migration during repeated deformation. Spectroscopic shifts should therefore be combined with swelling measurements, extraction experiments, and hydrated mechanical tests rather than treated as sufficient proof of a durable interface.

COF membranes that regulate interfacial water at zinc anodes provide adjacent mechanistic guidance [96]. They should be identified as protective interlayers unless the experiment also incorporates a hydrated polymer network. Similarly, a triazole-based covalent gel demonstrates that an organic gel can stabilize zinc plating, but it should not be relabelled a crystalline two-dimensional COF without structural evidence [97]. Such studies suggest how local coordination, water activity, and transport selectivity might be combined in hydrogel formulations. Their performance cannot be transferred quantitatively because polymer-bound water, free electrolyte, and confined framework water have different mobilities and reaction environments.

For ion transport, polar pore walls can alter solvation and ion partitioning, while macroporous hydrogel channels provide access between framework domains. These length scales must work together. A high salt concentration can increase charge-carrier density yet reduce mobility through ion association or network contraction. Increased filler loading can reinforce a gel while raising tortuosity or interrupting polymer continuity. Impedance-derived conductivity should be reported with thickness, electrode area, temperature, hydration history, and contact conditions. Transference numbers and diffusion measurements are especially useful when selective cation transport is claimed, because total conductivity alone does not identify which species carries current.

Fig. 5(f), adapted from an experimentally demonstrated alginate/PEDOT-containing hydrogel supercapacitor, provides a relevant device-level comparison [98]. Its electronically modified outer gel layers and ion-conducting separator demonstrate the spatial organization of distinct functional components within a hydrogel-based device. Importantly, the reported architecture does not incorporate MOF or COF materials; rather, it offers a conceptual basis for future designs in which framework nanosheets could be positioned near the electrode interfaces while retaining an electrically insulating gel core. In comparison, Fig. 5(d) illustrates a framework-containing hydrogel-to-aerogel processing strategy [90], whereas Fig. 5(g) presents an experimentally supported pore-charging mechanism in a liquid-electrolyte system [88]. Collectively, these examples provide complementary insights into material processing, device architecture, and charge-storage mechanisms without suggesting that an integrated framework–hydrogel energy-storage device has already been experimentally demonstrated.

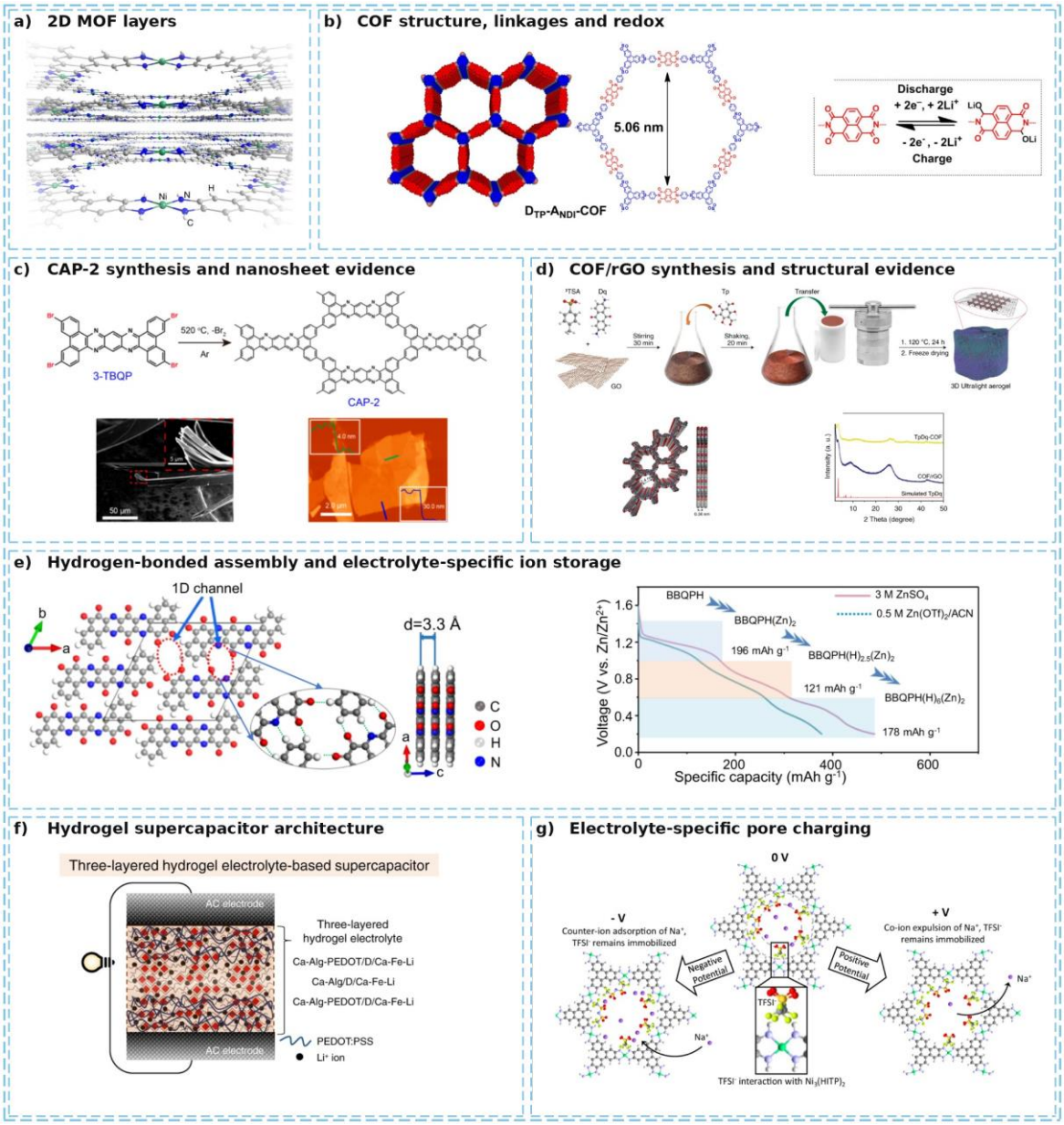


Fig. 5. Structural characteristics, synthesis strategies, charge-storage mechanisms, and experimental configurations of organic framework-based materials and hydrogel energy-storage systems. a) Layered coordination structure of the conductive metal–organic framework $Ni_3(HITP)_2$, highlighting its two-dimensional network architecture. Reproduced with permission from Ref. [85]. Copyright 2016, Nature. b) AA-stacked DTP–ANDI–COF structure, illustrating boronate-linked porous channels and diimide-based lithiation redox processes. Reproduced with permission from Ref. [89]. Copyright 2015, Nature. c) Synthesis and morphological characterization of CAP-2, including scanning electron microscopy (SEM), atomic force microscopy (AFM), and corresponding nanosheet height profiles. Selected panels from Ref. [92]. Copyright 2025, American Chemical Society. d) Synthesis of COF/rGO composites, molecular structure of TpDq-COF, and corresponding powder X-ray diffraction (PXRD) patterns, illustrating the framework-containing hydrogel-to-aerogel fabrication process through freeze-drying before electrode testing. Reproduced with permission from Ref. [90]. Copyright 2018, American Chemical Society. e) Hydrogen-bonded molecular assembly of BBQPH and corresponding discharge profiles, highlighting electrolyte-dependent Zn^{2+}/H^+ charge-storage mechanisms. Reproduced with permission from Ref. [94]. Copyright 2023, Nature. f) Schematic configuration of a supercapacitor incorporating activated-carbon electrodes and a three-layer alginate/PEDOT-containing hydrogel electrolyte, demonstrating spatial separation of electronic and ionic transport functions without incorporating MOF or COF materials. Reproduced with permission from Ref. [98]. Copyright 2022, Nature. g) Proposed Na^+ adsorption and expulsion mechanism in $Ni_3(HITP)_2$, involving immobilized $TFSI^-$ anions in aqueous 1 M NaTFSI electrolyte. Reproduced with permission from Ref. [88]. Copyright 2025, Nature. The composite incorporates selected panels from the original publications, retaining their internal labels, scale bars, axes, and legends, while external labels a)–g) identify the assembled panels. These studies provide complementary experimental evidence for framework structures, electrode processing, ion-storage mechanisms, and hydrogel device architectures but do not establish an experimentally demonstrated integrated framework–hydrogel energy-storage device.

2.5.5 Critical Comparison: Conductivity, Wet Stability, Processing, and Device-Level Evidence

Conductive MOFs offer electronically addressable coordination networks, COFs offer modular covalent pore chemistry, and conjugated polymers offer accessible electronic pathways with comparatively flexible processing. Hydrogen-bonded assemblies add reversible molecular organization, but their persistence depends strongly on solvent and packing. None provides an unconditional advantage. Interfacial engineering of MOF/graphene electrodes reinforces the importance of

contact chemistry and hybrid architecture [99], while COF supercapacitor literature shows how conductive supports and linkage selection affect utilization [100]. Comparisons should therefore identify the conducting phase, active redox phase, ion-bearing phase, and mechanical scaffold explicitly. A hybrid's performance should not be assigned entirely to its framework component.

Wet stability requires simultaneous retention of chemistry, accessible porosity, particle morphology, and interfacial contact. Dry-state crystallinity after cycling can coexist with leaching, local amorphization, or reconstruction. Relevant protocols include prolonged electrolyte soaking, leachate analysis, spectroscopy at controlled state of charge, and mechanical testing after electrochemical aging. The electrolyte used matters as much as nominal linkage strength: concentrated salts, acidic gels, and alkaline electrolytes impose different coordination and hydrolysis conditions. The COF electrochemistry literature provides a broad foundation for such chemistry-specific assessment, but demonstrations in fuel-cell membranes or nonaqueous batteries should remain separate from hydrogel-storage claims [84].

Electronic and ionic conductivities must also be separated experimentally. Blocking-electrode measurements can estimate ionic resistance, whereas electronic leakage requires an appropriate steady-state measurement and correction for polarization. Results obtained at different water contents should not be pooled into a single material ranking. In a mixed conductor, a smaller impedance semicircle may reflect improved electrode contact rather than faster transport inside framework pores. Hydrated microscopy, thickness-dependent impedance, and transport measurements before and after filler extraction can help discriminate these possibilities.

Processing and loading impose equally important limits. Solvothermal reactions, solvent exchange, catalyst removal, exfoliation, and freeze-drying can dominate cost and reproducibility. Scale-up should report isolated yield, batch variability, solvent consumption, and whether sheet dimensions survive mixing and gelation. At practical electrode thickness, increased loading can amplify ionic gradients and electronic resistance even when low-loading gravimetric performance appears excellent. Areal capacity, volumetric energy, active-material fraction, and complete-device mass are therefore needed alongside gravimetric metrics. The mass and volume of electrolyte, current collectors, separators, and encapsulation must be included when claiming device-level energy density.

A defensible evaluation sequence begins with isolated-sheet characterization, proceeds to the equilibrated composite, and finishes with a sealed two-electrode device. Each stage should retain a mass balance for active material, supporting conductor, and polymer. Dissolution controls should distinguish reversible ion binding from irreversible loss of organic molecules or metal nodes. Aging under open circuit is also necessary because electrochemical cycling alone can conceal calendar degradation and self-discharge. These measurements would clarify whether an apparently stable capacity originates from persistent framework chemistry or from compensating changes elsewhere in the composite.

Future progress depends on testing coupled failure modes rather than adding isolated functionality. Bending radius, deformation rate, cycle count, temperature, and water loss should accompany electrochemical retention data; a photograph of a bent device establishes configuration, not fatigue resistance. Particularly useful directions include oriented ion pathways, electronically connected framework electrodes with insulating gel cores, and chemically anchored interfaces that preserve pore accessibility. Biomedical applications additionally require formulation-specific extractables, cytotoxicity, and degradation-product evidence. Neither organic composition nor a natural-polymer matrix establishes biodegradability or biocompatibility. The immediate opportunity is reproducible architecture-specific validation that connects molecular design with sustained performance under realistic hydration, loading, and mechanical conditions.

2.6 Comparative Structure–Property Relationships and Design Principles in 2D Material–Hydrogel Systems

The electrochemical performance of 2D material–hydrogel systems is governed by the interplay among conductive networks, ion-accessible pathways, surface chemistry, interlayer spacing, and mechanical integrity. Figure 6 illustrates representative fabrication strategies and structure–property relationships in MBene-, MXene-, and MoS_2-based systems, while Table 1 provides a broader comparison with other 2D material–hydrogel architectures. These systems differ in how they distribute the conductive skeleton, ion-bearing phase, electrochemically active sites, and mechanically supporting network, ultimately influencing charge transport, electrolyte accessibility, and structural stability.

MBene-based systems. Figure 6(a,b) illustrates the design of a flexible MoB MBene-based interdigitated supercapacitor electrode and the chemical etching route used to obtain 2D MoB from the MoAlB precursor. Selective removal of the Al-containing layers facilitates the formation of 2D MBene structures, while the interdigitated electrode geometry provides a compact configuration for electrochemical energy storage [64]. The electrode architecture and processing conditions influence the accessibility of active surfaces and the effectiveness of current collection. In comparison, MBene colloids can regulate ion transport through nanosheet interactions and colloidal organization without necessarily forming a polymer-based hydrogel network, demonstrating an alternative approach to controlling transport in 2D materials [38].

MXene–carbon hybrid hydrogels. Figure 6(c,d) presents the configuration of a zinc-ion hybrid supercapacitor and the synthesis of an N-doped $Ti_3C_2T_x$–reduced graphene oxide (RGO) hybrid hydrogel. Following selective etching of Ti_3AlC_2 and nitrogen doping, $Ti_3C_2T_x$ is combined with graphene oxide (GO) and subjected to hydrothermal treatment in the presence of L-ascorbic acid. The resulting interconnected MXene–RGO architecture can improve electrical connectivity, reduce nanosheet restacking, and facilitate electrolyte-ion transport through the three-dimensional network [35]. These characteristics highlight the importance of combining

conductive 2D sheets with a supporting carbon framework to balance electronic transport and ion accessibility.

Cation-mediated MoS_2 hydrogel membranes. Figure 6(e,f) illustrates the cation-mediated assembly and formation of the F-MoS_2-H membrane, whereas Fig. 6(g,h) presents its structural characterization. Cation intercalation partially screens electrostatic repulsion between negatively charged MoS_2 nanosheets and promotes interlayer association and ionic gelation. The reported interlayer spacing decreases from approximately 15.0 Å in the hydrated MoS_2-H membrane to 9.8 Å following BMIM-BF_4-mediated assembly. Changes in zeta potential further support the role of ionic interactions in regulating nanosheet assembly [36]. Cross-sectional scanning electron microscopy (SEM) and grazing-incidence wide-angle X-ray scattering (GIWAXS) reveal differences in stacking morphology and nanosheet organization between the dried D-MoS_2 and F-MoS_2-H membranes. These results demonstrate that controlled interlayer interactions and water retention can influence membrane structure, swelling resistance, and ion-transport pathways [36].

Broader structure–property relationships. Related LDH/carbon-hydrogel systems [65] and reduced graphene oxide/polyacrylate cells [70] further illustrate how the integration of conductive phases with supporting networks influences electrochemical and mechanical behavior. Across these architectures, surface functionalization and interfacial bonding regulate hydration and nanosheet interactions, while increased interlayer spacing can improve ion accessibility only when adequate electronic connectivity is maintained. Likewise, polymeric or carbon-based reinforcement can enhance structural integrity but may restrict swelling and ion mobility when the network becomes excessively dense. Therefore, optimizing 2D material–hydrogel systems requires balancing electronic conductivity, ionic transport, active-site accessibility, and mechanical stability rather than maximizing any single structural parameter. Figure 6 and Table 1 collectively summarize these design principles without implying a universal performance ranking across materials with different compositions, electrolytes, electrode configurations, and testing conditions.

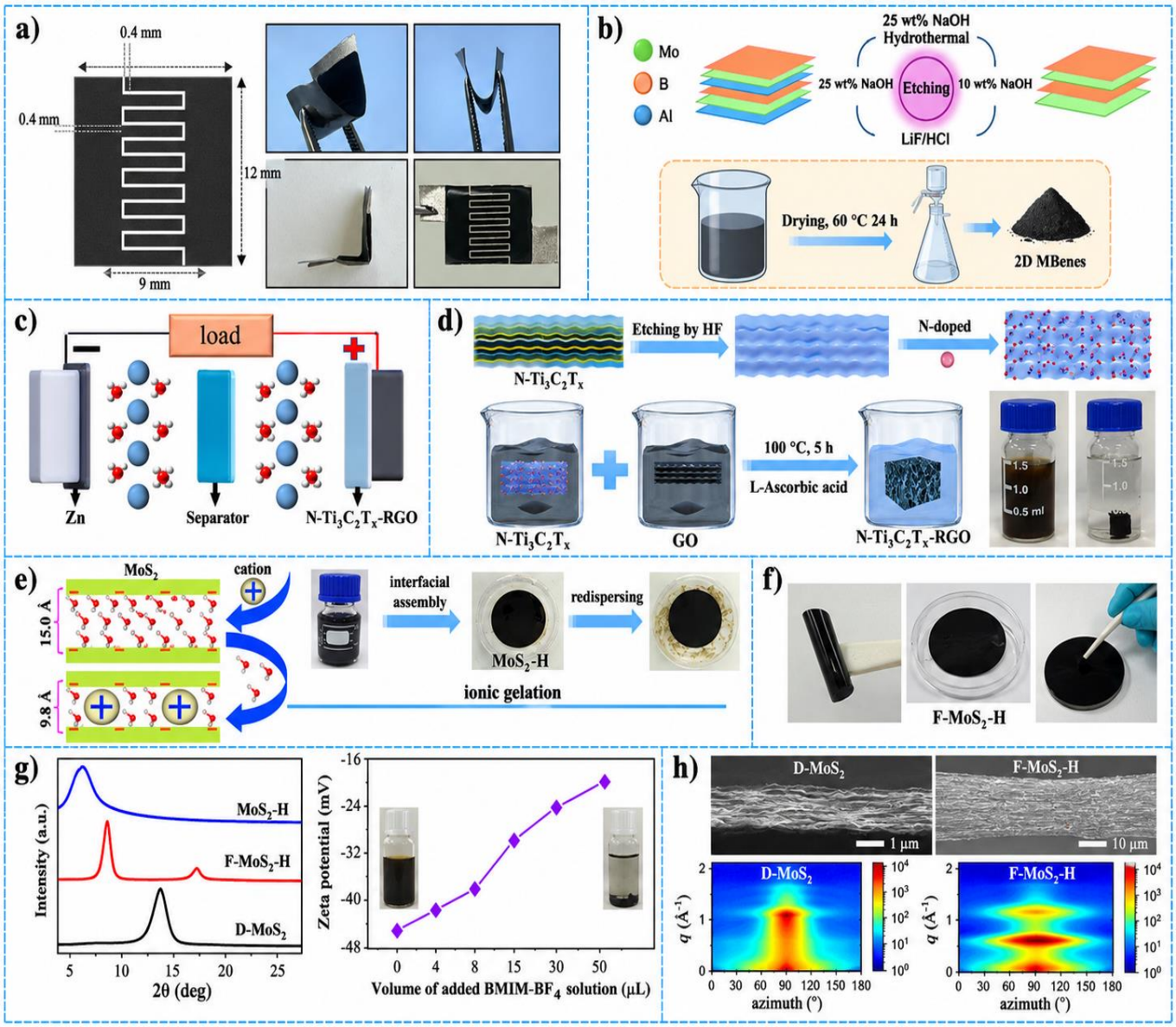


Fig. 6. Representative fabrication strategies, assembly mechanisms, and structural characterization of 2D material-based electrodes and hydrogel membranes for electrochemical energy storage. (a) Design and flexibility of a MoB MBene-based interdigitated electrode. (b) Chemical etching and preparation of 2D MoB MBenes from MoAlB. (c) Schematic configuration of a zinc-ion hybrid supercapacitor employing Zn and N-$Ti_3C_2T_x$–RGO electrodes. (d) Synthesis of N-doped $Ti_3C_2T_x$ and fabrication of the N-$Ti_3C_2T_x$–RGO hybrid hydrogel. (e) Cation-mediated intercalation and ionic gelation of MoS_2 nanosheets. (f) Photographs of the F-MoS_2-H membrane and its preparation. (g) X-ray diffraction (XRD) patterns and zeta-potential variations associated with BMIM-BF_4 addition. (h) Cross-sectional SEM images and 2D GIWAXS patterns of D-MoS_2 and F-MoS_2-H membranes. Panels (a,b) adapted from Ref. [64] under its applicable Creative Commons Attribution 4.0 International License (CC BY 4.0).; panels (c,d) and panels (e–h) Reproduced with permission from Ref. [35]. Copyright 2026 Elsevier B.V." and "Reproduced with permission from Ref. [36]. Copyright 2025 Elsevier B.V.

2.7 Structure–Property Relationships in Ion Transport and Charge Storage

The electrochemical performance of hydrogel–MXene composites depends on the simultaneous continuity of electron-conducting networks, ion-accessible hydrated domains, and mechanically resilient interfaces. Mobile ions migrate through interconnected aqueous pores, polymer-associated ionic regions, and interlayer galleries before reaching electrochemically active surfaces. Accordingly, high electronic conductivity alone cannot guarantee high capacitance: tightly restacked nanosheets may provide efficient electron transport while excluding electrolyte from substantial fractions of the electrode. Conversely, excessive swelling or porosity can increase ionic accessibility but decrease packing density, contact integrity, and volumetric energy storage. Rational design therefore requires coordinated optimization of gallery spacing, surface chemistry, pore connectivity, polymer cross-linking, and the electrode–electrolyte interface [5, 29, 41, 43, 101].

Charge storage can arise from electric-double-layer charging, reversible surface or near-surface redox reactions, and ion intercalation; battery-type insertion or conversion reactions may also contribute in some hybrid systems. These mechanisms cannot be distinguished solely from a high capacitance value. Cyclic voltammetry (CV), galvanostatic

charge–discharge (GCD), electrochemical impedance spectroscopy (EIS), and scan-rate-dependent kinetic analyses provide complementary information on transport limitations and charge-storage contributions. Moreover, compression, stretching, and bending may redistribute pore water, shift nanosheet contacts, or change diffusion distances. Measurements conducted only in an undeformed state therefore cannot establish reliability for wearable devices or soft robotic systems [25, 32, 45, 101, 102]. Figure 7 connects these considerations to three experimentally reported strategies: conductive-polymer confinement, aluminum-assisted hydrogel self-assembly, and ionic-liquid-mediated MXene intercalation.

Cheng et al.[42]. developed an organic–inorganic hydrogel that integrates poly(3,4-ethylenedioxythiophene)(styrene sulfonate) (PEDOT), carboxyl-functionalized titanium carbide MXene (C-MXene), and poly(vinyl alcohol) (PVA) cross-linked with glutaraldehyde (GA) [42]. As outlined in Fig. 7(a), PVA was first dissolved in a PEDOT dispersion under heating and stirring. Addition of C-MXene promoted interactions with PEDOT and altered PEDOT chain conformation; GA subsequently cross-linked PVA to establish a mechanically compliant network. Acetic acid treatment reduced the insulating PSS fraction, enhancing continuity of the conductive component. An all-hydrogel supercapacitor was then constructed by placing a PVA–H_3PO_4 gel electrolyte between two conductive hydrogel electrodes. This architecture deliberately separates the ion-bearing electrolyte from the electronically conductive, mechanically deformable electrodes while allowing intimate contact across their interfaces.

The resulting morphology illustrates how nanoscale confinement influences macroscopic behavior. Photographs in Fig. 7(b) demonstrate deformation by bending and twisting, while cross-sectional SEM in Fig. 7(c) shows a layered, porous framework rather than the particulate morphology characteristic of the original PEDOT dispersion. According to the reported interpretation, PEDOT transitions toward more extended chains and occupies spaces between stacked C-MXene nanosheets. Electrostatic interactions, hydrogen bonding, conductive-chain entanglement, and covalent cross-linking then support interconnected electron pathways and mechanical stress transfer [42]. This synergy is composition-sensitive: insufficient MXene limits conductive reinforcement, whereas excessive loading promotes aggregation and undermines homogeneous strain distribution. Thus, hydrogel elasticity and electrical conductivity must be evaluated together, instead of optimizing nanosheet content solely for its nominal conductivity.

A contrasting strategy uses aluminum-assisted reduction and self-assembly of MXene with graphene oxide (GO) to form a three-dimensional MXene@reduced graphene oxide (rGO) hydrogel [102]. In Fig. 7(d), the initial MXene/GO suspension remains fluid after Al powder is added, whereas introducing hydrochloric acid (HCl) triggers gelation. The acid participates in the Al-mediated process, but larger acid quantities accelerate gas generation and may damage network uniformity. The reported preparation therefore selected 80 µL HCl as a practical processing condition. Controls containing MXene without GO, or Al^{3+} ions instead of metallic Al, did not yield comparably self-supporting networks. These comparisons indicate that Al serves a role beyond simple ionic charge screening, while GO contributes to a continuous scaffold. Importantly, substituting Al^{3+} for Al powder caused flocculation rather than equivalent gel formation, highlighting the significance of the coupled reduction and assembly process [102].

Network integrity depended strongly on precursor proportions. In the reported compression observations, a specimen prepared with 0.15 mL GO collapsed under a 5 g load, whereas increasing GO to 0.30 mL improved shape retention. Raising the Al-powder content accelerated gelation but, at excessive concentrations, generated aggregation and residual particles. SEM observations in Fig. 7(e) show a comparatively compact MXene@rGO-Al-7.5 architecture before acid washing, followed by a more open network after removal of metallic Al and immersion in deionized water. GO-deficient assemblies displayed cracking, and excessive Al^{3+} promoted large agglomerates. These results underline an important structural trade-off: a sufficiently connected scaffold is needed to withstand handling, but excessively compact domains or processing residues can obstruct uniform electrolyte infiltration. The observations support improved accessibility as a plausible architectural benefit; they do not alone quantify the fraction of electrochemically utilized surface [102].

Diffraction and Raman analysis provide complementary evidence for assembly, reduction, and possible degradation. As shown in Fig. 7(f), the MXene-related (002) reflection in assembled gels shifts toward lower diffraction angles; reported peaks near $2\theta = 6.28°$ indicate expanded layer periodicity consistent with intercalation, although spacing changes alone do not identify the inserted species unambiguously. Reflections near 39° and 45°, assigned to metallic Al, disappear after acid washing, supporting its removal within the sensitivity of XRD. Additional reflections assigned to TiO_2 in the Al^{3+}-treated control raise concerns about concurrent MXene oxidation. Raman bands near 1375 and 1580 cm^{-1} correspond to defect-associated D and graphitic G vibrations, respectively. The reported G-band shift from 1591.75 to 1576.42 cm^{-1} and increase in I_D/I_G from 0.957 to 1.028 are consistent with structural changes accompanying GO reduction; neither metric alone proves complete chemical reduction. Taken together, SEM, XRD, and Raman measurements identify processing-dependent changes in porosity, layer arrangement, and carbon structure [102].

A third route targets the ion-accessible galleries themselves. Huang et al. investigated $Ti_3C_2T_x$ prepared using 1-butyl-3-methylimidazolium tetrafluoroborate ($BMImBF_4$), which functions as both an etching medium and an intercalating species [103]. The Ti_3AlC_2 MAX precursor in Fig. 7(g) exhibits densely stacked layers, while the processed MXene in Fig. 7(h,i) displays a more open, accordion-like morphology. XRD patterns in Fig. 7(j) show displacement of the (002) reflection to $2\theta = 6.63°$, corresponding to an apparent interplanar spacing of approximately 13.3 Å for Cu Kα radiation. For comparison, the supplied study places conventional HF-etched material near 8.6°–8.9°. Incorporation of $BMIm^+$ into expanded

interlayer regions is consistent with the accompanying structural evidence. Nevertheless, an enlarged interlayer distance is not a direct measurement of ion-diffusion rate and must be interpreted alongside electrochemical measurements [103].

Surface analysis adds a necessary chemical dimension. The FTIR spectrum in Fig. 7(k) contains bands reported at 603, 700–750, 1033, 1238, 1609, 1716, 2896, and 3256 cm^{-1}, assigned in the source to Ti–C, Ti–O, C–F, –COOH, C–O, C=O, C–H, and –OH-related vibrations, respectively [103]. Because some infrared features overlap, these assignments should be regarded as supportive rather than individually conclusive without complementary surface analysis. Raman features in Fig. 7(l) also indicate changes associated with oxidized surface species. Together, interlayer expansion and surface functionalization influence wetting, accessible charge-storage sites, and electrode stability. The ionic-liquid-treated MXene is not itself described here as a polymer hydrogel; its relevance to hydrogel composites lies in demonstrating how controlled gallery chemistry may mitigate restacking and facilitate ion access.

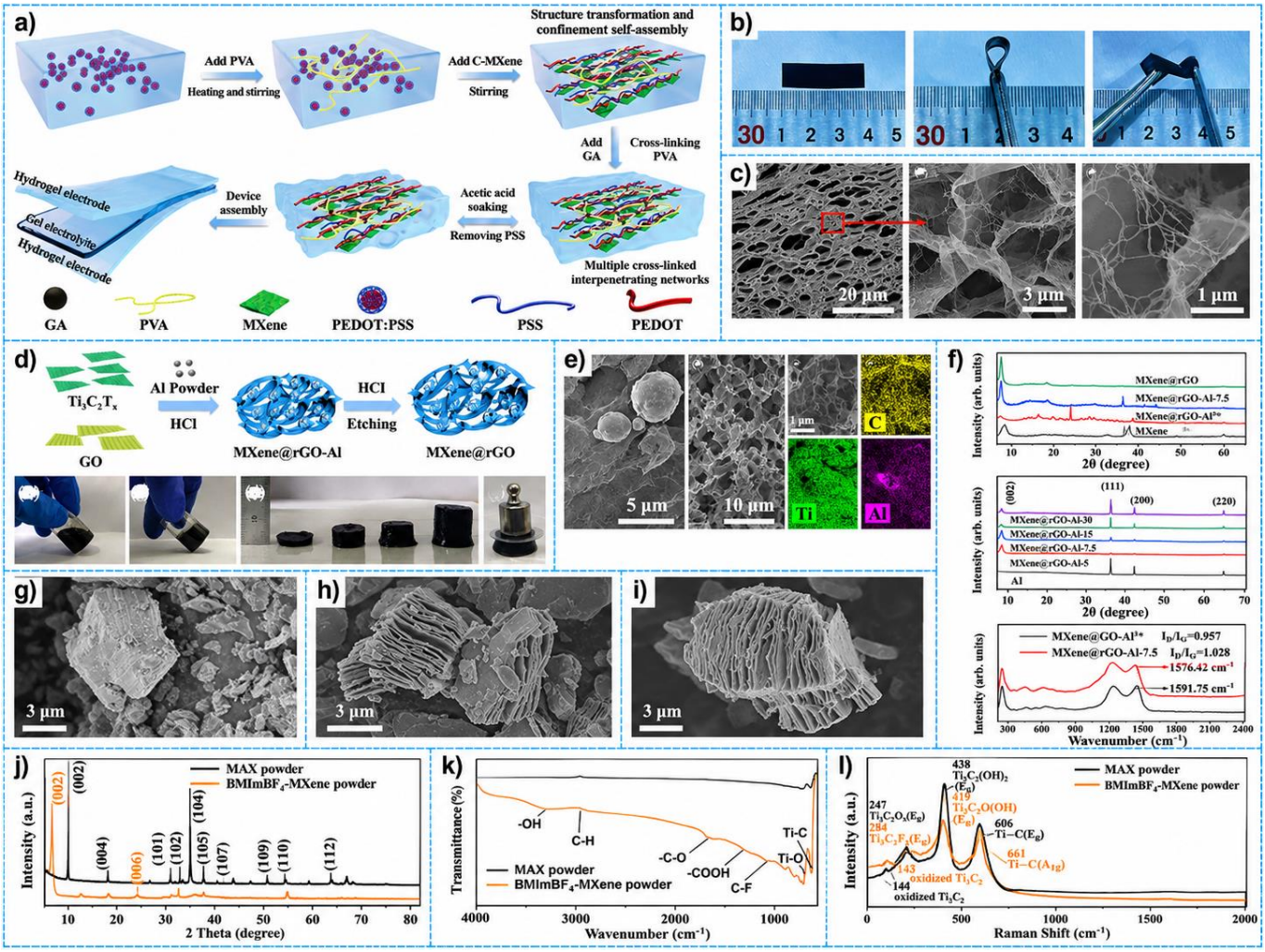


Fig. 7. Fabrication, morphology, assembly mechanisms, and structural characterization of MXene-based hydrogel electrodes. (a) Preparation of PEDOT/C-MXene/GA-PVA conductive hydrogel and assembly of an all-hydrogel supercapacitor; (b) photographs before and during bending or twisting; and (c) cross-sectional scanning electron microscopy (SEM) of its layered porous structure. Reproduced from Ref. [42]. Copyright 2024, the authors; published by Wiley-VCH under the Creative Commons Attribution 4.0 International License. (d) Aluminum-assisted assembly of MXene and graphene oxide into MXene@rGO-Al hydrogels, including photographic evidence of gel formation and load-bearing behavior; (e) corresponding microstructural observations; and (f) diffraction and spectroscopic characterization of the assembled networks. Reproduced from Ref. [102]. Copyright 2023, Elsevier; reproduction permission should be confirmed. (g) SEM of Ti_3AlC_2 MAX precursor; (h,i) SEM of ionic-liquid-processed $Ti_3C_2T_x$ MXene; (j) X-ray diffraction (XRD); (k) Fourier-transform infrared (FTIR) spectra; and (l) Raman spectra. Reproduced from Ref. [103]. Copyright 2026, the authors; published by the American Chemical Society under the Creative Commons Attribution 4.0 International License.

For galvanostatic discharge, gravimetric capacitance is commonly calculated using:

$$C_g = \frac{I\Delta t}{m\Delta V} \quad (6)$$

where I is discharge current, Δt is discharge time, m is active material mass, and ΔV is the corrected voltage window.

For thin films and microdevices, areal capacitance is often more meaningful:

$$C_A = \frac{I\Delta t}{A\Delta V} \quad (7)$$

Areal capacitance is especially relevant when the device footprint is constrained by the size of a wearable sensor or circuit module.

Volumetric capacitance is calculated as:

$$C_V = \frac{I\Delta t}{V_e \Delta V} \quad (8)$$

where V_e is the electrode or device volume, depending on the stated normalization basis.

The ideal stored energy of a capacitive device scales with capacitance and the square of voltage window:

$$E = \frac{1}{2} C (\Delta V)^2 \quad (9)$$

This relationship explains why extending the stable voltage window is highly effective, but only if safety and electrolyte stability are maintained.

Average power density is obtained by dividing the delivered energy by discharge time:

$$P = \frac{E}{\Delta t} \quad (10)$$

Power requirements should be evaluated according to application-specific load profiles, particularly for actuators, wireless communication systems, and wearable sensors operating under different duty cycles.

In Eq. (9), C and E must use compatible absolute or normalized bases. If C is total capacitance, E is energy in joules; if capacitance is mass- or volume-normalized, the resulting energy uses that same basis. Equation (10) yields average power or corresponding power density, depending on whether E represents total energy or normalized energy. These idealized expressions should not replace integration of a substantially nonlinear discharge profile. Extending the voltage window can increase energy, but electrolyte stability, parasitic reactions, leakage, and deformation-induced resistance changes constrain practical operation.

Mechanical history introduces another source of variability that cannot be inferred from the initial pore structure. Under repeated stretching or compression, the hydrated polymer network may redistribute liquid, change pore dimensions, and alter the orientation or overlap of conductive sheets. Some changes can reverse upon unloading, whereas accumulated microcracks, disrupted cross-links, and interfacial delamination may cause persistent increases in resistance. In situ or repeated-cycle EIS, mechanical hysteresis tests, and post-deformation microscopy should therefore be interpreted together. Reporting capacitance retention without the accompanying deformation amplitude, cycle count, recovery period, or moisture conditions can obscure the true operational durability of a flexible electrode [25, 32, 45].

Overall, Fig. 7 demonstrates complementary approaches to controlling structure–property relationships: confinement of conductive polymer chains between MXene sheets [42], formation of a self-supporting MXene/rGO network [102], and tuning of interlayer spacing through ionic-liquid processing [103]. Each addresses a different limitation, so their performances should not be ranked without matching electrode mass, device geometry, electrolyte, and operating conditions. Long-term testing should combine repeated mechanical loading with EIS, structural imaging, water-retention measurements, and capacitance tracking to distinguish reversible rearrangement from cracking, oxidation, or permanent loss of electronic contact [25, 32, 45]. Such coordinated assessment is essential for reliable hydrogel–MXene energy-storage systems.

Table 1 consolidates the two-dimensional families discussed in Section 2. Entries identify demonstrated roles and proposed transfers separately; constituent conductivity or calculated storage capability is not substituted for packaged-device performance.

Table 1. Representative two-dimensional materials for hydrogel-based energy storage, summarizing their compositions, preparation, integration strategies, properties, applications, and supporting evidence.

Family and representative composition	Properties and preparation	Hydrogel integration and functionalization	Advantages, limitations and applications	Recent evidence
MXenes; $Ti_3C_2T_x$, $V_4C_3T_x$; $M_{n+1}X_nT_x$	Conductive carbides/nitrides with terminations. Selective etching, washing and delamination; chemistry depends on route.	In situ polymerization, conductive cross-linking, PVA electrolytes, PPy electrodes; polar sheet–polymer association.	Conductive and electroactive networks; restacking, oxidation and reagent residues require control. Flexible capacitors and sensing.	[33, 34, 104]
MBenes and layered boride precursors; MoB, $MoAl_{1-x}B$, $Mo_4/_3B_2T_2$	Boride chemistry; selective MAB deintercalation or other controlled synthesis. Residual A element and layer number require verification.	Demonstrated boride electrode/PVA–H_2SO_4 gel and hydrated colloid; bulk polymer-gel filler integration remains prospective.	Composition-dependent redox and transport roles. Limited matched hydrogel fatigue/biological evidence; supercapacitors and Li/Na/Zn battery studies.	[37, 38, 64, 105]
Graphene derivatives; GO, rGO, N-doped carbon sheets	Conjugated carbon; oxygen/defect content controls dispersibility and conduction. Oxidation/exfoliation followed by controlled reduction or assembly.	Carbon-sheet gelation; polymer association; CNT or conductive-polymer spacers; separate polymer electrolyte adjacent to carbon electrode.	Versatile scaffolds and hybridization; GO conductivity, rGO restacking and persistent residues are formulation concerns. Capacitors and hybrid cells.	[35, 69-72]
TMDs; MoS_2; MoS_2/MoO_2 hybrids	Layered metal–chalcogen compounds; phase and defects matter. Sheet preparation, ionic gelation and hybrid assembly are study-specific.	MoS_2 conductive gel membrane; GO-gel coating; polymer-network remodeling and ion coordination in adjacent interface studies.	Electroactive and interfacial functions; transport, phase stability and scaffold dependence require control. Capacitive storage, Na anodes and sensors.	[36, 66, 74, 75]
LDHs; NiCo-LDH and Co–Ni hydroxide layers	Metal-hydroxide layers with interlayer species; hydrothermal growth and hierarchical nanosheet deposition.	rGO coating and carbon-hydrogel electrode pairing. LDH/carbon networks should be distinguished from LDH-filled polymer electrolytes.	Redox-active electrode chemistry; conductivity and electrolyte compatibility can limit utilization. Asymmetric capacitors and hybrid electrodes.	[65, 78, 79]
Graphitic/polymeric carbon nitride; g-C_3N_4-based sheets	Nitrogen-rich polymeric semiconductor; condensation, exfoliation and dopant/heterostructure engineering depend on the reported route.	MXene coupling, conductive-carbon support, oxide/LDH hybrids; transfer to load-bearing hydrogel storage remains to be demonstrated.	Tunable interfaces and photoactivity; conductivity and limited hydrogel-specific evidence constrain generalization. Photo-rechargeable and hybrid capacitors.	[67, 80-82]

T_x denotes route-dependent surface terminations. X is carbon and/or nitrogen in the MXene formula. The MBene row includes explicitly identified partially deintercalated and colloidal configurations; it does not equate all layered borides with isolated monolayers. CNTs are one-dimensional spacers, not a two-dimensional family.

3. 2D Material–Hydrogel Composite Design and Fabrication

Hydrogel–MXene composite design depends on the functional role of MXene as an electrochemically active material, conductive filler, mechanical reinforcement, photothermal actuator, cross-linking mediator, or interfacial adhesive component. Electrode applications require high active-material loading and continuous electronic pathways, whereas sensors benefit from optimized conductive percolation. Biomedical applications additionally require careful consideration of cytotoxicity and oxidation stability [32, 106-108]. An effective design strategy prioritizes application-specific failure mechanisms. Hydrogel toughness and reversible bonding mitigate electrode cracking, whereas MXene loading, nanosheet alignment, and conductive-polymer incorporation improve electronic transport. Environmental degradation can be addressed through organohydrogel, ionogel, antioxidant, or encapsulation strategies, reducing the need for excessive filler incorporation [31, 32, 106].

Fabrication methods must accommodate device architecture and operating conditions. Casting and in situ polymerization enable bulk hydrogel fabrication, while freeze-casting generates oriented pores and anisotropic transport pathways. Printing and patterned deposition facilitate microsupercapacitors, interdigitated electrodes, wireless devices, and smart textiles. Solvent exchange enhances

environmental tolerance, whereas layer-by-layer assembly enables spatial control over conductivity, hydration, and mechanical compliance [39, 45, 109]. MXene-based composites include all-MXene hydrogels, MXene/graphene oxide networks, MXene/polymer hydrogels, ionogels, organohydrogels, and multifunctional hybrid systems. Their selection depends on requirements for conductivity, processability, adhesion, stimulus responsiveness, biocompatibility, and controlled degradation [4, 110].

Table 2. *Representative design strategies for flexible energy-storage devices, including hydrogel electrolyte networks, all-MXene hydrogel/ionogels, MXene/GO or MXene/rGO hydrogels, MXene/polymer hydrogels, and organohydrogel/ionogels, highlighting their functions, advantages, and limitations, summarized from Refs. [2, 4, 21, 31, 32, 42, 44, 102, 106, 111-114], and [115].*

Material/design strategy	Main function in flexible energy storage	Key advantages	Main limitations	Representative refs.
Hydrogel electrolyte networks	Ion transport, leakage suppression, soft contact	High ionic conductivity, tissue-like compliance, shape conformability	Drying, freezing, narrow aqueous voltage window	[2, 21, 111]
All-MXene hydrogel/ionogel	Conductive and pseudocapacitive active network	High MXene loading, high electrical conductivity	Restacking, oxidation, limited structural integrity	[102, 112, 113]
MXene/GO or MXene/rGO hydrogel	3D conductive scaffold with improved porosity	Reduced restacking, improved mechanical stability and transport	Complex reduction and interfacial control	[42, 44, 116]
MXene/polymer hydrogel	Conductive, stretchable, adhesive, self-healing composites	Tunable chemistry, scalable wet processing, multifunctionality	Possible filler aggregation and polymer/filler mismatch	[32, 106, 114]
Organohydrogel/ionogel	Anti-drying, anti-freezing, stable ion transport	Improved environmental stability	Possible lower biocompatibility depending on solvent/ionic liquid	[31, 115]

3.1 MXene–Hydrogel Architectures

All-MXene hydrogels enable high active-material loading through concentration-induced gelation, ion-mediated assembly, or solvent exchange. Their continuous conductive networks support efficient electron transport, although nanosheet restacking, oxidation, and weak structural integrity remain significant limitations. Ionogel and organogel strategies improve solvent retention and potentially broaden electrochemical operating windows but introduce additional considerations concerning viscosity, cost, and biocompatibility [102, 112, 113]. MXene/graphene oxide (GO) and MXene/reduced graphene oxide (rGO) hydrogels exploit complementary nanosheet properties. Hydrophilic GO promotes aqueous dispersion and hydrogen bonding, whereas rGO enhances electrical conductivity and mechanical reinforcement. Hybrid frameworks suppress MXene restacking, improve ionic accessibility, and may reduce oxidation through controlled interfacial organization [42, 44, 116]. MXene/polymer hydrogels provide greater compositional versatility. Polyacrylamide and poly(acrylic acid) facilitate swelling and functionalization, poly(vinyl alcohol) improves toughness, poly(N-isopropylacrylamide) (PNIPAM) provides thermoresponsiveness, gelatin and polysaccharides improve biological compatibility, and PEDOT enhances electronic conductivity. However, excessive polymer incorporation can reduce active-material utilization and volumetric capacitance [4, 106, 114, 117].

Fig. 8 demonstrates how MXene integration with polymeric networks influences structural organization, electrical transport, mechanical reinforcement, and functional responsiveness. Incorporating PEDOT and K-modified MXene into PNIPAM improves electronic connectivity and interfacial interactions while introducing thermoresponsive behavior relevant to wearable devices and soft actuators [106]. Fig. 8(a)–(c) illustrates the fabrication and biomedical applicability of MnO_2–MXene–gelatin methacryloyl (MMG) hydrogels. Initially, Ti_3AlC_2 was etched using lithium fluoride (LiF)/hydrochloric acid (HCl) at 40–45 °C for 48 h, followed by washing to approximately pH 6, ultrasonication under an inert atmosphere, and centrifugation at 3000 rpm for 30 min. MnO_2 was subsequently grown using $KMnO_4$ and $MnSO_4{\cdot}H_2O$ at a 6:1 molar ratio with concentrated H_2SO_4, followed by hydrothermal treatment at 140 °C for 24 h. The resulting nanocomposite was washed and freeze-dried. As, Figure 8(a–c) illustrates the fabrication, structural characterization, and potential biomedical applications of MnO_2–MXene/GelMA (MMG) hydrogels. As shown in

Fig. 8(a), MnO_2–MXene nanocomposites with concentrations ranging from 0 to 5000 μg mL^{-1} were incorporated into gelatin methacryloyl (GelMA), photoinitiator (LAP), and water precursors at a mass ratio of 15:1:250. Photocrosslinking was performed using a 60 W light source (405 nm) at an irradiance of approximately 40 mW cm^{-2}, while digital light processing (DLP) printing employed approximately 250 mW cm^{-2}, a layer thickness of 50 μm, and exposure times of 6–8 s (Fig. 8(b)). The resulting hydrogels were characterized using tensile testing, rheological measurements, electrical impedance analysis, and microscopy to evaluate their mechanical, electrical, and morphological properties. Figure 8(c) further illustrates their proposed application as multifunctional cardiac patches for myocardial repair, emphasizing mechanical support, electrical coupling, reactive oxygen species (ROS) scavenging, and

cellular responses under oxidative and ischemic conditions [118].

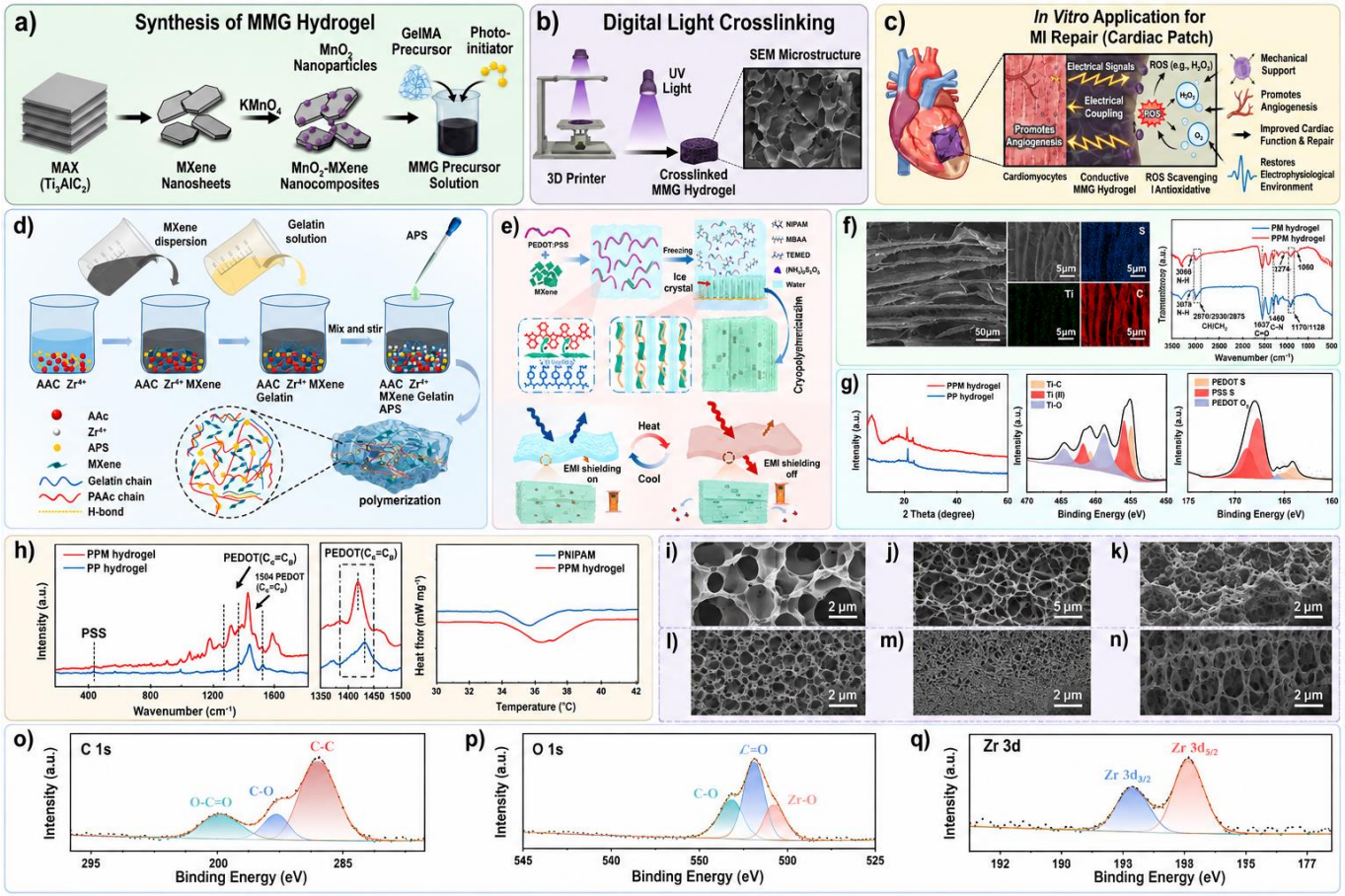


Fig. 8. Synthesis, fabrication, structural characterization, stimuli-responsive behavior, and multifunctional applications of MXene-based hydrogels. (a) Schematic preparation of MnO_2–MXene nanocomposites and their incorporation into gelatin methacryloyl (GelMA) precursors. (b) Digital light processing (DLP)-assisted photocrosslinking of MMG hydrogels, accompanied by scanning electron microscopy (SEM) images illustrating their microstructural morphology. (c) Proposed application of MMG hydrogels as conductive cardiac patches for myocardial repair, highlighting mechanical support, electrical coupling, reactive oxygen species (ROS) scavenging, and microenvironmental regulation, together with assessments of endothelial-cell and cardiomyocyte responses, electrical connectivity, sarcomeric organization, and cell viability under oxidative and ischemic conditions. Reproduced with permission from Ref. [118]. Copyright 2026, Royal Society of Chemistry. (d) Schematic illustration of MPG hydrogel synthesis and the proposed polymerization mechanism. (e) Schematic fabrication of thermoresponsive MXene/PEDOT/PNIPAM (PPM) hydrogels, illustrating reversible transitions between electromagnetic interference (EMI) shielding and electromagnetic-wave transparency upon heating and cooling. (f) SEM morphology, energy-dispersive X-ray spectroscopy (EDS) elemental mapping of C, S, and Ti, and Fourier-transform infrared (FTIR) spectra of PM and PPM hydrogels. (g) X-ray diffraction (XRD) patterns and high-resolution Ti 2p and S 2p X-ray photoelectron spectroscopy (XPS) spectra. (h) Raman spectra and differential scanning calorimetry (DSC) thermograms characterizing the structural and thermoresponsive properties. Panels (e–h) reproduced from Ref. [106]. Copyright 2025, the authors; published by Wiley-VCH under the Creative Commons Attribution 4.0 International License. (j–n) Morphological characterization of MPG hydrogels prepared with MXene concentrations of (j) 2, (k) 4, (l) 6, (m) 8, and (n) 10 mg mL^{-1}. (o–q) High-resolution XPS spectra of the optimized MPG hydrogel, showing the (o) C 1s, (p) O 1s, and (q) Zr 3d core-level regions. Panels (d, j–q) reproduced with permission from Ref. [119]. Copyright 2026, Royal Society of Chemistry.

Figure 8(d) presents the one-pot free-radical polymerization and hierarchical network formation of MPG hydrogels. Acrylic acid undergoes polymerization to form a covalently cross-linked poly(acrylic acid) (PAAc) framework, while gelatin contributes reversible hydrogen-bonding and electrostatic interactions. Simultaneously, Zr^{4+} ions coordinate with carboxyl groups in PAAc and oxygen-containing surface functionalities of MXene nanosheets, establishing an interconnected covalent–physical–coordination network. These complementary interactions facilitate stress redistribution, reversible energy dissipation, and structural reinforcement. The resulting dual-locking mechanism improves mechanical strength, deformation tolerance, electrical stability, and resistance to excessive swelling, highlighting the importance of synergistic interfacial interactions in mechanically robust and conductive hydrogel architectures [119].

Figure 8(e–h) demonstrates the synthesis, structural characteristics, and thermoresponsive behavior of $Ti_3C_2T_x$ MXene/PEDOT/poly(N-isopropylacrylamide) (PNIPAM) hydrogels, designated PPM. As illustrated in Fig. 8(e), MXene nanosheets with an approximate thickness of 1.8 nm and lateral dimensions of 1.2–2 μm were integrated with PEDOT and PNIPAM through directional freezing and in situ polymerization. Interfacial interactions between the conductive components suppress MXene aggregation, whereas anisotropically aligned pores facilitate directional water transport, accelerating thermally induced swelling–deswelling transitions. Scanning electron microscopy (SEM) and energy-dispersive X-ray spectroscopy (EDS) reveal an oriented porous morphology and the distribution of C, S, and

Ti throughout the hydrogel network (Fig. 8(f)). Fourier-transform infrared (FTIR) spectroscopy identifies characteristic polymer vibrations, including a shift of the N–H stretching band from 3078 to 3068 cm^{-1}, suggesting enhanced hydrogen-bonding interactions among the constituent components. X-ray diffraction (XRD) reveals the MXene (002) reflection at 2θ = 6.71°, consistent with increased interlayer spacing following polymer incorporation. Furthermore, high-resolution Ti 2p and S 2p X-ray photoelectron spectroscopy (XPS) spectra provide evidence of the MXene and PEDOT components and their interfacial chemical environments (Fig. 8(g)). Raman spectroscopy reveals an approximately 9.83 cm^{-1} redshift of the PEDOT Cα=Cβ symmetric stretching band initially located at 1427 cm^{-1}, consistent with increased quinoid character and modified polymer-chain conformation. Differential scanning calorimetry (DSC) further characterizes the hydrogel's thermal response (Fig. 8(h)). Collectively, these structural and spectroscopic findings support the development of a thermoresponsive conductive network capable of reversible, water-mediated modulation between electromagnetic interference (EMI) shielding and electromagnetic-wave-transparent states [106].

3.2 Hydrogel Architectures Incorporating MBenes and Other 2D Materials

Fabrication must specify electrode, electrolyte, and reinforcement roles. Boride electrodes with separate PVA/H_2SO_4 gel [64] and MBene colloidal electrolytes [38] cannot be treated as identical polymer composites. MXene@PPy/rGO assembly [71], MXene/rGO gelation [35], MoS_2 ionic gelation [36], and MoS_2-assisted polymerization [66] provide distinct processing routes. Transfer requires controls for precursor stability, dispersion, initiator compatibility, washing, and post-gelation sheet identity. Polymer-only and filler-only controls should establish whether improved connectivity compromises separator insulation.

3.3 Interfacial Engineering, Dispersion, and Network Formation

Interfacial interactions between MXene nanosheets and polymer chains critically influence electrical percolation, mechanical reinforcement, and fatigue resistance in conductive hydrogels. Hydrogen bonding between MXene surface terminations and polymeric amide, hydroxyl, carboxyl, or sulfonate groups strengthens interfacial compatibility, while multivalent-ion coordination and dynamic covalent interactions enhance structural integrity and self-healing. Conductive polymers further bridge neighboring MXene sheets, improving electronic continuity and processability [24, 106, 120]. Three-dimensional network architectures can be engineered through directional freezing, shear alignment, extrusion printing, magnetic assistance, or ultrasonic alignment. Unlike approximately isotropic random networks, oriented structures provide direction-dependent electrical conduction, ionic transport, and mechanical reinforcement. However, excessive nanosheet alignment may restrict cross-plane transport, necessitating optimization of structural anisotropy [39, 45].

Crosslinking density governs the balance between mechanical stability and ionic mobility. Highly cross-linked networks exhibit improved stiffness, dimensional stability, and swelling resistance but may restrict ion diffusion and self-healing. Conversely, loosely cross-linked hydrogels provide greater stretchability and electrolyte uptake while remaining susceptible to creep, dehydration, and structural deformation. Dual-network, interpenetrating-network, and reversible crosslinking strategies integrate load-bearing and energy-dissipating mechanisms. MXene nanosheets can additionally function as conductive reinforcements and dynamic crosslinking sites, coupling mechanical and electrochemical performance [25, 32, 108]. Quantitative control of crosslinking density is therefore essential for optimizing competing material properties. Sacrificial bonds and reversible coordination interactions provide additional energy-dissipation pathways without permanently disrupting conductive networks, enabling improved fatigue resistance and mechanical recovery [24, 25, 32].

Fig. 9 illustrates how hierarchical structural design, interfacial coordination, and dynamic crosslinking improve hydrogel conductivity, mechanical durability, self-healing, and multifunctionality. The MXene–tannic acid–polyacrylamide (MTP) hydrogel electrolyte was fabricated through one-pot thermal polymerization. Initially, $ZnSO_4{\cdot}7H_2O$ was dissolved in 8 mL of MXene–tannic acid (MT) dispersion to obtain a 2 M electrolyte. Subsequently, 1.2 g acrylamide (AM), 4 mg N,N'-methylenebisacrylamide (MBA), and 40 mg ammonium persulfate (APS) were added. Following degassing, polymerization proceeded at 60 °C for 2 h. MXene loadings of 10, 24, 50, 80, and 100 mg produced MTP-1 to MTP-5, respectively. Pure PAM, MXene–PAM (MP), and tannic acid–PAM (TP) controls were prepared similarly [121].

For device assembly, Zn^{2+}-preintercalated V_2O_5 (Z-VO) was synthesized by dispersing 0.364 g V_2O_5 in 30 mL water, adding 2 mL H_2O_2 (30 wt.%), and subsequently introducing 0.75 mmol $ZnSO_4$. Hydrothermal treatment at 120 °C for 24 h produced Z-VO, which was mixed with acetylene black and polyvinylidene fluoride (PVDF) at a 7:2:1 mass ratio in N-methyl-2-pyrrolidone (NMP). The slurry was deposited onto titanium foil and vacuum-dried at 60 °C for approximately 12 h, yielding an active-material loading of approximately 3.0 mg cm^{-2}. Fig. 9(a) presents the spider-web-inspired MTP architecture, comprising a three-dimensional PAM framework reinforced by tannic acid-functionalized MXene. The interconnected network provides Zn^{2+} adsorption sites and transport channels. As illustrated in Fig. 9(b), conventional aqueous $ZnSO_4$ electrolytes can promote dendritic deposition, hydrogen evolution, and concentration polarization. Although PAM electrolytes partially immobilize free water, limited interfacial contact and sluggish Zn^{2+} transport may persist. In contrast, MXene surface groups and tannic acid phenolic functionalities regulate Zn^{2+} migration and water coordination, suppressing parasitic reactions and promoting more uniform zinc deposition. Conductive hydrogels are increasingly investigated for neuronal interfaces, physiological monitoring, and bioelectronic stimulation [122, 123]. Conventional polymer hydrogels generally exhibit insufficient electronic conductivity

compared with conductive polymers, carbon materials, and metals [124-126].

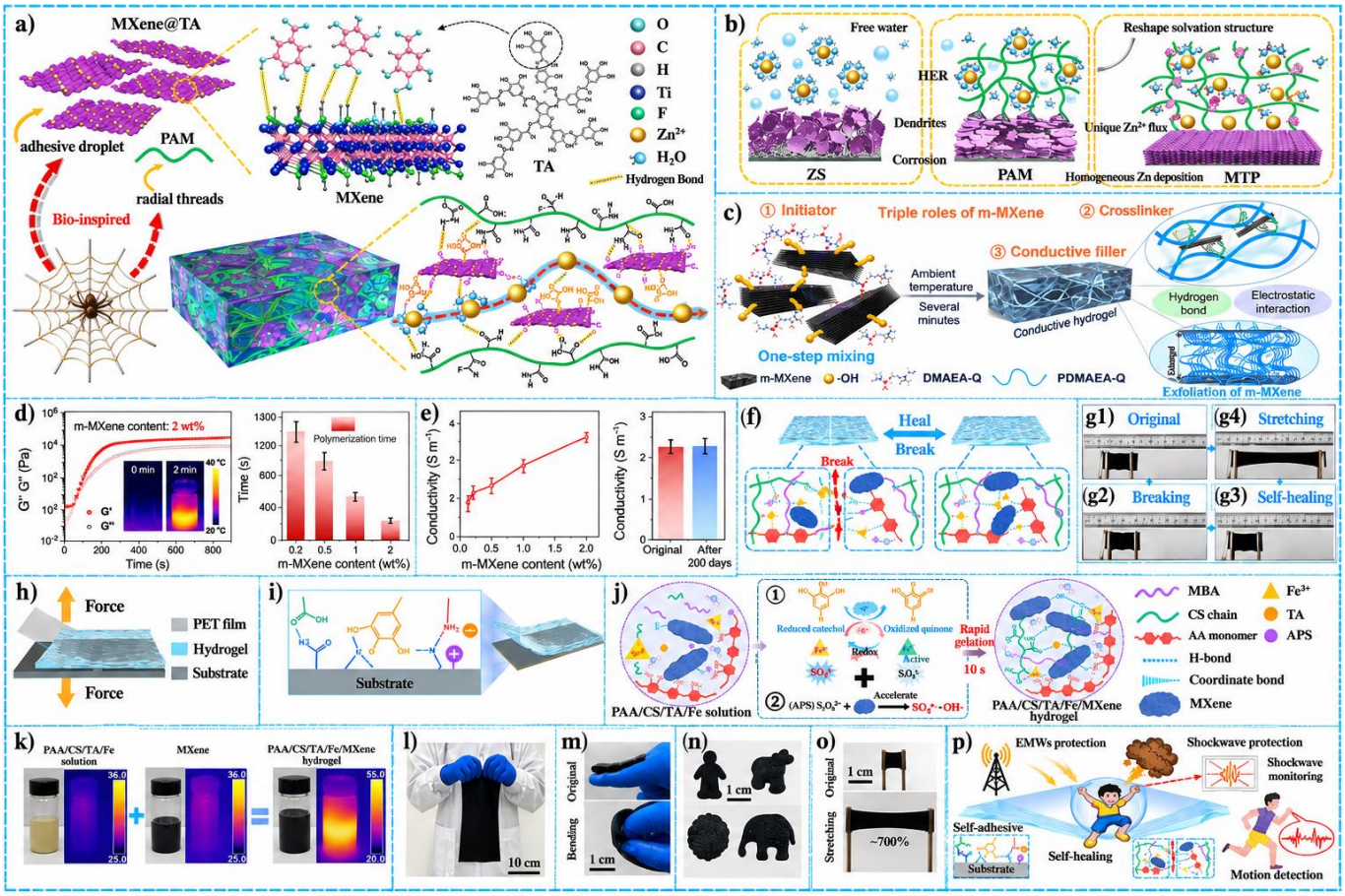


Fig. 9. Synthesis, structural design, gelation behavior, electrical stability, self-healing, adhesion, and multifunctional properties of MXene-based hydrogels. (a) Schematic illustration of the spider-web-inspired hierarchical structures of MXene@TA-PAM (MTP), MXene-PAM (MP), TA-PAM (TP), and pure PAM hydrogels; (b) schematic comparison of Zn anodes in ZS, PAM hydrogel, and MTP hydrogel electrolytes. Reproduced from Ref. [121]. Copyright 2026, the authors, published by Springer Nature under a Creative Commons Attribution 4.0 International License. (c) Schematic illustration of conductive MD-hydrogel synthesis with varying m-MXene concentrations, highlighting the enhancement in electrical conductivity with increasing m-MXene content; (d) infrared thermal imaging and rheological characterization of exothermic gel formation, demonstrating rapid gelation through changes in storage (G') and loss (G") moduli during polymerization with 2 wt.% m-MXene; (e) long-term electrical stability of the MD-hydrogel, demonstrating sustained functionality for over 200 days. Reproduced with permission from Ref. [32]. Copyright 2022, Elsevier. (f) Schematic illustration of the self-healing mechanism of the PCM-3 hydrogel; (g1) original hydrogel before cutting; (g2) hydrogel after cutting; (g3) rejoined hydrogel after self-healing; (g4) stretched hydrogel following self-healing; (h) schematic illustration of the 180° peel test for evaluating adhesion; (i) proposed adhesion mechanism of the PCM-3 hydrogel; (j) schematic illustration of the room-temperature preparation mechanism of poly(acrylic acid) (PAA)/chitosan (CS)/Ti_3C_2 MXene (PCM) hydrogels; (k) infrared thermal images showing the hydrogel molding process; (l) photograph of a large-area PCM hydrogel; (m) photograph demonstrating the dynamic adhesion of an ultraflexible PCM hydrogel to a finger during bending; (n) shape adaptability and (o) stretchability of PCM hydrogels; and (p) schematic representation of their integrated self-healing, adhesion, electromagnetic and shockwave protection, and wide-frequency sensing capabilities. Reproduced with permission from Ref. [127]. Copyright 2026, Wiley-VCH.

MXene incorporation improves electronic transport and sensing functionality [128-130], supporting applications in intelligent theranostics, prosthetics, and wearable human–machine interfaces [131-133]. Huang et al. [32] developed conductive MD hydrogels using multilayer MXene (m-MXene) as a polymerization initiator and conductive component. MXene-generated hydroxyl species initiated vinyl polymerization without external heating, irradiation, or additional co-initiators. Fig. 9(c) illustrates this rapid synthesis and the influence of MXene concentration on conductivity. Fig. 9(d) demonstrates exothermic gelation through infrared thermal imaging and rheological changes in storage and loss moduli. The hydrogel containing 2 wt.% m-MXene achieved 3.8 S m^{-1} conductivity, while the 0.5 wt.% formulation retained electrical functionality for more than 200 days, as shown in Fig. 9(e). Similarly, Xue et al. [106] developed MXene/PEDOT/PNIPAM hydrogels exhibiting 11.76 S m^{-1} conductivity and stable sensing over 300 loading–unloading cycles.

Self-healing in PCM hydrogels arises from reversible hydrogen bonding and metal-coordination interactions, as illustrated in Fig. 9(f). Fig. 9(g1)–(g4) sequentially demonstrates the intact, cut, rejoined, and stretched hydrogel states. Dynamic bond reformation restores mechanical continuity and conductive pathways, enabling electrical recovery after damage. Catechol, amino, and carboxyl functionalities additionally contribute to interfacial adhesion [127]. Adhesion was evaluated through the 180° peel-test configuration illustrated in Fig. 9(h). Maximum adhesion energies were 189.2, 19.27, 150.72, and 25.09 J m^{-2} for aluminum, polytetrafluoroethylene (PTFE), stainless steel, and glass, respectively. As depicted in Fig. 9(i), hydrogen bonding, covalent interactions, electrostatic attraction, π–π

interactions, and metal–ligand coordination collectively support adhesion to diverse substrates. These reversible interfacial interactions enhance mechanical durability and operational reliability [127]. Fig. 9(j) illustrates room-temperature PCM hydrogel polymerization through Fe^{3+}/Fe^{2+} and catechol/quinone redox interactions, which facilitate APS activation and radical generation. MXene further promotes APS decomposition, accelerating acrylic acid polymerization, while Fe^{3+} coordination strengthens network formation. Infrared thermal images in Fig. 9(k) demonstrate gelation within tens of seconds, enabling fabrication of the large-area hydrogel shown in Fig. 9(l) [127]. The resulting interconnected network supports dynamic adhesion during finger bending, as demonstrated in Fig. 9(m), alongside shape adaptability and stretchability in Fig. 9(n) and (o). PCM formulations incorporated MXene contents of 0%, 0.58 wt.%, 1.17 wt.%, 1.75 wt.%, and 2.04 wt.%, with PCM-3 selected for subsequent characterization except electromagnetic interference (EMI) shielding. Increasing MXene content shifted the (002) X-ray diffraction reflection toward higher angles, indicating reduced interlayer spacing and stronger network interactions. The hydrogel achieved an electrical conductivity of 0.33 S m^{-1}. As summarized in Fig. 9(p), the synergistic combination of conductive MXene networks, reversible crosslinks, and interfacial adhesion supports self-healing, EMI shielding, shockwave mitigation, and wide-frequency motion sensing. These findings establish interfacial engineering and three-dimensional network control as essential strategies for mechanically resilient, multifunctional hydrogel–MXene systems [127].

3.4 Functional, Self-Healing, and Degradable Composite Hydrogels

Self-healing is essential for hydrogel–MXene energy-storage systems because mechanical damage disrupts electronic and ionic transport pathways. Effective recovery requires the reconstruction of polymer–polymer bonds, polymer–MXene interactions, and nanosheet contacts. Healing kinetics depend on polymer-chain mobility, hydration, temperature, and reversible-bond dynamics. Near-infrared (NIR)-responsive MXenes can accelerate network reconstruction through localized photothermal heating, although excessive heating may induce dehydration or tissue damage [31, 108, 134]. Self-healing performance should be assessed through both mechanical and electrochemical recovery. Visual reconnection or tensile-strength restoration does not necessarily indicate recovered conductivity or capacitance. Accordingly, healing efficiency should consider mechanical strength, elongation, electrical resistance, impedance, capacitance retention, and electrolyte leakage before damage, after damage, and following recovery. Experimental conditions, including healing duration, contact pressure, temperature, hydration, and external stimulation, should also be documented [108, 134, 135].

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Interfacial adhesion is equally important because mechanically robust hydrogels may still detach from skin, textiles, or current collectors. Catechol functionalities, hydrogen bonding, ionic coordination, and electrostatic interactions enhance adhesion while maintaining electrode–electrolyte contact during deformation [2, 136]. Environmental stability further requires anti-drying and anti-freezing strategies. Incorporating glycerol, ethylene glycol, ionic liquids, salts, or hygroscopic polymers can improve solvent retention and freezing resistance, although these additives may influence viscosity, ionic mobility, biocompatibility, and electrochemical stability [31, 48, 115]. For transient and sustainable applications, hydrogel–MXene composites must maintain functionality during operation while undergoing controlled degradation afterward. Hydrolysis, oxidation, ionic dissolution, and polymer-chain cleavage govern degradation behavior. Biomedical implementation additionally requires consideration of degradation-product toxicity and biological clearance [14, 15, 137]. Fig. 10 illustrates the integration of dynamic cross-linking, photothermal responsiveness, interfacial adhesion, environmental adaptability, and controlled degradability. These characteristics enable mechanical and electrical recovery while supporting temporary biomedical and wearable applications. The principal challenge is balancing operational stability with programmed material degradation [15, 108, 134, 135, 138]. Fig. 10(a) illustrates the fabrication and therapeutic mechanism of a self-evolving FAHP hydrogel developed for infected diabetic wounds. The hydrogel comprises hydrazide-functionalized hyaluronic acid (HHA), aldehyde-functionalized poly(aspartic acid) (APA), Ag^+ ions, and hydrolyzable Fe_3N/Fe_3O_4 nanoheterojunction (FNO nHJ) photosensitizers, integrated through dynamic Schiff-base and coordination cross-linking. During early healing, fluid absorption and the procoagulant effects of Ag^+ and aldehyde groups promote hemostasis. Under infection conditions, FNO-mediated self-oxygenated photodynamic therapy generates reactive oxygen species (ROS), including singlet oxygen ($^1O_2$) and superoxide radicals ($\cdot O_2^-$), alongside localized hyperthermia. These effects cooperate with Ag^+ to eliminate resistant bacteria and biofilms, while disulfide-mediated glutathione (GSH) depletion enhances antibacterial activity.

During subsequent healing, the polymer network promotes in situ conversion of Ag^+ into silver nanozymes, which regulate excessive ROS and generate oxygen through catalytic reactions. Concurrently, degradation-derived ammonia supports angiogenesis, cellular proliferation, and

neutralization of the mildly acidic microenvironment. Iron oxide-mediated immunomodulation further promotes M2 macrophage polarization. Enhanced wound repair was demonstrated in a methicillin-resistant *Staphylococcus aureus* (MRSA)-infected diabetic rat model, with transcriptomic analysis supporting regenerative pathway involvement [139].

Fig. 10(b)–(e) presents the PFCD hydrogel electrolyte developed for flexible neutral zinc–air batteries (FNZABs). Its design integrates adhesion, ionic conductivity, oxygen permeability, mechanical durability, and environmental stability. As illustrated in Fig. 10(b), these functions arise from complementary polymer interactions. Fig. 10(c) depicts three-step synthesis involving a poly(acrylic acid) (PAA)–Fe^{3+}–CEC coordination network, β-cyclodextrin (β-CD)-mediated hydrogen bonding, and potassium persulfate ($K_2S_2O_8$)-initiated thermal polymerization. CEC, prepared through Michael addition between acrylic acid and chitosan amino groups, introduces hydroxyl and carboxyl functionalities. These polar groups generate localized electric fields that induce temporary dipoles in oxygen molecules, facilitating oxygen transport. Hydrophilic –COOH, –OH, and $–NH_2$ groups additionally support water retention and interconnected transport pathways. Fig. 10(d) illustrates oxygen migration toward the gas–liquid–solid triple-phase interface, where the oxygen reduction reaction (ORR) occurs. Fig. 10(e) presents the assembled battery comprising a zinc-foil anode, porous cathode, and PFCD electrolyte. The integrated network improves interfacial stability, oxygen transport, mechanical recovery, and zinc-deposition regulation [140].

MXene-containing hydrogels also exhibit substantial deformation tolerance and reusable sensing functionality. Yang et al. [134] developed poly(acrylic acid–stearyl methacrylate)/MXene [P(AA–SMA)M] hydrogels incorporating reversible hydrophobic interactions and metal–ligand coordination. Fig. 10(f) highlights their mechanical performance, including stretchability approaching 2600%. The pristine hydrogel achieved approximately 2580% elongation, while healed samples retained elongations of 1900%, 1030%, and 843% after the first, tenth, and twentieth healing cycles, respectively. As illustrated in Fig. 10(g), tensile strength gradually decreased from 142 to 102 kPa during repeated recovery, demonstrating substantial residual mechanical integrity. These properties enabled wearable sensing applications involving handwriting and gesture recognition [134]. Ge et al. [108] further investigated double-cross-linked MXene–PAA hydrogels with enhanced gelation and photothermal self-healing. Fig. 10(h) illustrates network formation, while Fig. 10(i) compares healing efficiencies among different formulations. Reversible interactions involving MXene nanosheets, glycerol, and PAA chains promote network reconstruction, as depicted in Fig. 10(j). During deformation, dynamic hydrogen bonds dissipate mechanical energy despite irreversible rupture of N,N′-methylenebisacrylamide (MBAA)-cross-linked polymer chains. NIR-induced photothermal heating further accelerates healing, with additional recovery characteristics presented in Fig. 10(k). These combined mechanisms support mechanically adaptive sensing and actuation systems [108].

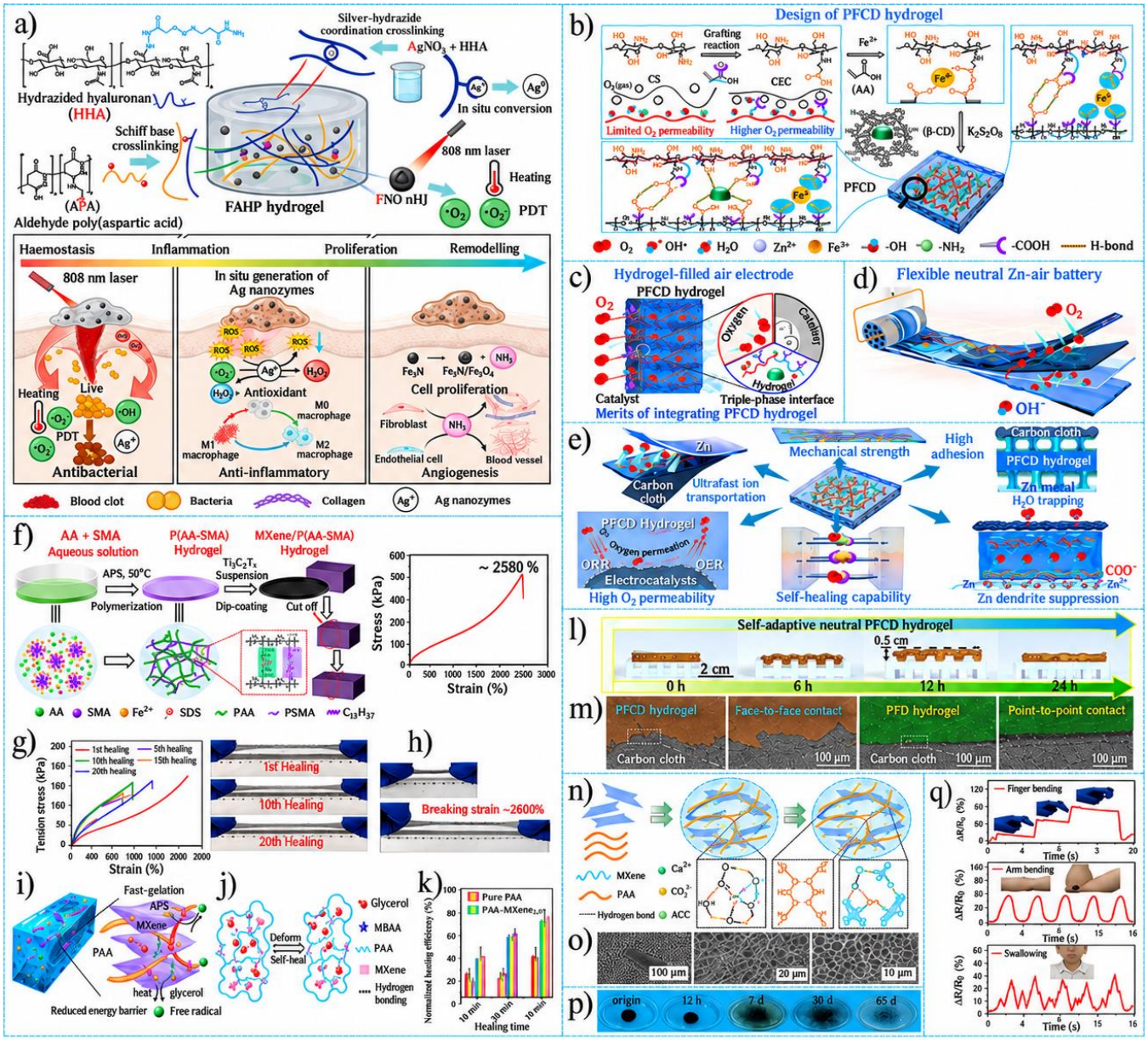


Fig. 10. Schematic illustrations of hydrogel fabrication, functional mechanisms, self-healing behavior, self-adaptive performance, and degradability. (a) Formation of the FAHP hydrogel and its mechanism for promoting infected diabetic wound healing. Reproduced with permission from Ref. [139]. Copyright 2026, Wiley-VCH. (b) Comprehensive schematic illustrating the integrated design, synthesis, and functional advantages of the PFCD hydrogel for advanced flexible neutral Zn–air batteries (FNZABs); (c) three-step synthesis of the PFCD hydrogel; (d) oxygen reduction reaction (ORR) at the hydrogel triple-phase interface; (e) structural configuration of the assembled FNZAB prototype, highlighting the synergistic performance advantages of the PFCD hydrogel. (f) Mechanical performance of a double-cross-linked MXene hydrogel for strain sensing, demonstrating a tensile strength of approximately 525 kPa and stretchability approaching 2600%; (g) durability and self-healing behavior of the P(AA–SMA)M hydrogel film after 20 fracture–healing cycles. Adapted with permission from Ref. [134]. Copyright 2023, American Chemical Society. (h) Gelation process of the double-cross-linked MXene hydrogel; (i) comparison of the self-healing efficiencies of different hydrogel compositions; (j) MXene-assisted self-healing mechanism; (k) additional characterization of hydrogel self-healing properties. Adapted with permission from Ref. [108]. Copyright 2021, American Chemical Society. (l) and (m) Mechanical characterization and evaluation of the self-adaptive behavior of the PFCD hydrogel. Panels (b)–(e) and (l)–(m) are reproduced from Ref. [140]. Copyright 2026, the authors, published by Wiley-VCH. (n) Preparation of delaminated MXene nanosheets; (o) final freeze-dried structure of the MXene–PAA–ACC hydrogel; (p) SEM characterization and phosphate-buffered saline immersion behavior, illustrating its three-dimensional interconnected porous structure, swelling, and degradation behavior; (q) photographs of hydrogel-based epidermal sensors attached to different body regions, demonstrating real-time resistance responses to human movements, including swallowing and finger and elbow bending. Reproduced with permission from Ref. [15]. Copyright 2021, American Chemical Society.

The PFCD hydrogel additionally demonstrates reversible shape adaptation and electrode conformity. Fig. 10(l) illustrates its mechanical adaptability: samples maintained in differently shaped polytetrafluoroethylene (PTFE) molds for 6–12 h conformed to the imposed geometries before recovering their original configurations after 12 h on a flat surface. Dynamic hydrogen bonds facilitate molecular rearrangement, whereas Fe^{3+} coordination maintains structural integrity. Fig. 10(m) further demonstrates intimate electrode–electrolyte contact, which improves interfacial adhesion and effective electrochemical contact area compared with conventional point-contact hydrogel interfaces [140]. Controlled degradability

provides an additional route toward sustainable wearable electronics [14, 137]. Combining reversible self-healing with programmed degradation can extend functional service life while reducing persistent electronic waste [122, 141, 142]. Li et al. [15] developed degradable epidermal sensors using $Ti_3C_2T_x$ MXene, amorphous calcium carbonate (ACC), and PAA. Fig. 10(n) illustrates delaminated MXene preparation, while Fig. 10(o) presents the resulting freeze-dried MXene–PAA–ACC hydrogel architecture. Fig. 10(p) demonstrates its interconnected porous morphology and degradation behavior in phosphate-buffered saline (PBS), where complete degradation occurred within 65 days. Hydrophilic carboxyl functionalities promote water penetration and disruption of dynamic cross-links, facilitating network dissolution. Fig. 10(q) demonstrates wearable sensing of swallowing and finger and elbow movements, with response times below 20 ms. Similarly, Li et al. [135] reported antibacterial, biodegradable MXene hydrogel sensors that completely disintegrated within 45 days in PBS (pH 7.2) at room temperature. Degradation resulted from water-induced polymer plasticization and disruption of supramolecular interactions. Sun et al. [138] developed MXene/poly(vinyl alcohol)/bacterial cellulose (PBM) hydrogels through freeze–thaw processing. Reversible hydrogen bonding and borate ester linkages enabled self-healing and degradability, with degradation occurring within 53 min in 3% H_2O_2 and 56 days in PBS. Collectively, these studies demonstrate that reversible cross-linking, interfacial adhesion, solvent management, and controlled degradation are interconnected design requirements. Optimizing these characteristics enables hydrogel–MXene systems to maintain mechanical and electrical functionality during operation while supporting recoverability, environmental adaptability, and application-specific end-of-life behavior.

4. Design Requirements for Soft Energy Storage

Soft mechatronic energy-storage devices must satisfy interconnected material, device, and system requirements. Essential material properties include conductivity, mechanical compliance, water retention, biocompatibility, and chemical stability. Device performance depends on capacitance, operating voltage, energy and power density, cycling durability, encapsulation, and deformation tolerance. System integration further requires compatibility with sensors, actuators, energy harvesters, wireless modules, and power-management circuits [5, 16, 20]. Conventional electrochemical measurements must be complemented by mechanical and environmental testing under bending, twisting, stretching, compression, perspiration, dehydration, temperature cycling, and repeated adhesion. Robotic applications require actuation-cycle evaluation, whereas biomedical systems additionally demand saline resistance, cytocompatibility, sterilization tolerance, and degradation assessments [2, 13, 17]. Testing sequences should distinguish residual damage from dynamic performance changes. Mechanical cycling followed by electrochemical characterization identifies permanent degradation, while simultaneous testing reveals operational instability. Dehydration–rehydration and repeated attachment–detachment experiments further assess environmental recovery and adhesive fatigue, providing realistic service-life estimates [2, 13, 17].

4.1 Mechanical Compliance Under Deformation

Mechanical compliance requires sustained flexibility, fatigue resistance, and electrochemical stability rather than high single-cycle stretchability. Device durability depends on the entire multilayer structure, including electrodes, electrolytes, current collectors, separators, encapsulation, adhesives, and interconnects, because localized failure can compromise overall functionality [17, 25, 123]. Bending performance depends on device thickness, curvature, strain distribution, and mechanical cycling. Accordingly, characterization should report bending radius, thickness, strain amplitude, cycle number, recovery time, humidity, and electrical retention. Microscopic evaluation of electrode edges and interconnects helps identify localized damage [16, 17]. Complete-device testing is essential because independently flexible components may experience interfacial shear, delamination, or modulus mismatch after assembly. Neutral-axis optimization, compliant interconnects, and suitable encapsulation can mitigate these effects [16, 17, 123]. Compression may redistribute ions and expel water, whereas torsion can disrupt electrode–electrolyte contact. Dynamic crosslinks, gradient-modulus structures, kirigami designs, serpentine interconnects, and adhesive encapsulation improve deformation tolerance [45, 108].

The approximate surface bending strain is:

$$\varepsilon_{\text{bend}} = \frac{t}{2r+t} \tag{11}$$

where t is device thickness and r is the inner bending radius. The expression assumes a mid-thickness neutral plane; for $t \ll r$ it approaches t/(2r). Multilayer devices require their actual neutral-axis position.

Capacitance retention after mechanical cycling is:

$$\eta_{\text{ret}} = \frac{C_N}{C_0} \times 100\% \tag{12}$$

where C_n and C_o represent capacitance after (N) cycles and initially, respectively, under identical conditions.

The strain-sensing gauge factor is:

$$GF = \frac{\Delta R/R_0}{\varepsilon} \tag{13}$$

where ΔR is resistance change, R_o is initial resistance, and ε is applied strain. Hysteresis and signal drift should also be considered.

4.2 Electrochemical Performance and Cycling Stability

Electrochemical metrics should reflect application-specific requirements. Gravimetric capacitance facilitates material comparison, whereas areal and volumetric performance better represent footprint-constrained wearable devices. Flexible batteries additionally require capacity-retention measurements under deformation [5, 16, 143]. Comprehensive evaluation should combine electrochemical output, mechanical durability, environmental stability, and integration readiness. Wearable systems prioritize areal capacitance, energy density, adhesion, and bending durability, while robotic

systems emphasize pulse-power delivery, compression tolerance, and actuation stability [5, 16, 143].

Aqueous hydrogels provide high ionic conductivity and compliance but exhibit voltage limitations associated with water electrolysis. Asymmetric electrodes, water-in-salt electrolytes, ionogels, redox-active gels, and organohydrogels can extend operating voltage, although viscosity, cost, safety, and biocompatibility trade-offs remain [115, 125]. Cycling degradation may originate from MXene oxidation, polymer aging, electrolyte evaporation, or interfacial delamination. Impedance spectroscopy helps distinguish electrolyte resistance, charge-transfer limitations, and contact deterioration [30, 41, 45].

Battery-type specific capacity is:

$$Q_{sp} = \frac{I\,\Delta t}{3.6\,m} \quad (14)$$

where I is current in A, Δt is discharge time in s, and m is active-material mass in g, giving Qsp in mAh g^{-1}; omitting 3.6 instead gives charge in C g^{-1}.

Coulombic efficiency is:

$$CE = \frac{Q_{\mathrm{dis}}}{Q_{\mathrm{ch}}} \times 100\% \quad (15)$$

where Qdis and Qch are discharge and charge capacities.

For symmetric supercapacitors, equivalent series resistance is estimated as:

$$ESR = \frac{\Delta V_{\mathrm{IR}}}{2I} \quad (16)$$

where ΔVIR is the instantaneous potential change for reversal from +I to −I, a total current step of 2I. For a step from zero current to I, use ΔVIR/I instead.

These integrated criteria enable reliable assessment of hydrogel–MXene energy-storage devices under realistic mechanical and electrochemical operating conditions.

4.3 Encapsulation, Environmental Stability, and System Integration

Encapsulation is essential for hydrogel–MXene energy-storage devices because it regulates water retention, oxygen penetration, electrolyte leakage, mechanical compliance, and operational stability. Suitable encapsulation materials include elastomers, biodegradable polymers, ion-selective membranes, and multilayer barriers, selected according to device functionality and intended lifetime. For biodegradable systems, encapsulation must additionally permit controlled water penetration after the designated service period [13, 14, 137]. Encapsulation should function as an active structural component rather than passive packaging. Although impermeable barriers improve hydration retention, they may restrict programmed degradation, whereas permeable coatings can compromise shelf life. Effective MXene protection requires comprehensive edge sealing, interconnect isolation, and minimization of processing-induced defects [13, 14, 30].

MXene oxidation remains a major limitation because it depends on oxygen exposure, hydration, pH, temperature, nanosheet dimensions, defects, and surface terminations. Polymer wrapping, antioxidant additives, graphene hybridization, oxygen-barrier coatings, controlled storage temperature, and pH regulation can mitigate oxidation while preserving ionic accessibility [30, 41, 45]. System-level integration requires reliable electrical and mechanical coupling between energy-storage units and functional components. Printed MXene electrodes can be integrated with wireless coils, sensors, photodetectors, and triboelectric nanogenerators. However, autonomous operation additionally requires rectification, voltage regulation, charge balancing, and protection circuits to ensure stable energy delivery [18, 20, 40].

Fig. 11 highlights the importance of evaluating electrochemical performance alongside mechanical flexibility, adhesion, and environmental durability. High capacitance alone cannot guarantee practical functionality if dehydration, interfacial delamination, or mechanical mismatch compromises device operation. Therefore, stable ionic transport and electrode contact under realistic deformation are essential performance criteria [2, 5, 16, 17]. Fig. 11(a) illustrates a flexible zinc-ion battery (ZIB) incorporating a three-dimensional P(AM-AA)/Ti-MXene hybrid hydrogel electrolyte (HHGE). The cyclic voltammetry (CV) profile of the Zn//V_2O_5@CNTs pouch cell at 2 mV s^{-1}, shown in Fig. 11(b), exhibits characteristic Zn^{2+} insertion/extraction behavior. The anodic feature near 1.28 V separates into two peaks, indicating multistep Zn^{2+} extraction, while peaks at approximately 0.9 and 0.57 V correspond to V^{5+}/V^{4+} and V^{4+}/V^{3+} redox transitions, respectively. Overlapping CV profiles indicate favorable electrochemical reversibility [126].

Mechanical characterization in Fig. 11(c) demonstrates the reinforcement provided by Ti-MXene. The 0.2MX/P(AM-AA) hydrogel achieved a tensile strength of 1.6 MPa, modulus of 0.018 MPa, and toughness of 0.103 MJ m^{-3}. Corresponding values were 1.1 MPa, 0.014 MPa, and 0.68 MJ m^{-3} for 0.1MX/P(AM-AA); 0.9 MPa, 0.01 MPa, and 0.35 MJ m^{-3} for 0.3MX/P(AM-AA); and 0.4 MPa, 0.004 MPa, and 0.2 MJ m^{-3} for pristine P(AM-AA). Hydrogen bonding promotes reinforcement, whereas excessive MXene loading induces aggregation and structural defects [126]. Fig. 11(d) and (e) demonstrate shape recovery and support of a 500 g load, respectively. An acrylamide/acrylic acid ratio of 80:20 balances flexibility and interfacial interactions. Furthermore, 24 h immersion in 2 M $ZnSO_4$ facilitates Zn^{2+} coordination with polymer functional groups, strengthening reversible crosslinks without excessive brittleness. Adhesion measurements in Fig. 11(f) demonstrate the interfacial performance of C-PAM@TA hydrogels. The hydrogel achieved 49.5 kPa adhesion strength on zinc foil over a 2 × 2 cm^2 contact area. As illustrated in Fig. 11(g) and (h), it supported 1 kg loads during vertical and horizontal adhesion tests on different substrates, outperforming C-PAM, which supported only 500 g vertically. Fig. 11(i) further demonstrates adhesion under a 1 kg load at −18 °C, consistent with its measured freezing point of −24.0 °C. Phenolic hydroxyl groups contribute to interfacial bonding and improved electrode contact.

Electrochemical cycling in Fig. 11(j) demonstrates an initial areal capacity of 122.87 μAh cm^{-2}, retaining 87.4 μAh cm^{-2} after 7000 cycles at 1 mA cm^{-2}, with Coulombic efficiency approaching 100%. In comparison, the C-PAM-based cell retained 59.8 μAh cm^{-2} after 5000 cycles. At −18 °C, capacity decreased from 57.50 to 27.02 μAh cm^{-2} after 1000 cycles, demonstrating retained low-temperature functionality [128].

Fig. 11(k) illustrates device flexibility at different bending angles. At 2 mA cm^{-2}, the battery initially delivered 100.22 μAh cm^{-2}, exhibiting a slight capacity increase during bending from 45° to 180°. Following recovery to its flat configuration, capacity reached 115.28 μAh cm^{-2}, indicating favorable deformation tolerance [128]. Additional bending performance is presented in Fig. 11(l)–(p) [129]. Fig. 11(l) illustrates different battery configurations, while Fig. 11(m) and (n) present charge–discharge behavior, capacity retention, and Coulombic efficiency. Initial specific capacity was 94.17 mAh g^{-1}, increasing slightly during bending and reaching 108.50 mAh g^{-1} upon returning to the flat state, with Coulombic efficiency approaching 100%. Electrochemical impedance spectroscopy in Fig. 11(o) indicates reduced interfacial resistance during bending, consistent with improved electrode–electrolyte conformity. After 1800 bending cycles at 180° and 3 A g^{-1}, the battery retained 94.67 mAh g^{-1}, as demonstrated in Fig. 11(p). Finally, Fig. 11(q) demonstrates the deformation tolerance of a flexible Zn//V_2O_5 pouch cell incorporating a PAM–PEG/Ti-MXene (0.1 wt.%) interpenetrating polymer network electrolyte. With a 2 × 2 cm^2 electrode area, the device maintained nearly unchanged galvanostatic charge–discharge profiles at 1 A g^{-1} during bending up to 180°, alongside approximately 100% Coulombic efficiency. Collectively, these findings demonstrate that encapsulation, oxidation resistance, interfacial adhesion, and mechanical compliance must be optimized alongside electrochemical performance to achieve reliable hydrogel–MXene energy-storage systems suitable for integrated wearable and soft mechatronic applications.

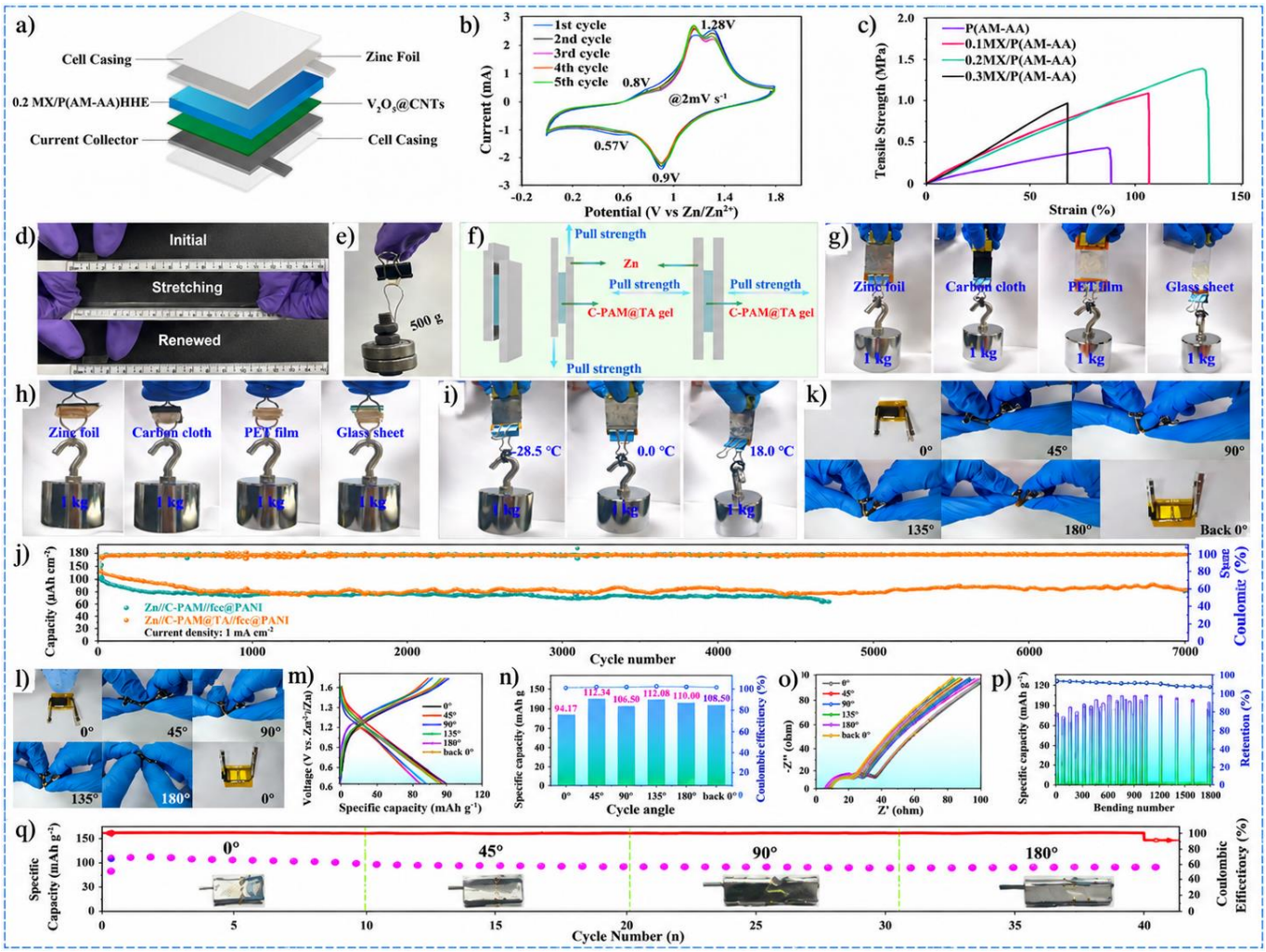


Fig. 11. Electrochemical performance, mechanical properties, adhesion, flexibility, and cycling stability of MXene-based hydrogel electrolytes for flexible zinc-ion batteries (ZIBs). (a) Schematic illustration of the fabricated ZIB pouch-cell architecture incorporating 0.2MX/P(AM-AA) hybrid hydrogel electrolyte (HHGE); (b) cyclic voltammetry (CV) profile of the Zn//V_2O_5@CNTs cell at a scan rate of 2 mV s^{-1}; (c) comparison of the tensile strengths of P(AM-AA) and Ti-MXene/P(AM-AA) hydrogels; (d) shape recovery capability; and (e) load-bearing capacity of the 0.2MX/P(AM-AA) hydrogel. Reproduced with permission from Ref. [126]. Copyright 2025, Elsevier. (f) Schematic illustration of vertical and horizontal lap-shear adhesion tests; (g) photographs demonstrating the vertical lap-shear adhesion of C-PAM@TA hydrogels to various materials while supporting a 1 kg weight; (h) photographs illustrating horizontal lap-shear adhesion to different substrates under a 1 kg load; (i) temperature-dependent vertical lap-shear adhesion of C-PAM@TA hydrogels while lifting a 1 kg weight; (j) cycling stability of flexible ZIBs at a current density of 1 mA cm^{-2} under ambient conditions; and (k) photographs demonstrating the flexibility of a Zn//C-PAM@TA//fcc@PANI ZIB at different bending angles. Reproduced with permission from Ref. [128]. Copyright 2024, Elsevier. (l) Photographs illustrating the flexibility of a Zn//MPZC//MXene/PANI battery at various bending angles; (m) galvanostatic charge–discharge (GCD) profiles of flexible ZIBs at different bending angles and a current density of 3 A g^{-1}; (n) capacity retention and Coulombic efficiency at various bending angles; (o) electrochemical impedance spectroscopy (EIS) spectra under different bending conditions; and (p) capacity retention and Coulombic efficiency during repeated bending cycles up to 180°. Reproduced with permission from Ref. [129]. Copyright 2025, Elsevier. (q) GCD profiles of

a flexible ZIB incorporating a PAM–PEG/Ti-MXene (0.1 wt.%) interpenetrating polymer network (IPN)-based HHGE under various bending conditions. Reproduced with permission from Ref. [130]. Copyright 2025, Elsevier.

4.4 Material-Specific Trade-Offs: Conductivity, Oxidation, Processability, and Safety

Conductivity, stability, and exposure must be assessed together. MXene/PVA electrolytes [33] require electronic isolation, whereas MXene/PPy electrodes [34] require persistent conducting contacts. Porous MXene/PNIPAAm aging [104] and MBene colloidal electrolytes [38] demonstrate formulation-specific stability questions. Low-temperature rGO/PAM [73] and ion-coordinated PVA–MoS_2 [75] connect hydration with function. Degradable MXene/PVA/bacterial-cellulose gels [138] do not establish universal biological safety. Comparisons should report residual reagents, released species, aging, and complete-device failure under the intended exposure.

5. Hydrogel-Based Flexible Energy Storage Devices

Hydrogel nanomaterial-based energy-storage devices integrate mechanical flexibility with electrochemical functionality through multifunctional polymer networks. Hydrogels can function as leakage-resistant electrolytes, conductive scaffolds incorporating MXenes, graphene derivatives, conductive polymers, or redox-active particles, and integrated components providing separation, adhesion, sensing, and encapsulation. Such multifunctionality reduces interfacial complexity and improves mechanical compatibility in soft mechatronic systems [1, 4, 5]. Design requirements vary with device architecture. Supercapacitors primarily require accessible electrochemically active surfaces and stable voltage windows, whereas batteries demand suppression of dendrite formation, electrode-volume changes, and parasitic reactions. Microsupercapacitors prioritize miniaturization, printing resolution, and interconnect reliability, while self-powered and biomedical systems require efficient energy management, safety, and predictable operational lifetimes. Accordingly, hydrogel–MXene architectures should be optimized according to application-specific requirements [16, 20, 115].

5.1 Flexible Supercapacitors and Microsupercapacitors

Flexible supercapacitors are particularly suitable for soft mechatronic applications because of their high power density, rapid charge–discharge capability, and cycling durability. MXene electrodes provide high volumetric capacitance, hydrogel electrolytes maintain deformable electrode–electrolyte interfaces, and flexible polymeric or textile substrates provide mechanical support. However, sustained electrical contact and ionic accessibility during repeated deformation remain essential challenges [5, 39, 144]. Practical device demonstrations should extend beyond powering light-emitting diodes because such tests provide limited information about real-world operating requirements. More representative evaluations include powering wearable strain sensors during movement, buffering photovoltaic or triboelectric energy, supporting intermittent wireless transmission, and supplying soft actuators during cyclic operation. These demonstrations establish a clearer relationship between electrochemical properties and functional autonomy [18, 20, 40].

Microsupercapacitors (MSCs) provide additional advantages through miniaturized interdigitated electrode configurations that shorten ionic diffusion pathways and facilitate integration with sensors and electronic circuits. Inkjet printing, screen printing, extrusion printing, transfer printing, stamping, and laser patterning enable controlled MXene deposition onto flexible substrates. However, electrode quality strongly depends on ink viscosity, surface tension, dispersion stability, drying behavior, and post-deposition conductivity [19, 39, 109]. Additive-free MXene inks offer high active-material fractions but remain susceptible to oxidation, nozzle clogging, nonuniform drying, and insufficient substrate adhesion. Incorporating polymeric or hydrogel additives improves rheological properties and processability, although excessive loading may reduce electrical conductivity. Consequently, systematic characterization should consider ink rheology, nanosheet stability, printing resolution, and electrochemical performance [20, 131-133]. Electrical integration is equally important for practical MSC applications. Series-connected cells increase output voltage but introduce potential cell mismatch, whereas parallel configurations increase capacity and current delivery. Conductive connections employing liquid metals, printed MXene, serpentine conductors, or elastomer-supported metallic films must maintain low resistance during repeated deformation [19, 145, 146].

Areal energy density is expressed as:

$$E_A = \frac{E}{A} \tag{17}$$

where E_a represents areal energy density, (E) denotes stored energy, and (A) is the footprint area of the cell or integrated device.

For series-connected MSC arrays, the ideal output voltage is:

$$V_{\text{series}} = \sum_i V_{\text{cell},i} \tag{18}$$

where Vcell,i represents the voltage of the individual cell. Practical output may be reduced by voltage mismatch and leakage.

Fig. 12 illustrates the relationship between MXene processing, electrode patterning, and MSC fabrication. Printing resolution determines electrode dimensions and accessibility, while integrated interconnects and electrolyte isolation influence operational stability under deformation [19, 39, 109, 145, 146].

MXene electrodes have been investigated in aqueous, solid-state, and microsupercapacitor systems because of their favorable electrical properties and processing versatility [5]. Depending on device configuration, MXenes can be fabricated as freestanding films, slurry-coated electrodes, or patterned conductive layers. Symmetric and asymmetric electrode architectures may incorporate aluminum [109], graphite [147], or nickel foam [148] as conductive substrates. Aqueous supercapacitors commonly employ pouch-cell configurations [148], whereas solid-state devices utilize sandwich architectures

containing gel electrolytes between electrodes [147]. MSC fabrication requires precise deposition of nanometer- to micrometer-thick MXene films [109, 131-133, 149]. Fig. 12(a) summarizes digital methods, including inkjet and three-dimensional printing, alongside nondigital approaches such as screen and transfer printing [150]. Mask-assisted filtration, spray coating, and laser patterning further enable controlled electrode fabrication. Their effectiveness depends on MXene dispersion quality, ink formulation, and compatibility with the selected substrate.

Fig. 12(b) illustrates stamping-assisted fabrication of all-MXene MSCs. Interdigitated, yin–yang, and spiral stamp geometries were designed using computer-aided design software and fabricated through three-dimensional printing. $Ti_3C_2T_x$ or Ti_3CNT_x inks were subsequently deposited onto hydrophilic, textured stamp surfaces and transferred onto paper-based substrates. Silver-wire connections and gel electrolytes completed the devices, followed by room-temperature drying. This approach enables customizable electrode geometries and simplified assembly. Fig. 12(c) illustrates ADS-MXene(CMC) ink preparation for electrohydrodynamic (EHD) jet printing. Approximately 1 μm MXene flakes were mixed with ethanol-dispersed ADOPA ligands at 25 °C for 1 h. Following centrifugation, the treated material was redispersed in a carboxymethyl cellulose-containing solvent mixture comprising diethylene glycol, ethylene glycol, ethanol, and deionized water [151]. The resulting ink exhibited approximately 3400 S cm^{-1} conductivity, 4 × 10^3 cP viscosity, and oxidation resistance for three months under accelerated aging at 55 °C and 85% relative humidity.

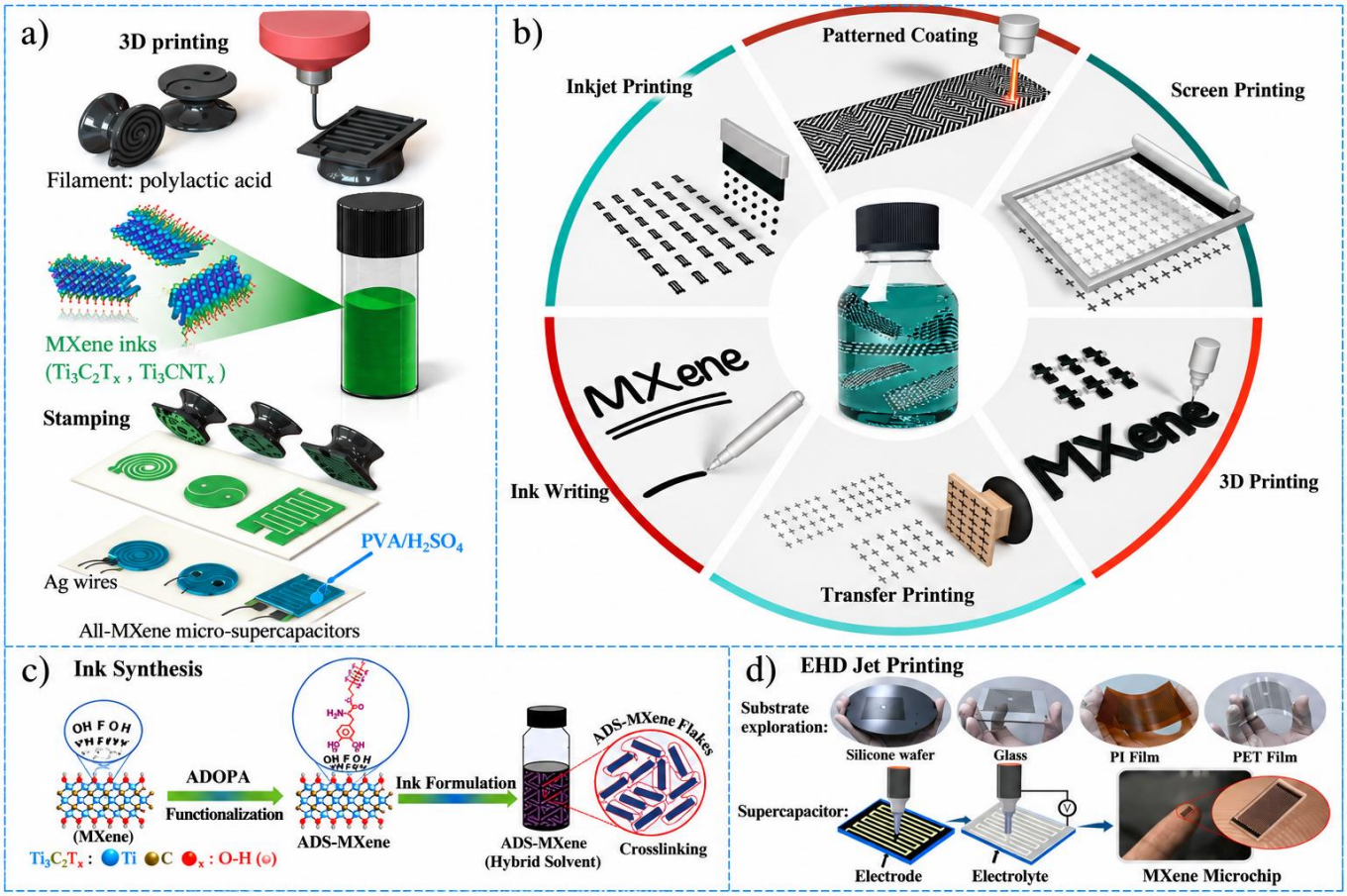


Fig. 12. Advanced MXene patterning, printing techniques, and fabrication strategies for microsupercapacitors (MSCs). (a) Schematic illustration of MXene film patterning techniques, including screen printing, 3D printing, transfer printing, ink writing, inkjet printing, and laser patterning of preformed MXene films, highlighting their versatility in fabricating precise and adaptable coatings through direct patterning and additive manufacturing. Reproduced with permission from Ref. [39]. Copyright 2020, Wiley-VCH. (b) Fabrication of all-MXene-based MSCs using a stamping technique to simplify device assembly and facilitate integration into flexible and scalable energy storage systems. Reproduced with permission from Ref. [109]. Copyright 2018, Wiley-VCH. (c) Schematic illustration of printable ADS-MXene(CMC) ink preparation through chemical etching, MXene flake surface functionalization, and ink formulation using carboxymethyl cellulose (CMC) in a hybrid solvent; (d) electrohydrodynamic (EHD) jet printing of symmetric interdigitated high-definition microsupercapacitors (HD-MSCs) on various substrates, including silicon wafers, glass, polyethylene terephthalate (PET), and polyimide (PI) films, followed by electrolyte deposition on glass to fabricate an MXene-based microchip. Reproduced with permission from Ref. [151]. Copyright 2026, Elsevier.

Fig. 12(d) demonstrates high-resolution EHD jet printing of interdigitated MSCs on silicon wafers, glass, PET, and polyimide substrates. ADS-MXene(CMC) ink was deposited onto gold current collectors to minimize interfacial defects and microcracking. A solid-state gel electrolyte was subsequently printed onto the patterned electrodes and ultraviolet-cured. This integrated process enables precise electrode–electrolyte fabrication and compact MXene-based microchip development.

Table 3. *Overview of representative microsupercapacitor fabrication techniques, including direct laser processing, screen printing, 3D printing, inkjet printing, spray coating, laser engraving, laser scribing, and mask-assisted filtration, highlighting their material formulations, electrochemical performance, energy-storage characteristics, and potential applications in flexible electronics, wearable devices, and self-powered sensing systems., summarized from Refs. [18-20, 40, 125, 152-155], and [156].*

Fabrication technique	Ink/material formulation	Electrochemical characteristics	Target application	Ref.
Direct laser processing	MXene/black phosphorus hybrid ink	0.8 V; capacitance 896.8 F cm^{-3}; energy density 15.9 μWh cm^{-2}	Self-powered intelligent sensor system	[18]
Screen printing	MXene-LTO and MXene-LFP inks	0.6 V; capacitance 1108 mF cm^{-2}; energy density 13.8 μWh cm^{-2}	Self-sustained pressure sensor	[19]
Extrusion 3D printing	MXene aqueous inks	0.6 V; capacitance 900 mF cm^{-2}; energy density 9.7 μWh cm^{-2}	Wireless sensing devices	[20]
Screen printing	$Ti_3C_2T_x$/Co-Al LDH in NMP	1.45 V; capacitance 28.5 mF cm^{-2}; energy density 8.84 μWh cm^{-2}	Force detection sensors	[156]
Inkjet printing	MXene/PH1000 ink	0.6 V; capacitance 754 F cm^{-3}; energy density 9.4 mWh cm^{-3}	Temperature monitoring	[155]
Spray coating	$Ti_3C_2T_x$ with LiCl additive	1.3 V; capacitance 802.3 F cm^{-3}; energy density 188 μWh cm^{-3}	Pressure sensing	[125]
Laser engraving	MXene/graphene oxide aqueous solution	1.35 V; capacitance 83.96 mF cm^{-2}; energy density 21.3 μWh cm^{-2}	Pressure sensor systems	[152]
Laser scribing	MXene aqueous ink	1.2 V; capacitance 84.07 mF cm^{-2}; energy density 16.8 μWh cm^{-2}	Photodetector	[40]
Mask-assisted filtration	$Ti_3C_2T_x$ doped with Mn ions	1.0 V; capacitance 87 mF cm^{-2}; energy density 11.8 mWh cm^{-3}	EMI shielding	[154]

Collectively, these fabrication strategies demonstrate that MXene ink chemistry, electrode architecture, printing precision, and interconnect stability govern the electrochemical performance and integration potential of flexible MSCs. Optimizing these interdependent parameters is essential for developing compact, mechanically reliable energy-storage systems capable of supporting autonomous soft mechatronic devices.

5.2 Flexible Batteries and Hybrid Energy Systems

Flexible batteries and hybrid energy-storage systems are essential for autonomous soft mechatronic platforms requiring sustained energy delivery, mechanical compliance, and efficient power management. Hydrogel electrolytes improve battery safety, suppress leakage, and maintain electrode contact during deformation, while MXene-based microsupercapacitors (MSCs) buffer intermittent harvested energy and supply rapid power pulses for sensing, wireless communication, and actuation [1, 5, 18-20].

Aqueous zinc-ion batteries (ZIBs) are particularly attractive because of their relatively low cost, safety, and compatibility with hydrogel electrolytes. Nevertheless, zinc dendrite formation, hydrogen evolution, cathode dissolution, and electrolyte dehydration compromise cycling stability [1, 125]. Other battery chemistries, including lithium-, sodium-, and multivalent-ion systems, may experience passivation, corrosion, gas evolution, and undesirable polymer–electrolyte interactions. Although MXene incorporation improves conductivity, ion transport, and interfacial stability, uncontrolled surface reactions can accelerate degradation. Consequently, flexible batteries require electrochemical and mechanical reliability assessments under realistic operating conditions [123, 125, 143, 157]. MXenes function as conductive hosts, protective interlayers, current collectors, surface coatings, and ion-storage electrodes. Their hydrophilic surfaces and tunable interlayer galleries facilitate ionic transport, particularly for strongly solvated multivalent ions [158, 159]. Hydrogel separators additionally prevent direct electrode contact while maintaining ionic conductivity, puncture resistance, and dimensional stability. Self-healing and antifreezing formulations further improve environmental tolerance [1, 115].

Hybrid architectures combine complementary energy technologies. Triboelectric nanogenerators harvest mechanical motion, photovoltaic cells convert light, and thermoelectric devices utilize temperature gradients, whereas batteries provide sustained energy and supercapacitors deliver rapid power [18, 19, 160]. MXenes can serve as electrodes, interconnects, electromagnetic shields, and sensors, while hydrogels provide ionic conduction, adhesion, and compliant interfaces [3, 19, 20]. Effective integration requires quantitative evaluation of harvested energy, rectification efficiency, storage capacity, leakage, self-discharge, and load requirements. Operational demonstrations should therefore assess realistic duty cycles rather than brief device activation [18, 20]. Fig. 13 presents flexible microbatteries, interconnected MSC arrays, and integrated energy-harvesting systems, emphasizing voltage matching, mechanical reliability, and coordinated energy distribution [18-20, 160].

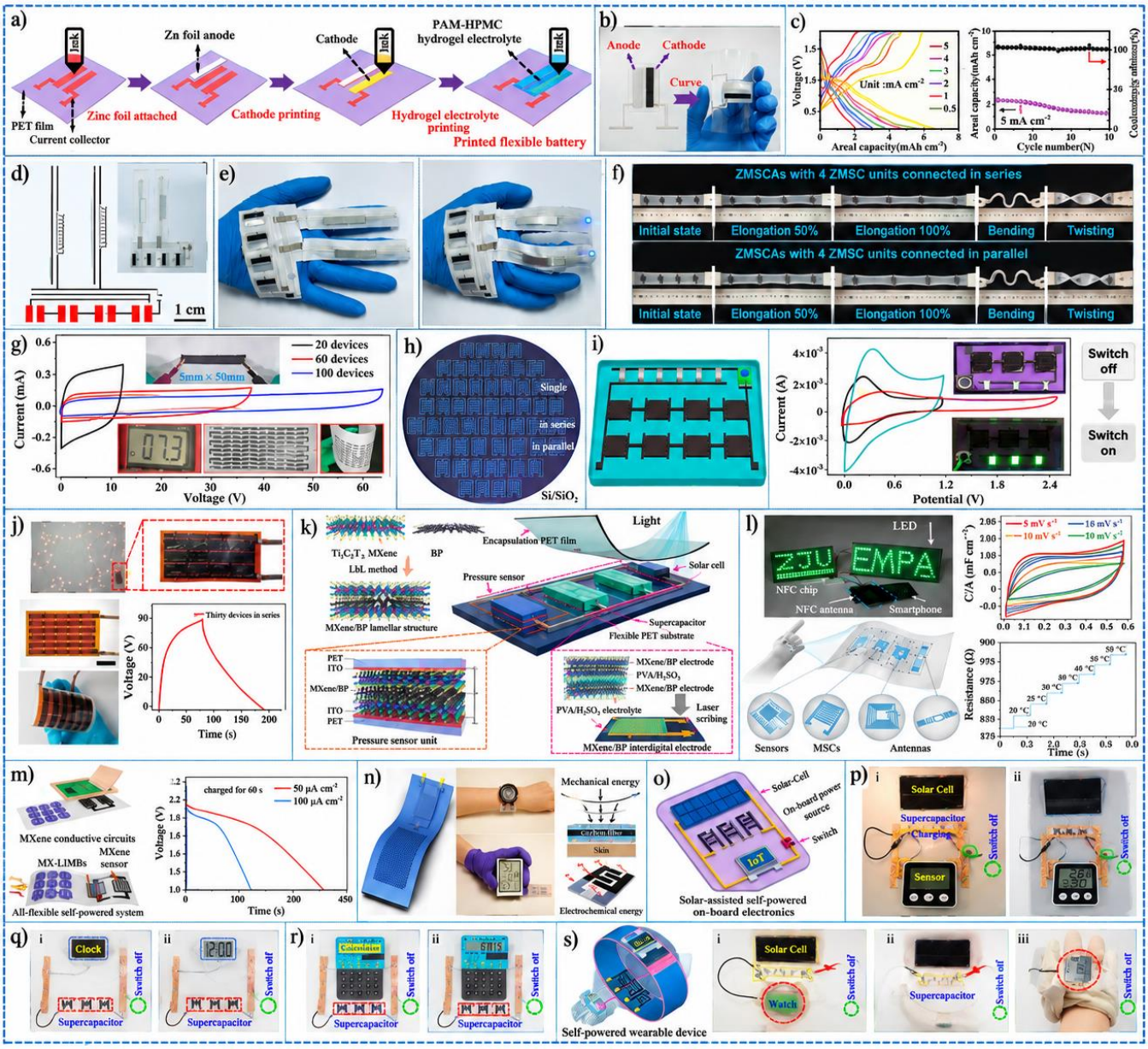


Fig. 13. Fabrication, electrochemical performance, electrical interconnection, energy harvesting integration, and self-powered applications of flexible zinc-ion microbatteries (FZIMBs) and MXene-based microsupercapacitors (MSCs). (a) Schematic illustration of the printing process of FZIMBs; (b) optical photographs of the fabricated FZIMBs; (c) galvanostatic charge–discharge (GCD) profiles at current densities of 0.5–5 mA cm^{-2} and cycling performance at 5 mA cm^{-2}; (d) and (e) demonstrations of practical FZIMB applications. Reproduced with permission from Ref. [161]. Copyright 2024, Elsevier. (f) Integration of stretchable zinc-ion MSCs using liquid metal bridge/island (LMBI) interconnects to maintain mechanical durability and electrochemical stability under bending and stretching. Source cited as Ref. [145]. Copyright 2021, Elsevier. (g) Fabrication of 100 tandem linear MXene MSCs through screen printing, achieving an operating voltage of 60 V; (h) configuration of 30 MSCs connected in series; (i) parallel and series configurations of MXene-based MSCs, highlighting enhanced charge transfer through highly conductive MXene pathways. Reproduced with permission from Ref. [162]. Copyright 2020, Elsevier; and Ref. [146]. Copyright 2022, American Chemical Society. (j) Integration of MXene-based MSCs with solar cells for efficient energy harvesting and stable electrical output. Reproduced with permission from Ref. [18]. Copyright 2021, Wiley-VCH. (k) Integration of MXene-based materials and hydrogels with pressure sensors for flexible sensing applications. Panels (g) and (k) are reproduced with permission from Ref. [19]. Copyright 2021, Wiley-VCH. (l) Room-temperature printing of MXene inks for multifunctional flexible electronics integrating wireless charging, energy storage, and rapid temperature sensing. Reproduced from Ref. [20]. Copyright 2022, the authors, published by Springer Nature. (m) Integration of MXene-based MSCs with a triboelectric nanogenerator for wearable self-powered energy harvesting and storage. Ref. [160] for panel (m); and (n) Schematic illustration of a 3D-printed supercapacitor serving as an onboard power source for miniature electronic devices; (o) solar-powered sensing system incorporating a 3D-printed supercapacitor; (p) demonstration of a clock powered by a 3D-printed supercapacitor; (q) illustration of a calculator powered by a 3D-printed supercapacitor; (r) schematic illustration of a self-powered wearable device incorporating a 3D-printed flexible supercapacitor; and (s) demonstration of a customized wearable watch powered by a 3D-printed supercapacitor. Reproduced with permission from Ref. [163]. Copyright 2026, Elsevier.

Fig. 13 highlights the importance of integrating energy harvesting, storage, and utilization within mechanically adaptable platforms. Reliable operation requires coordinated charging characteristics, voltage regulation, durable electrical interconnections, and stable energy delivery during repeated deformation [18-20, 160]. Fig. 13(a)–(c) illustrates three-dimensional (3D)-printed flexible zinc-ion microbatteries (FZIMBs) incorporating printable hydrogel electrolytes (PHHEs). Cathode ink comprising $NH_4V_4O_{10}$, carbon nanotubes (CNTs), and 3 wt.% polyvinylidene fluoride (PVDF) was

deposited onto conductive silver-patterned polyethylene terephthalate (PET) substrates. A 0.03 mm zinc-foil anode was subsequently attached, followed by cathode printing at an active-material loading of 14.47 mg cm^{-2}. Ultraviolet curing of the printed PHHE completed device fabrication, as shown in Fig. 13(a) and (b) [161].

The galvanostatic charge–discharge (GCD) results in Fig. 13(c) demonstrate areal capacities of 6.45, 5.23, 4.89, 4.00, 3.48, and 2.43 mAh cm^{-2} across current densities of 0.5–5 mA cm^{-2}. At 5 mA cm^{-2}, the battery retained 1.77 mAh cm^{-2} after 30 cycles, corresponding to 72.84% capacity retention and 99–100% Coulombic efficiency. These results demonstrate favorable rate capability and electrochemical reversibility, supporting the integration of printed hydrogel electrolytes into flexible microbattery architectures [161]. Fig. 13(d) and (e) demonstrates practical wearable sensing using four series-connected FZIMBs, a pressure-sensitive component, and light-emitting diodes (LEDs). When attached to the hand, finger bending completes the electrical circuit and activates the LEDs. Increasing deformation reduces sensor resistance, thereby increasing LED brightness. This response illustrates the integration of energy storage and movement-responsive sensing within a mechanically flexible platform [161].

Electrical interconnection is essential for achieving the voltage and capacity required by miniaturized electronic systems [164, 165]. Series-connected MSCs increase output voltage, while parallel arrangements enhance storage capacity and current delivery. Highly conductive MXene pathways reduce contact resistance and facilitate compact device integration [146, 162]. Fig. 13(f) illustrates liquid-metal bridge/island (LMBI) interconnects that accommodate mechanical deformation while preserving electrochemical functionality [145]. High-voltage integration is demonstrated in Fig. 13(g), where 100 screen-printed MSCs achieved 60 V [19]. Similarly, Fig. 13(h) presents 30 series-connected MSCs on a polyimide substrate generating 90 V [153]. Fig. 13(i) further illustrates series and parallel configurations, emphasizing the importance of reliable conductive connections for scalable energy-storage arrays. Self-powered microelectronic systems increasingly combine MSCs with photovoltaic devices, wireless charging technologies, and nanogenerators [18, 20, 166-170]. Fig. 13(j) illustrates solar-cell integration for energy harvesting and storage, while Fig. 13(k) demonstrates coupling with flexible pressure sensors. Fig. 13(l) presents room-temperature MXene printing for multifunctional electronics incorporating wireless charging, energy storage, and rapid temperature sensing.

Fig. 13(m) illustrates MXene-based MSC integration with a triboelectric nanogenerator. Under periodic mechanical excitation at 5 Hz, the integrated system reached approximately 0.6 V within 30 min. Although this confirms the feasibility of mechanical energy harvesting and electrochemical storage, relatively limited conversion efficiency highlights the need to improve energy-transfer and charging performance [160]. Fig. 13(n)–(q) demonstrates the application of three-dimensional printed $Ti_3C_2T_x$/polyaniline (PANI) MSCs as compact power sources. A miniature photovoltaic panel rated at 2.5 V and 60 mA directly charged three series-connected MSCs operating at 2.4 V, without an intermediate voltage-conversion circuit. Under artificial illumination, full charging occurred within approximately 5 s. The stored energy subsequently powered a digital watch and calculator, demonstrating practical energy delivery for miniature electronics [163]. Finally, Fig. 13(r) and (s) illustrates wearable integration of the flexible MSC into a wrist-mounted platform, where miniature solar charging enabled digital-watch operation. This demonstration combines scalable printing, rapid charging, mechanical adaptability, and reliable energy delivery [163]. Collectively, these developments demonstrate that autonomous soft mechatronic systems require coordinated optimization of battery capacity, supercapacitor power delivery, energy harvesting, electrical interconnections, and mechanical durability. Integrating these functions within compatible hydrogel–MXene architectures offers a pathway toward flexible, self-powered wearable electronics and adaptive sensing platforms.

5.3 Device-Level Comparison of Different 2D Material–Hydrogel Systems

Device comparisons must include both electrodes, electrolyte, collectors, and packaging. Boride interdigitated devices [64], LDH/carbon-hydrogel capacitors [65], MoS_2 membranes [36], rGO/polyacrylate cells [70], and MXene/rGO hybrid capacitors [35] use different reactions and component roles; MXene/PVA electrolytes add another configuration [33]. Report delivered discharge energy, footprint, thickness, loading, voltage, and mass basis. Compare deformation at matched strain or radius, hydration, and cycling conditions, including impedance and self-discharge. Battery capacity and capacitor capacitance must remain distinct.

6. Soft Robotics and Self-Powered Wearables

Soft robotics and wearable electronics require energy-storage systems that maintain electrochemical functionality under continuous mechanical deformation. Conventional rigid batteries restrict movement and concentrate mechanical stress, whereas liquid electrolytes present leakage and packaging challenges. Hydrogel–MXene composites offer flexible, stretchable, adhesive, and multifunctional architectures, enabling distributed energy storage across robotic structures, wearable textiles, and conformable electronic interfaces [2, 3, 106]. Energy consumption varies considerably among soft mechatronic components. Sensors generally require low continuous power, wireless communication modules demand intermittent energy pulses, and actuators may require short-duration high-current outputs. Hybrid architectures combining flexible batteries with locally integrated microsupercapacitors can therefore provide sustained energy storage and rapid power delivery. Patterned hydrogel–MXene electrodes additionally facilitate distributed integration without substantially increasing mechanical stiffness, supporting compact and mechanically adaptive electronic systems [16, 20].

6.1 Energy Storage for Soft Robotic Actuation

Energy-storage placement critically influences robotic deformation, mechanical stability, and actuation efficiency. Positioning storage devices near the neutral mechanical plane minimizes bending-induced strain, whereas embedding conductive layers reduces external wiring but may alter stiffness distribution. Gradient hydrogels and patterned MXene electrodes enable localized energy storage in mechanically favorable regions while maintaining flexibility and actuation amplitude [45, 108]. Different actuation mechanisms impose distinct electrical and thermal requirements. Pneumatic systems require pumps and valves, dielectric elastomers demand high operating voltages, thermoresponsive hydrogels require heating, and electrochemical actuators depend on ionic transport. MXene-containing hydrogels additionally support photothermal and Joule-heating actuation through their optical absorption and conductive networks. However, thermal regulation, electrical isolation, and stable electrode interfaces remain essential for reliable multifunctional operation [106, 108]. Mechanical durability is equally important because robotic components experience repeated compression, abrasion, puncturing, and localized damage. Self-healing hydrogel–MXene networks can reconstruct conductive pathways, while adhesive interfaces minimize delamination. Healing performance should also be evaluated following electrochemical cycling because interfacial reactions and electrolyte changes may influence the reversible interactions responsible for structural recovery [108, 134].

Fig. 14 demonstrates the integration of flexible energy storage, distributed sensing, and electrochemical monitoring within mechanically adaptive platforms. These applications require coordinated optimization of mechanical compliance, electrode–electrolyte contact, electrical stability, and energy delivery. Consequently, storage placement and material architecture must balance deformation tolerance with the power requirements of integrated sensing and actuation systems [16, 20, 45, 106, 108]. The mechanical compliance and sensing characteristics of PSC hydrogels enabled the development of self-powered strain sensors integrated with flexible batteries. Fig. 14(a) illustrates their conformable attachment to different body regions. Fig. 14(b) presents reproducible resistance responses during finger bending between 0° and 90°, demonstrating angle-dependent sensitivity and stable signal generation. Real-time resistance variations during finger movement are further demonstrated in Fig. 14(c). Beyond large joint movements, the sensor detected subtle physiological activities, including swallowing and throat vibrations during pronunciation of “time,” “test,” “wide,” “temperature,” “battery,” and “sensor,” as illustrated in Fig. 14(d). The integrated Zn‖PSC‖V_2O_5 battery retained approximately 93% capacity following bending while maintaining stable Coulombic efficiency. Puncture and compression tests additionally demonstrated improved operational safety by preventing electrode contact and internal short circuits, unlike liquid-electrolyte batteries that exhibited leakage and voltage deterioration [171]. PSC hydrogels were further employed to monitor internal battery deformation associated with reversible zinc deposition and stripping. Fig. 14(e) illustrates the experimental configuration, incorporating a hydrogel sensing component, electrochemical workstation, source measurement unit, and X-ray-transparent sample holder. The sensor was positioned alongside the energy-storage unit to measure resistance variations induced by internal pressure changes. Simultaneously, transmission-mode operando X-ray diffraction (XRD) monitored electrode structural evolution during electrochemical cycling, enabling correlation between mechanical responses and zinc deposition behavior [171].

The corresponding operando XRD results in Fig. 14(f) revealed minimal structural changes in the V_2O_5 cathode, with lattice-parameter variations below 0.8%. No characteristic $Zn_xV_2O_5{\cdot}nH_2O$ reflection appeared within the 3°–5° region, while the V_2O_5 (200) and (110) reflections remained near 7.2° and 12.2°, respectively. In contrast, the Zn (002) reflection at approximately 16.5° gradually weakened during stripping and recovered during subsequent plating, confirming reversible zinc deposition. The synchronized PSC sensor exhibited a maximum relative resistance variation of approximately 0.19%. Increasing resistance during charging reflected deformation induced by zinc deposition and associated changes in ionic transport distances. This coupled electrochemical–mechanical response enabled real-time monitoring of internal pressure fluctuations and electrode-volume evolution [171]. Self-powered pressure-sensing matrices provide additional capabilities for spatially resolved mechanical monitoring. Fig. 14(g) illustrates a flexible Zn–MnO_2 battery-based sensor comprising orthogonally patterned electrode layers and an ionic gel-paper separator. The integrated 16 × 16-pixel sensing array incorporates a column-selection multiplexer, analog-to-digital converter, digital-to-analog converter, and field-programmable gate array. These components enable coordinated pixel addressing, signal acquisition, and reconstruction of pressure distributions, supporting applications in wearable electronics, robotics, and physiological monitoring [172].

The dynamic sensing characteristics are demonstrated in Fig. 14(h), including distinguishable responses during single- and double-click mouse operations, simulated tremor monitoring, and human gait detection. Mechanical stimulation at 6 Hz generated recognizable signals relevant to simulated Parkinsonian tremors, while shoe-insole integration enabled differentiation among walking, brisk walking, running, and jumping through variations in signal amplitude and frequency. Robotic fingertip integration was also considered for remote monitoring in challenging environments. Fig. 14(i) further presents a flexible visual touch device combining a paper-based battery, pressure sensor, and light-emitting diode. The device remained functional when wrapped around a cylindrical surface, with illumination intensity varying according to applied pressure [172].

Hydrogel electrolyte engineering additionally improves zinc-based energy-storage reliability by regulating water activity and ion transport. Molecular simulations indicated that multicomponent polymer networks establish extensive hydrogen-bonding interactions with water, modifying Zn^{2+} solvation and suppressing undesirable interfacial reactions [173]. Fig. 14(j) compares Zn//Zn symmetric cells containing

PAM, PAM/LA, and PAM/LA/PSBMA hydrogel electrolytes. At 10 mA cm^{-2} and 1 mAh cm^{-2}, the respective cycling lifetimes were approximately 10, 74, and 290 h, with the optimized electrolyte exhibiting an average zinc depth of discharge of 1.7%. Fig. 14(k) presents the corresponding zinc-anode morphologies following cycling. The PAM electrolyte produced pronounced dendritic deposits, whereas PAM/LA reduced dendrite formation and interfacial degradation. The PAM/LA/PSBMA electrolyte generated a comparatively uniform zinc surface with limited fine dendritic structures. At a lower current density of 1 mA cm^{-2} and an areal capacity of 2 mAh cm^{-2}, the optimized symmetric cell maintained stable cycling for 681 h, demonstrating improved zinc-deposition reversibility and interfacial stability [173].

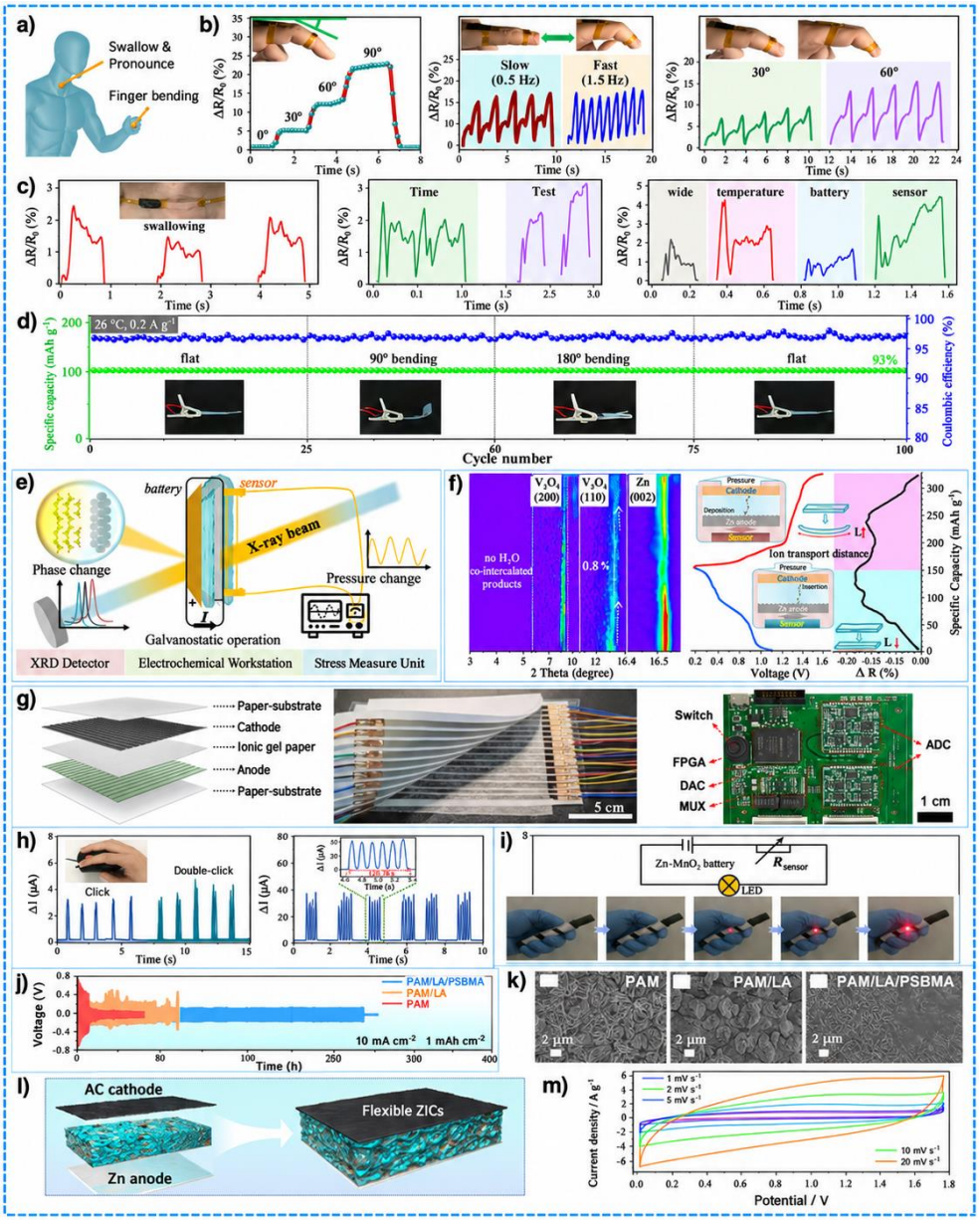


Fig. 14. Multifunctional applications, sensing performance, mechanical responsiveness, and electrochemical characterization of self-powered microsensor devices and hydrogel-based energy storage systems. (a) Schematic illustration of microsensor placement on the human body; (b) step-response resistance profiles at different bending angles; (c) real-time resistance changes of the self-powered sensor attached to fingers during movement; (d) resistance responses during swallowing and pronunciation of the words "time," "test," "wide," "temperature," "battery," and "sensor," together with the specific capacity and Coulombic efficiency of the integrated PSC-gel microdevice at various bending angles; (e) schematic illustration of the microsensor for detecting stress variations induced by Zn deposition and stripping; and (f) contour plot showing variations in the 2θ position and intensity of the (002) reflection obtained from operando X-ray diffraction (XRD) measurements during Zn stripping/plating in a Zn|PSC|V_2O_5 full cell, accompanied by corresponding battery voltage and sensor resistance responses. Reproduced with permission from Ref. [171]. Copyright 2024, American Chemical Society. (g) Exploded schematic, photograph, and readout circuitry of the self-powered pressure-sensing matrix; (h) current responses to single- and double-click mouse operations, simulated knocking associated with early-stage Parkinson's disease, and gait monitoring using a shoe-insole-integrated pressure sensor; and (i) demonstration of a self-powered flexible visual touch device. Reproduced with permission from Ref. [172]. Copyright 2024, Elsevier. (j) Cycling stability of Zn//Zn symmetric cells incorporating PAM, PAM/LA, and PAM/LA/PSBMA hydrogel electrolytes; (k) scanning electron microscopy (SEM) images of Zn foils recovered from disassembled Zn//Zn symmetric cells after cycling; (l) schematic illustration of zinc-ion capacitors (ZICs) incorporating the PAM/LA/PSBMA hydrogel electrolyte; and (m) cyclic voltammetry (CV) profiles of the corresponding ZICs. Reproduced with permission from Ref. [173]. Copyright 2024, American Chemical Society.

Fig. 14(l) illustrates a flexible zinc-ion capacitor comprising an activated-carbon cathode, zinc-metal anode, and multifunctional PAM/LA/PSBMA hydrogel electrolyte–separator. Within an operating voltage window of 0–1.75 V, the optimized device exhibited enhanced electrochemical accessibility, longer discharge duration, reduced internal resistance, and the highest ionic conductivity of 9.48 mS cm^{-1} among the investigated formulations. These improvements were attributed to water retention and stable electrode–electrolyte interactions within the polyzwitterionic network. The nearly rectangular cyclic voltammetry profiles shown in Fig. 14(m) indicate rapid charge-storage kinetics, favorable electrochemical reversibility, and predominantly capacitive behavior [173]. Collectively, these findings demonstrate that hydrogel-based energy storage and sensing are complementary technologies for soft robotics and wearable electronics. Integrating compliant architectures, damage tolerance, stable ionic transport, and distributed monitoring supports the development of autonomous, multifunctional soft mechatronic systems capable of maintaining electrochemical and sensing performance during mechanical operation.

6.2 Bioinspired Energy Storage Architectures

Bioinspired architectures provide effective strategies for improving electrochemical performance through structural optimization rather than chemical modification alone. Biological systems offer hierarchical designs that integrate mechanical flexibility, efficient transport, and structural durability. Skin-inspired multilayers provide compliant interfaces and protected conductive regions, muscle-like aligned fibers facilitate anisotropic deformation and mechanical reinforcement, vascular networks support distributed ionic transport, and leaf-inspired porous architectures enhance interfacial accessibility. These principles can be incorporated into hydrogel–MXene composites through layered casting, directional freezing, three-dimensional printing, microfluidic processing, and gradient filler distribution [174-177]. Biomorphic architectures are particularly advantageous for energy-storage devices subjected to repetitive deformation. Aligned MXene networks promote directional electronic conductivity, while compliant polymer regions accommodate mechanical strain and dissipate energy. Vascular-like electrolyte channels facilitate ionic migration within thicker structures, whereas epidermal multilayers protect electrochemically active components from moisture, perspiration, and biological interfaces. Such hierarchical organization enables improved coordination of conductivity, mechanical toughness, and biocompatibility compared with homogeneous composite networks [45, 106].

Nevertheless, reproducible fabrication remains challenging because structural characteristics depend strongly on processing conditions. Variations in freezing rate, solvent composition, printing speed, ink viscosity, and MXene concentration can influence pore dimensions, nanosheet orientation, and interfacial integrity. Consequently, scalable manufacturing methods, including roll-to-roll printing, digital light processing, direct ink writing, and continuous coating, require standardized characterization of pore-size distribution, orientation factors, and electrical percolation thresholds to ensure consistent device performance [20, 39].

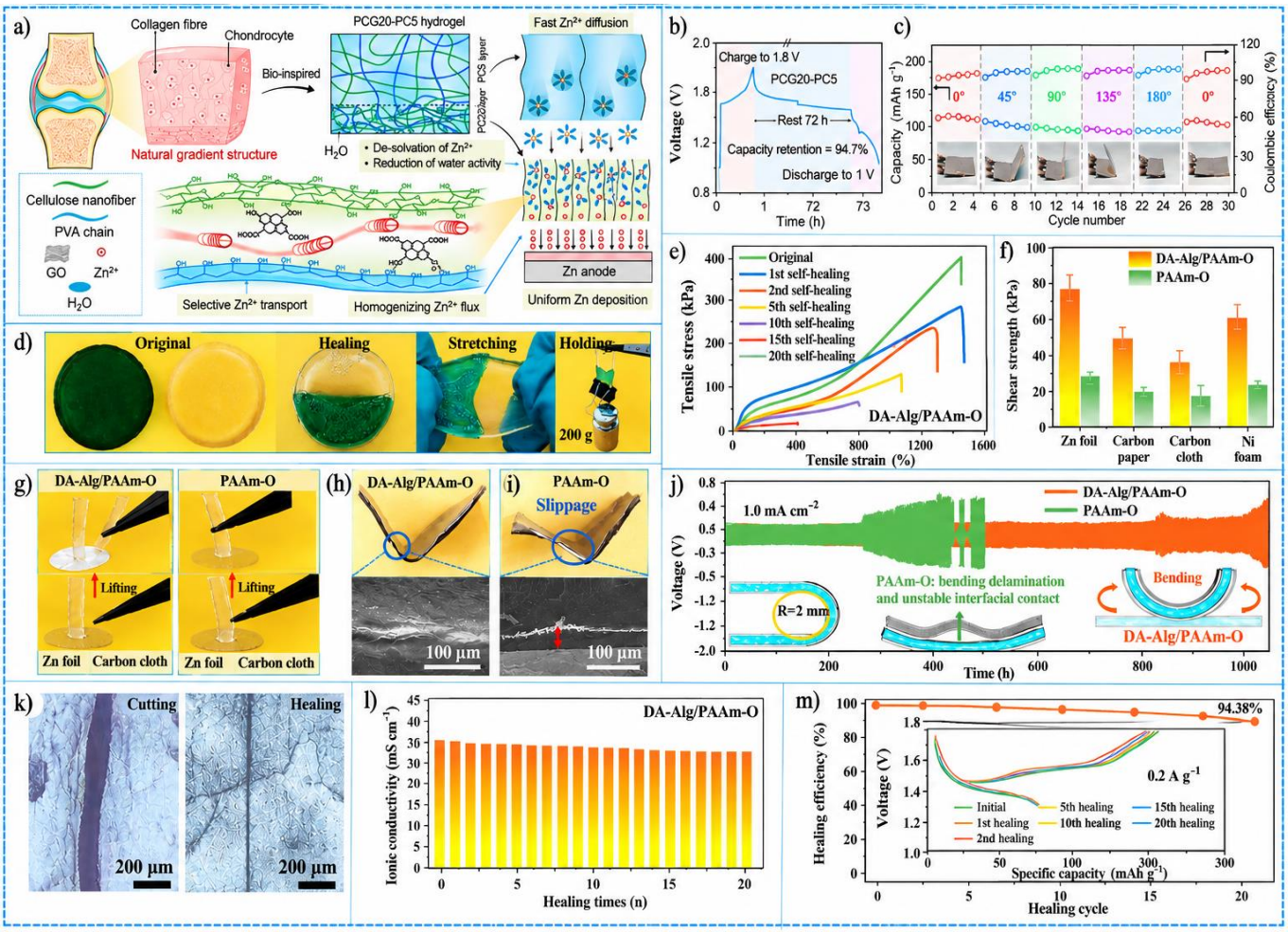


Fig. 15. Structural design, electrochemical performance, mechanical properties, adhesion, and self-healing characteristics of gradient-networked hydrogel electrolytes and gel polymer electrolytes (GPEs) for flexible zinc-ion batteries (ZIBs). (a) Schematic illustration of a natural cartilage-inspired gradient-networked hydrogel electrolyte designed to stabilize the Zn anode; (b) capacity retention of the PCG20–PC5-based Zn–MnO_2 ZIB after charging to 1.8 V, resting for 72 h, and discharging to 1 V; and (c) capacity retention of the PCG20–PC5-based Zn–MnO_2 pouch battery at different bending angles. Reproduced with permission from Ref. [178]. Copyright 2025, American Chemical Society. (d) Schematic illustration of the DA-Alg/PAAm-O GPE structure and the self-healing states of the PAAm GPE; (e) tensile stress–strain profiles of DA-Alg/PAAm-O GPEs at different healing times; (f) shear strength between substrates bonded using the GPEs; (g) comparison of the adhesive properties of PAAm-O and DA-Alg/PAAm-O GPEs; (h) Zn plating/stripping performance using the DA-Alg/PAAm-O GPE; (i) Zn plating/stripping performance using the PAAm-O GPE; (j) voltage–time profiles of Zn symmetric cells incorporating DA-Alg/PAAm-O and PAAm-O GPEs at a current density of 1 mA cm^{-2} and a bending radius of 2 mm, with an inset showing flexible Zn//Zn devices under bending conditions; (k) optical microscopy images illustrating the self-healing process of the DA-Alg/PAAm-O GPE; (l) ionic conductivity of the DA-Alg/PAAm-O GPE after repeated cutting–healing cycles; and (m) self-healing efficiency of aqueous zinc-ion batteries (AZIBs) during repeated cutting–healing cycles, with an inset showing the corresponding charge–discharge profiles at different healing stages. Reproduced with permission from Ref. [179]. Copyright 2024, Elsevier.

Fig. 15 illustrates how bioinspired structural gradients, reversible cross-linking, and interfacial adhesion contribute to the mechanical and electrochemical stability of flexible hydrogel electrolytes. These strategies address coupled transport and mechanical limitations through controlled network architecture rather than relying exclusively on increased conductive filler loading [39, 174-177]. Cartilage-inspired gradient hydrogels provide a representative approach to stabilizing zinc-metal anodes. Natural articular cartilage possesses a hierarchical structure in which densely packed collagen fibers and proteoglycans near the articular surface absorb mechanical impacts, whereas relatively loose regions near the subchondral bone facilitate nutrient transport and cellular metabolism. This spatial organization combines mechanical protection with efficient mass transport. Inspired by these characteristics, gradient hydrogel electrolytes incorporating poly(vinyl alcohol) (PVA) and cellulose nanofibers (CNFs) were developed to improve Zn^{2+} transport and interfacial stability. Their abundant zincophilic hydroxyl groups additionally support network reinforcement and interactions with zinc ions, as illustrated in Fig. 15(a) [178].

The electrochemical applicability of the PCG20–PC5 gradient electrolyte was evaluated in flexible Zn–MnO_2 batteries. Fig. 15(b) presents capacity retention following charging to 1.8 V, resting for 72 h, and discharging to 1 V, providing evidence of charge-storage stability. Fig. 15(c) demonstrates capacity retention under different bending angles, with only minor variations in electrochemical performance. These findings indicate that gradient-network organization can maintain ionic transport and electrode contact during deformation, supporting flexible energy-storage applications [178]. Dynamic polymer interactions offer complementary advantages for self-healing gel polymer electrolytes (GPEs). Fig. 15(d) illustrates the DA-Alg/PAAm-O network and its mechanical recovery

behavior. The separated electrolyte could reconnect and subsequently withstand stretching while supporting a 200 g load. Tensile stress–strain measurements in Fig. 15(e) demonstrate a healing efficiency of 99.9% after the first healing cycle and 72.4% after five cycles, compared with 73.3% for PAAm-O after five cycles. Pristine PAAm exhibited negligible recovery. This behavior was attributed primarily to reversible hydrophobic associations that promote polymer-network reconstruction following damage [179]. Interfacial adhesion further determines electrolyte reliability during mechanical deformation. As presented in Fig. 15(f), DA-Alg/PAAm-O exhibited adhesion strengths of approximately 30–80 kPa on carbon cloth, carbon paper, zinc foil, and nickel foam, exceeding the 10–28 kPa recorded for PAAm-O. Catechol functionalities within DA-Alg and hydrophobic interactions between polymer chains and substrates contributed to stronger interfacial bonding. Fig. 15(g) demonstrates improved adhesion to electrode substrates, whereas the weaker PAAm-O interfaces were more susceptible to separation. These characteristics help maintain electrochemically active contact during bending [179].

The electrochemical consequences of interfacial engineering were investigated through zinc plating and stripping. Fig. 15(h) and (i) compare the behavior of DA-Alg/PAAm-O and PAAm-O electrolytes, respectively. Stronger adhesion in the modified network reduces interfacial displacement and supports more uniform zinc deposition. Fig. 15(j) further demonstrates that DA-Alg/PAAm-O-based Zn//Zn symmetric cells maintained stable cycling beyond 1000 h at a bending radius of 2 mm, current density of 1 mA cm^{-2}, and areal capacity of 0.2 mAh cm^{-2}. In comparison, PAAm-O-based cells operated for approximately 400 h under identical conditions, highlighting the importance of interfacial stability [179]. Self-healing behavior was further characterized through repeated mechanical damage. Fig. 15(k) presents optical microscopy observations showing progressive crack closure and restoration of structural continuity within 24 h. Reversible hydrophobic associations among OMA segments and hydrogen bonding involving DA functional groups facilitated this reconstruction. Fig. 15(l) demonstrates approximately 90.1% ionic-conductivity retention after 20 cutting–healing cycles, with remaining losses primarily associated with dehydration. Importantly, Fig. 15(m) shows that assembled aqueous zinc-ion batteries maintained approximately 94.4% capacity-based healing efficiency after 20 cycles, accompanied by nearly overlapping charge–discharge profiles. These results demonstrate simultaneous mechanical and electrochemical recovery [179]. Collectively, bioinspired gradient architectures, dynamic bonding, and reinforced electrode–electrolyte interfaces provide complementary approaches to improving mechanical adaptability, ionic transport, and long-term electrochemical stability. Their integration offers a promising foundation for durable, self-healing, and conformable energy-storage systems suitable for adaptive soft mechatronic devices.

Recent advances in bio-inspired and multifunctional material engineering provide valuable strategies for developing next-generation flexible energy systems. Wang et al. [180] demonstrated how bio-inspired self-assembly and ordered interfacial architectures can improve the mechanical robustness and ionic transport of composite solid electrolytes. Similarly, Song et al. [181] highlighted biomimetic structural, surface, and interfacial engineering approaches to overcome electrode instability, electrolyte corrosion, and limited cycling durability in zinc-based energy storage devices. Extending these concepts beyond energy storage, Afzal et al. [105] reviewed the multifunctional integration of two-dimensional materials, including MXenes, graphene, and transition-metal dichalcogenides, into triboelectric nanogenerators for enhanced energy harvesting and self-powered sensing. Collectively, these studies emphasize the importance of hierarchical structural design, interfacial optimization, and multifunctional material integration for advancing mechanically adaptable and self-powered energy technologies

6.3 Wearable Sensing and Self-Powered Interfaces

Hydrogel–MXene composites offer considerable potential for wearable sensing because they combine mechanical conformability, electrical conductivity, and efficient signal transduction. Applied strain and pressure modify conductive pathways between MXene nanosheets, producing measurable resistance or capacitance variations [182, 183]. Temperature influences polymer-chain mobility and ionic transport, while hydrated interfaces facilitate physiological signal acquisition. Integrating local microsupercapacitors further enables intermittent sensing and communication without bulky external power sources [2, 3, 107]. Multifunctional wearable devices increasingly integrate energy storage, sensing, and adhesion within shared conductive networks. Although this approach reduces thickness and improves conformability, it introduces interactions between electrochemical and sensing functions. Mechanical deformation may alter capacitance, while charging and discharging can influence sensing signals through ionic redistribution or Joule heating. Therefore, reliable operation requires optimized device architecture, signal calibration, and power-management circuitry [3, 107, 184].

Electronic skin requires distributed sensing, mechanical softness, and stable responses during repeated movement. MXene-based hydrogels provide adjustable sensitivity and deformation tolerance, although highly sensitive percolation networks may exhibit restricted sensing ranges. Microstructured surfaces, multilayer architectures, and mixed-dimensional conductive networks help balance sensitivity, stretchability, and operational stability [107, 136, 184]. Smart textiles introduce additional requirements, including breathability, washability, abrasion resistance, and compatibility with conventional textile manufacturing. MXene coatings enable conductive fiber integration, whereas hydrogels improve interfacial contact. However, dehydration and laundering can compromise water-rich networks, necessitating encapsulation or organohydrogel and ionogel formulations. Performance assessment should consequently include repeated folding, friction, washing, and humidity cycling [39, 185]. For human–machine interfaces, reproducible gesture recognition is essential for interpreting finger bending, wrist motion, swallowing, speech, handwriting, and

rehabilitation movements. Hydrogel–MXene sensors integrated with local energy storage provide lightweight sensing platforms, although hydration-dependent drift and mechanical fatigue require appropriate calibration [135, 184].

Fig. 16 illustrates how mechanically adaptable batteries and fiber-shaped energy-storage devices can support wearable sensing and self-powered electronic interfaces. These applications require stable energy delivery, mechanical compliance, and reliable sensing under repeated deformation. Hydration, perspiration, adhesion fatigue, and environmental exposure must also be considered when assessing long-term performance [2, 3, 107, 135, 184, 185]. Bioinspired flexible batteries provide an effective strategy for accommodating complex mechanical deformation. Fig. 16(a) illustrates a scorpion-tail-inspired Zn‖MnO_2 battery comprising rigid electrochemical segments connected by flexible regions resembling intersegmental chitinous membranes. Finite element analysis in Fig. 16(b) demonstrated that bending at 15° and twisting at 10° concentrated strain within flexible connections, protecting active segments. Even under 60° bending, the architecture redistributed mechanical stress and minimized structural damage [186].

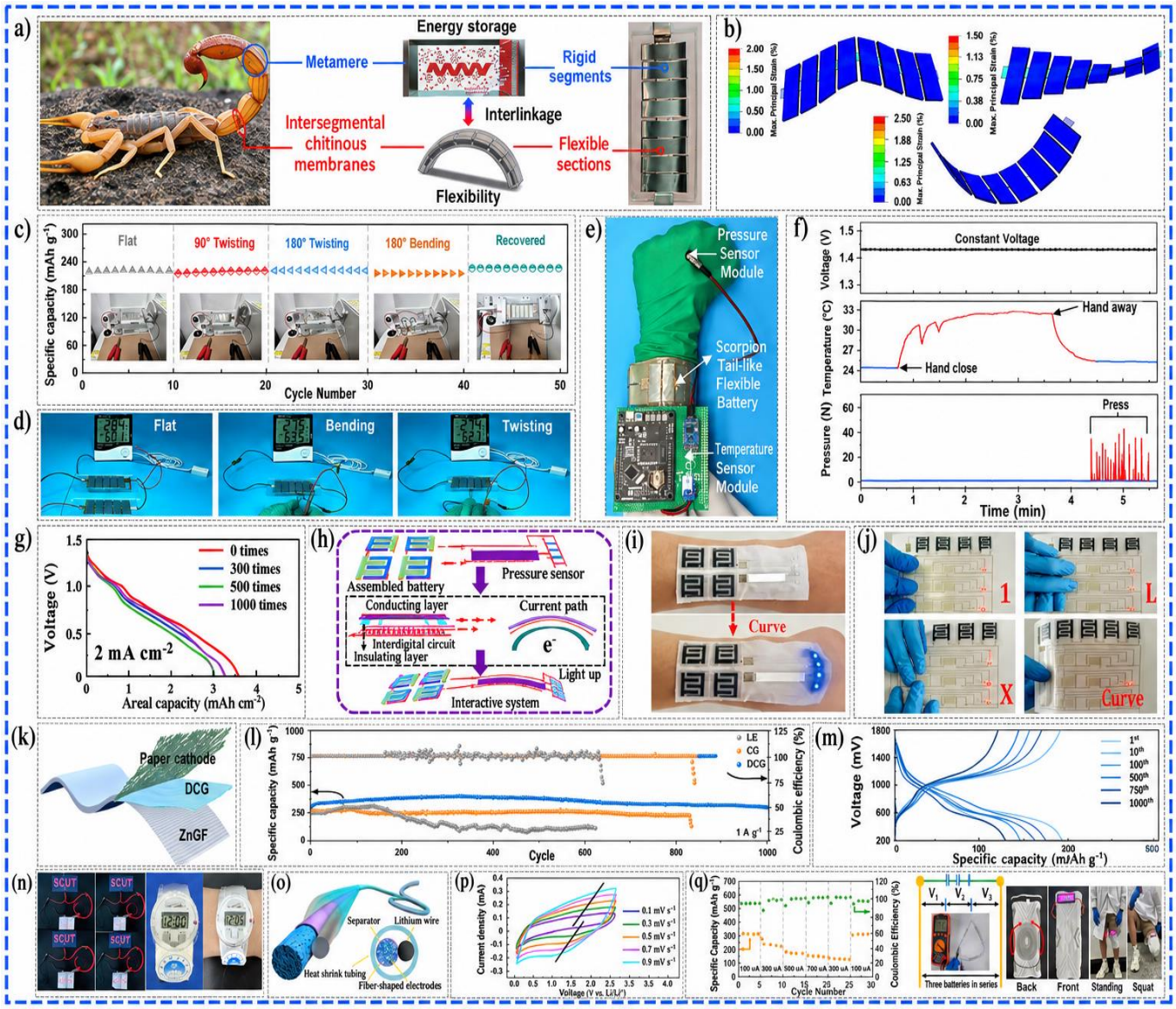


Fig. 16. Structural design, mechanical flexibility, electrochemical performance, and wearable applications of bioinspired flexible Zn-based batteries and $Ti_3C_2T_x$ CF-based fiber-shaped energy storage devices. (a) Schematic illustration of a scorpion tail-inspired flexible Zn||MnO_2 cell, featuring rigid energy storage segments resembling scorpion metameres and flexible interconnections mimicking intersegmental chitinous membranes; (b) finite element analysis of the maximum principal strain distribution under different deformation conditions; (c) cycling performance of bioinspired flexible Zn||MnO_2 cells under various mechanical deformations, with corresponding photographic insets; (d) photographs demonstrating the ability of flexible Zn||MnO_2 cells to power a thermo-hygrometer under different deformation conditions; and (e) photograph of a reinforced flexible Zn||MnO_2 cell worn on the wrist to power a multimodal integrated sensor chip for real-time monitoring, accompanied by time-dependent voltage profiles and recorded temperature and pressure signals. Reproduced from Ref. [186]. Copyright 2026, the authors, published by Springer Nature under a Creative Commons Attribution 4.0 International License. (f) Discharge profiles of a Zn-based alloy under repeated bending cycles; (g) schematic illustration of the operating principle and photograph of the integrated device; (h) electronic skin-like interactive system incorporating the integrated device; and (i) and (j) interactive integrated device combining a pressure-sensing array with a light-emitting diode (LED) array. Reproduced from Ref. [187]. Copyright 2026, the authors, published by SCIEPublish under a Creative Commons Attribution 4.0 International License. (k) Schematic illustration of a flexible Zn||V_2O_5 pouch cell incorporating a DCG electrolyte; (l) long-term cycling performance of Zn||V_2O_5 batteries using LE, CG, and DCG electrolytes within a voltage window of 0.2–1.8 V at a current density of 1 A g^{-1} and a V_2O_5 mass loading of approximately 5 mg cm^{-2}; (m) selected galvanostatic charge–discharge (GCD) profiles of the

Zn||V_2O_5 battery incorporating DCG at 1 A g^{-1}; and (n) practical demonstrations of flexible Zn||V_2O_5 pouch cells incorporating DCG electrolytes to power LEDs and a digital watch. Reproduced from Ref. [188]. Copyright 2025, the authors, published by the Royal Society of Chemistry under a Creative Commons Attribution 3.0 Unported License. (o) Schematic illustration of a fiber-shaped lithium-ion half-cell employing $Ti_3C_2T_x$ CF as the working electrode and Li metal as the counter/reference electrode; (p) cyclic voltammetry (CV) profiles recorded at scan rates of 0.1–0.9 mV s^{-1} within a potential window of 0.01–3.0 V versus Li/Li^+; and (q) rate capability at applied currents of 100, 300, 500, 700, and 900 μA, followed by recovery at 100 μA, together with Warburg factors derived from low-frequency electrochemical impedance spectroscopy (EIS) fitting before and after 1000 cycles, output voltage characteristics of series-connected fiber-shaped half-cell units demonstrating linear voltage scaling, and photographs of fiber-shaped half-cell assemblies integrated into a knee pad to power an LED strip under bending deformation. Reproduced with permission from Ref. [189]. Copyright 2026, Elsevier.

Electrochemical characterization confirmed reversible redox behavior and approximately 73% capacity retention after 100 cycles. As shown in Fig. 16(c), the device retained 98%, 96%, 88%, and 99% capacity during 90° twisting, 180° twisting, 180° bending, and subsequent recovery, respectively. Impedance and morphological analyses indicated stable interfaces following deformation. Fig. 16(d) demonstrates two interconnected batteries powering a thermo-hygrometer and digital timer. Wrist-mounted integration in Fig. 16(e) further enabled temperature and pressure monitoring at an operating voltage of approximately 1.42 V, demonstrating reliable energy delivery during movement [186].

Hydrogel-based battery integration additionally supports skin-compatible electronic interfaces. Polyacrylamide–polyvinylpyrrolidone (PAM–PVP) dual-network electrolytes provide microporous structures, adhesion, and mechanical flexibility. Fig. 16(f) presents discharge behavior under repeated bending, demonstrating the importance of electrochemical stability during deformation. Fig. 16(g) illustrates the integrated battery–sensor configuration, while Fig. 16(h) depicts its electronic skin-like application. The pressure-sensing and LED arrays shown in Fig. 16(i) and (j) provide visual feedback during finger bending through resistance-dependent brightness changes. These characteristics support interactive wearable sensing platforms [187]. High-capacity wearable zinc batteries further require rapid ionic transport and stable electrode interfaces. Fig. 16(k) illustrates a flexible Zn‖V_2O_5 pouch cell incorporating an ultrathin DCG electrolyte, zinc-textile anode, and V_2O_5 paper cathode, with excess zinc maintained below 50%. The electrolyte preserved characteristic electrochemical reactions while improving interfacial stability.

At 1 A g^{-1}, the DCG-based battery retained 85.2% capacity after more than 1000 cycles, whereas liquid-electrolyte and CG-based devices failed after 630 and 811 cycles, respectively, as demonstrated in Fig. 16(l). The galvanostatic profiles in Fig. 16(m) indicated reduced polarization and improved reaction kinetics. Consequently, the DCG electrolyte enabled a reversible specific capacity of 418.6 mAh g^{-1}, compared with 322.2 and 331.7 mAh g^{-1} for liquid and CG electrolytes, respectively. Fig. 16(n) demonstrates two series-connected flexible batteries powering 39 LEDs arranged in an “SCUT” pattern despite bending, puncturing, and shearing, without observable flames or thermal runaway. Wrist-mounted configurations additionally powered an electronic watch continuously for more than 12 h, demonstrating mechanical durability and practical energy delivery [188]. Fiber-shaped MXene electrodes provide another route toward textile-integrated energy storage. Fig. 16(o) illustrates a lithium-ion half-cell employing $Ti_3C_2T_x$-based carbon fibers (CFs). The hierarchical porous architecture supports lithium intercalation and surface-mediated charge storage. Initial irreversible lithium consumption, associated with solid-electrolyte-interphase formation and reactions involving MXene surface terminations, contributes to reduced first-cycle Coulombic efficiency. Subsequent stabilization indicates improved interfacial reversibility [189]. Cyclic voltammetry profiles in Fig. 16(p), recorded between 0.1 and 0.9 mV s^{-1}, exhibit increasing currents while maintaining similar shapes. Charge-storage kinetics can be described using $i = av^b$, where i represents peak current, v denotes scan rate, and a and b are fitting parameters. These characteristics indicate favorable electrochemical reversibility and efficient transport within the fiber architecture [189].

Fig. 16(q) presents specific capacities of 337.3, 246.0, 195.9, 167.5, and 144.3 mAh g^{-1} at 100, 300, 500, 700, and 900 μA, respectively. Capacity recovered to approximately 92.5% after returning to 100 μA, outperforming denser vacuum-dried controls. One, two, and three series-connected units produced 3.046, 6.07, and 9.10 V, respectively, with voltage superposition efficiency exceeding 99%. Knee-pad integration maintained LED illumination during bending, demonstrating wearable compatibility. Nevertheless, practical implementation requires additional evaluation of encapsulation, perspiration resistance, laundering durability, breathability, friction, and environmental stability [189]. Collectively, these developments demonstrate that reliable wearable autonomy requires coordinated optimization of electrochemical performance, structural flexibility, sensing stability, and environmental durability. Integrating these characteristics provides a foundation for multifunctional electronic skin, smart textiles, and self-powered human–machine interfaces.

Fig. 17 illustrates flexible energy-storage architectures integrated with wearable health-monitoring systems. Although these devices primarily demonstrate electrochemical and mechanical performance rather than biodegradation, they establish requirements for sustainable bioelectronics, including stable conductivity, conformability, encapsulation, and reliable operation during deformation [14, 15, 122, 137, 141, 142]. Complete-device biodegradability additionally requires assessment of residual conductors, packaging, degradation products, and electrochemical changes during dissolution [14, 15, 137, 142].

The flexible Fe_2O_3@Ti_3C_2||Ti_3C_2 asymmetric supercapacitor (FTASC), incorporating a poly(vinyl alcohol)/sulfuric acid (PVA/H_2SO_4) gel electrolyte, is illustrated in Fig. 17(a). Cyclic voltammetry (CV) and galvanostatic charge–discharge (GCD) profiles in Fig. 17(b) indicate reversible capacitive behavior. The device delivered 123 F g^{-1} at 0.5 A g^{-1}, slightly below its aqueous 1 M H_2SO_4 counterpart because gel-mediated ionic transport was more restricted. Fig. 17(c) shows

electrochemical activation during approximately 100 cycles, followed by stable performance: 94% capacitance retention and 96% Coulombic efficiency after 10,000 cycles at 2 A g^{-1}. Approximately 85% capacitance was retained after 10,000 bending cycles, with stable responses under bending, twisting, and compression [190].

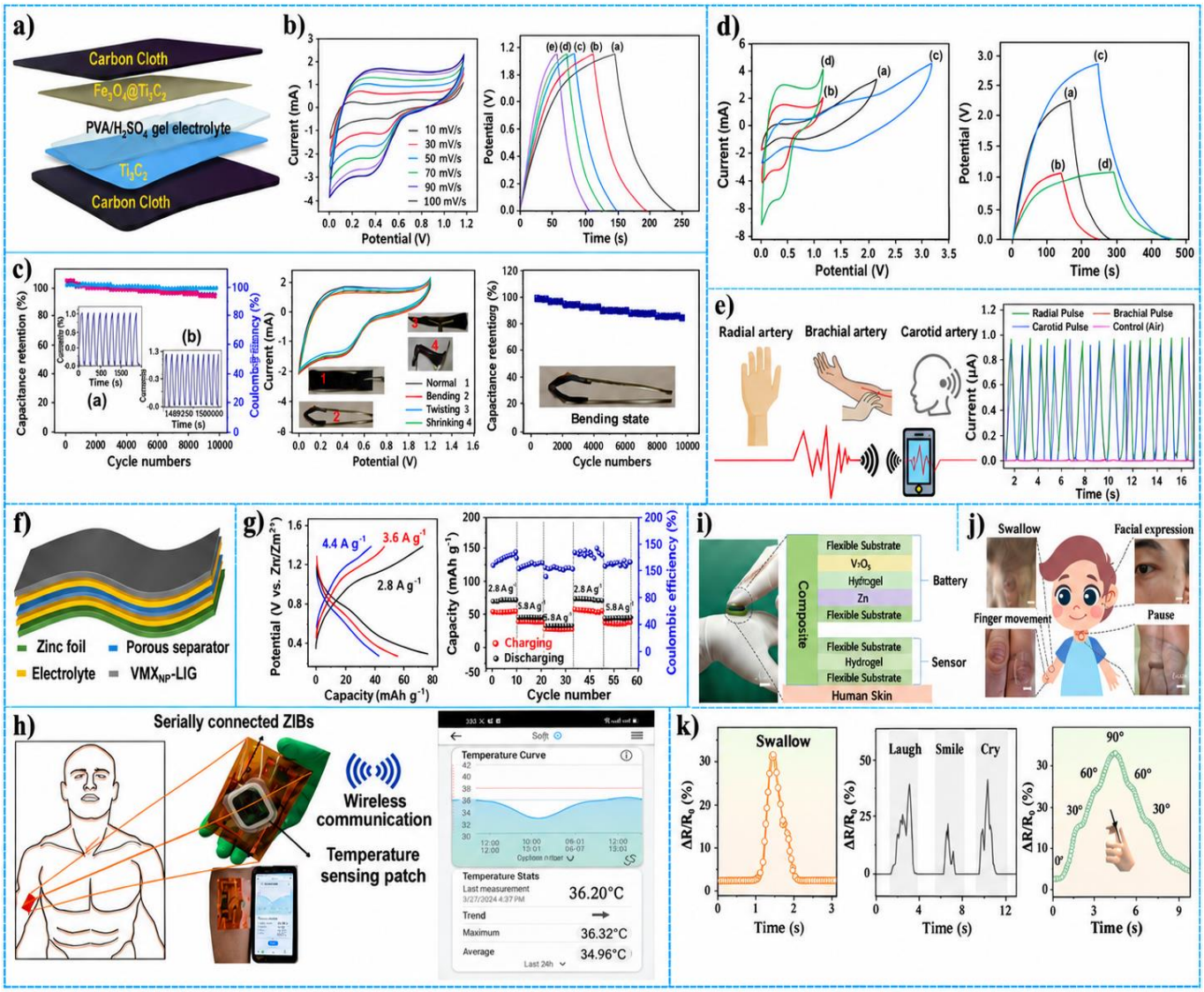


Fig. 17. Electrochemical performance, mechanical flexibility, and wearable health-monitoring applications of flexible energy storage and sensing devices. (a) Schematic illustration of the assembled flexible device; (b) cyclic voltammetry (CV) profiles recorded at various scan rates and galvanostatic charge–discharge (GCD) curves at current densities of 0.5, 1, 1.5, 2, and 2.5 A g^{-1}; (c) cycling stability and Coulombic efficiency, including the first and last 10 cycles, CV responses of the flexible Ti_3C_2-based asymmetric supercapacitor (FTASC) at different bending positions and angles, and cycling stability over 10,000 cycles under bending conditions; (d) CV and GCD characteristics of flexible interdigitated Fe_2O_3@Ti_3C_2||Ti_3C_2 FTASCs connected in series and parallel configurations; and (e) schematic illustration and real-time wireless monitoring of radial, brachial, and carotid pulse signals transmitted to a mobile device using an FTASC-powered health-monitoring system. Reproduced from Ref. [190]. Copyright 2025, the authors, published by Springer Nature under a Creative Commons Attribution 4.0 International License. (f) Schematic illustration of a flexible zinc-ion battery (ZIB) incorporating a VMXNP-LIG300 cathode and PVA/$ZnSO_4$ gel electrolyte, accompanied by representative charge–discharge profiles at different current densities; (g) rate performance of the VMXNP-LIG300 cathode at varying current densities; and (h) schematic illustration of a wearable temperature-sensing patch powered by three series-connected ZIBs, enabling wireless transmission of real-time body temperature data to a smartphone, together with 24-h temperature profiles recorded from two healthy male participants. Reproduced with permission from Ref. [191]. Copyright 2025, Wiley-VCH. (i) Schematic illustration of a fully integrated wearable sensing and energy storage device conformally attached to the skin; (j) schematic representations and photographs of wearable sensors attached to different body regions for physiological and motion monitoring; and (k) real-time sensing responses associated with swallowing, facial expressions, and finger movements. Reproduced with permission from Ref. [192]. Copyright 2025, Elsevier.

Because the FTASC relies on a hydrated electrolyte, its measured cycling durability should be distinguished from environmental stability during prolonged wear. Changes in gel hydration or electrode contact may alter resistance even if the active material remains electrochemically reversible. Therefore, bending retention, charge–discharge efficiency, and operational duration provide complementary rather than interchangeable measures of device reliability. Fig. 17(d) demonstrates scalable FTASC interconnection: one, two, and three series-connected units provided approximately 1.1, 2.2, and 3.3 V, respectively, alongside parallel configurations. In Fig. 17(e), the integrated platform monitored radial, brachial, and carotid pulses, recording approximately 72 beats min^{-1}. Two units charged to 2.2 V powered a 13 μW sensor for

approximately 500 s and supported wireless smartphone transmission [190].

The flexible zinc-ion battery shown in Fig. 17(f) comprised a VMXNP-LIG300 cathode, zinc anode, and PVA/$ZnSO_4$ gel electrolyte. GCD plateaus near 1.15 and 0.7 V reflected reversible Zn^{2+} insertion/extraction within the V_2O_5/V_2C cathode. Operating at 0.3–1.6 V, the battery delivered approximately 77 mAh g^{-1} at 2.8 A g^{-1}. Rate measurements in Fig. 17(g) showed capacity recovery after current-density changes, indicating favorable reversibility [191]. Fig. 17(h) depicts a temperature-monitoring patch powered by three series-connected batteries, transmitting 24 h measurements from two healthy male participants aged 36 and 30 years to smartphones. The battery retained approximately 80% capacity under severe bending; its V_2C/V_2O_5 heterostructure and laser-assisted fabrication supported stability and conformal integration [191]. Fig. 17(i) illustrates a skin-conformal Zn–PAA–C hydrogel strain sensor integrated with a flexible hydrogel battery, while Fig. 17(j) shows placement on different body regions. Fig. 17(k) presents distinct responses to swallowing, facial expressions, and finger movements. Swallowing induced 25%–45% skin strain and approximately 12%–30% relative resistance variation; finger and wrist bending produced angle-dependent responses up to 90°. Repeated movement demonstrated reproducible monitoring without external power [192].

7. Biomedical and Biodegradable Energy Storage

Biomedical energy-storage systems require stringent material safety, mechanical compatibility, and operational reliability because they may directly contact skin, mucosal surfaces, wounds, or internal tissues. Hydrogels are particularly attractive because their hydrated polymer networks resemble biological tissues and minimize mechanical mismatch. However, comprehensive biocompatibility evaluation must consider all device constituents, including residual monomers, cross-linkers, initiators, electrolytic salts, solvents, MXene nanosheets, oxidation products, encapsulation materials, adhesives, and degradation byproducts [2, 13, 111]. Unlike conventional wearable electronics, biomedical devices prioritize biological safety, predictable operational lifetime, sterilization compatibility, and electrochemical stability in physiological environments. Transient devices may undergo controlled degradation following therapeutic intervention or wound monitoring, whereas implantable systems may require wireless charging or energy harvesting to minimize surgical replacement. Consequently, material selection must balance electrochemical performance with biological compatibility and programmed device degradation [14, 15, 137].

7.1 Biocompatible Hydrogels for Bioelectronics

Biocompatibility assessment requires evaluation beyond conventional in vitro cell-viability measurements. Although cellular assays provide preliminary information, they cannot adequately characterize immune responses, protein adsorption, chronic inflammation, or mechanical irritation. Epidermal devices require testing for skin irritation, perspiration exposure, adhesive removal, and microbial contamination, whereas mucosal and implantable interfaces demand additional evaluation of saline swelling, enzymatic degradation, sterilization resistance, and prolonged material leaching [13, 193, 194]. Biological compatibility also depends on exposure duration and physiological conditions. Materials suitable for temporary skin contact may exhibit adverse responses during prolonged implantation because hydration, pH, ionic strength, enzymatic activity, and mechanical motion influence chemical stability and constituent release. For MXene-containing systems, nanosheet release, oxidation products, residual etchants, and photothermal heating require careful evaluation alongside polymer degradation. These considerations distinguish short-term wearable comfort from long-term biomedical safety [13, 193, 194].

MXene concentration, flake dimensions, oxidation state, surface terminations, and exposure duration further influence biological interactions. Polymer functionalization and encapsulation can restrict direct tissue exposure while retaining conductivity, although excessive encapsulation may compromise ionic transport or sensing sensitivity. Consequently, biomedical devices should identify whether MXene nanosheets remain exposed, partially embedded, or completely isolated within the polymer network [111, 195]. For electrophysiological applications, hydrated interfaces reduce contact impedance and movement-related artifacts. Adhesive hydrogels preserve conformal contact during physiological motion, while MXene conductive networks support electrical signal acquisition. When energy storage and sensing are integrated within the same biomedical platform, leakage currents, localized heating, and unintended electrochemical reactions must be controlled through appropriate insulation, current limitation, and device-level isolation [2, 193].

7.2 Biodegradable Device Components and Encapsulation

Biodegradable energy-storage systems require coordinated control of operational stability, environmental protection, and programmed degradation. During service, electrodes, electrolytes, and encapsulation layers must maintain electrical continuity, ionic transport, and mechanical integrity. Following the intended operating period, degradation may be initiated through water penetration, hydrolysis, oxidation, enzymatic activity, or pH changes. Encapsulation therefore regulates the transition from stable operation to controlled material disintegration, preventing premature failure while enabling the intended transient functionality [14, 137, 142]. Hydrogel electrolytes offer considerable potential for biodegradable devices because natural and hydrolyzable synthetic polymers can undergo controlled degradation. However, electrode materials and current collectors introduce additional challenges. Carbon-based conductors may remain environmentally persistent, transient metals release ionic species during dissolution, and MXene nanosheets may undergo oxidation or fragmentation, with their long-term biological and environmental fate remaining insufficiently established [14, 15].

Sustainable fabrication also requires minimizing hazardous processing chemicals, energy consumption, and material

waste. Water-based inks, alternative etching approaches, low-temperature processing, recyclable substrates, and life-cycle assessment are therefore important considerations for environmentally responsible hydrogel–MXene devices [58, 137, 196, 197]. End-of-life requirements depend on application. Disposable wearable systems prioritize environmental compatibility and safe disposal, implantable devices require controlled degradation and biological clearance, and soft robotic systems may benefit more from repairability and modular replacement. Accordingly, sustainability assessment should consider fabrication hazards, operational lifetime, reusability, degradation kinetics, and residual material composition [14, 15, 137].

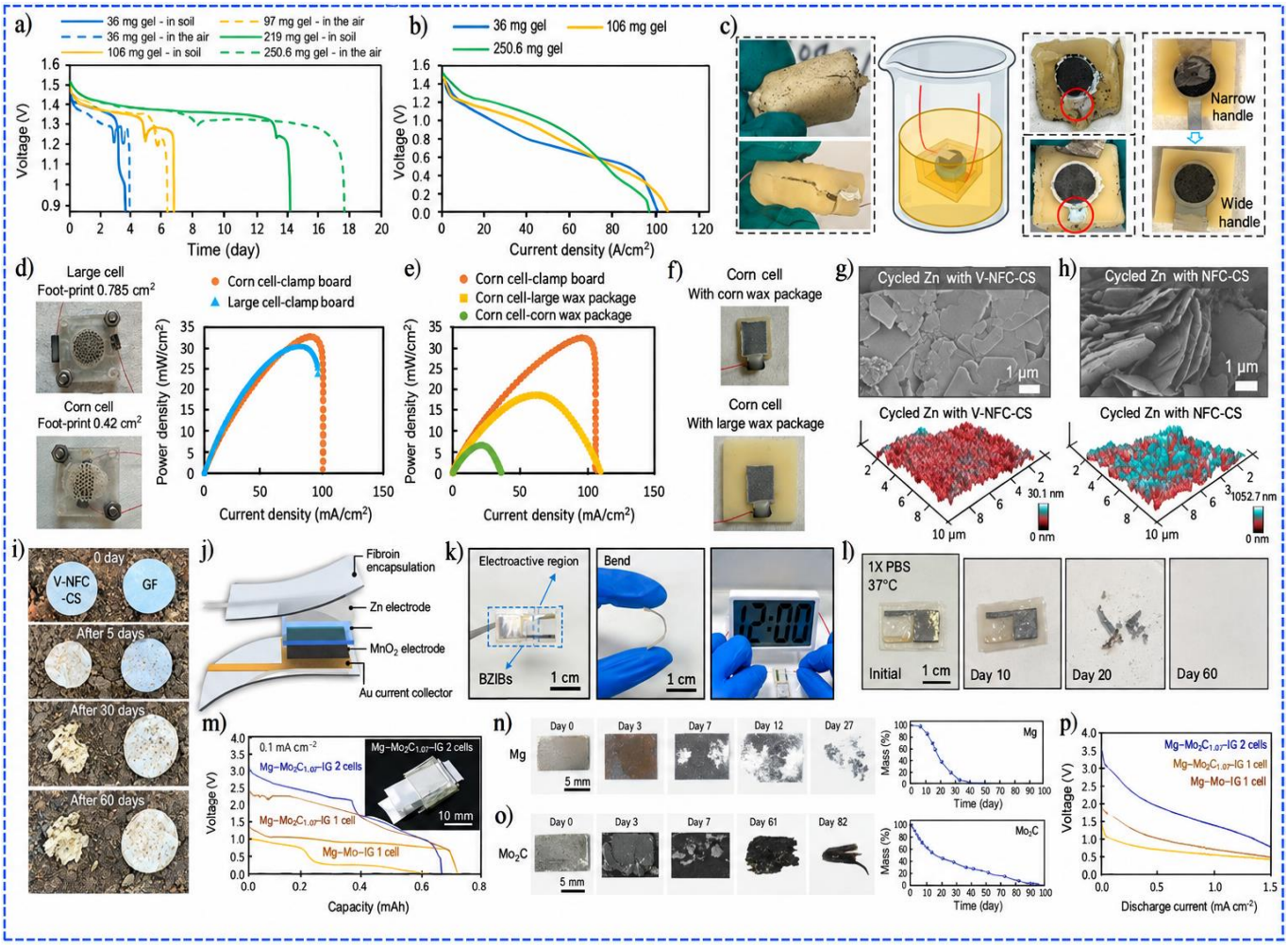


Fig. 18. Electrochemical performance, structural configurations, packaging strategies, degradation behavior, and biodegradability of flexible and biodegradable energy storage devices. (a) Discharge profiles of wax-encapsulated batteries containing different gel masses at a constant current of 30 μA under air and soil conditions; (b) current–voltage (I–V) characteristics of the corresponding batteries; (c) evaluation of wax-encapsulation integrity after two weeks of discharge, illustrating package cracking, additional dip-coating for enhanced protection, Zn handle consumption, narrow Zn handles and wide handles wrapped with Cu wire, and the application of silver paste and an additional wax coating; (d) top-view photographs of a battery with a footprint of 0.785 cm² and a corn cell with a footprint of 0.42 cm², both assembled using clamp boards; (e) power curves of batteries and corn cells with different packaging configurations; and (f) active-component layers of corn cells assembled on corn-sized and larger wax pads. Reproduced from Ref. [198]. Copyright 2025, the authors, published by the Royal Society of Chemistry under a Creative Commons Attribution 3.0 Unported License. (g) Scanning electron microscopy (SEM) images of Zn electrodes after galvanostatic charge–discharge (GCD) measurements using V-NFC-CS and NFC-CS separators; (h) atomic force microscopy (AFM) images of the corresponding Zn electrodes and photographs of V-NFC-CS and glass fiber (GF) separators after different soil-burial periods; and (i) distribution of relaxation times (DRT) profiles of Zn||Zn cells during the fourth GCD cycle. Reproduced from Ref. [199]. Copyright 2025, the authors, published by Springer Nature. (j) Exploded schematic illustrating the components of a biodegradable zinc-ion battery (BZIB), highlighting its biodegradability and potential application in promoting neuronal differentiation; (k) photographs of the BZIB in flat and bent configurations and during operation of an electronic watch; and (l) progressive dissolution of the BZIB following immersion in 1 × phosphate-buffered saline (PBS, pH 7.4) at 37 °C. Reproduced with permission from Ref. [200]. Copyright 2025, Elsevier. (m) Discharge profiles of solid-state Mg–Mo_2C1.0g–IG 2, Mg–Mo_2C1.0g–IG 1, and Mg–Mo 1 cells at a current density of 0.1 mA cm^{-2}, with an inset photograph of the Mg–Mo_2C1.0g–IG 2 cells; (n) time-lapse images illustrating degradation of the Mg component; (o) time-lapse images showing degradation of the Mo_2C component; and (p) corresponding mass changes of Mg and Mo_2C during degradation, together with voltage–discharge current characteristics of Mg–Mo_2C1.0g–IG 2, Mg–Mo_2C1.0g–IG 1, and Mg–Mo 1 cells measured at a scanning rate of 10 μA cm^{-2} s^{-1}. Reproduced from Ref. [201]. Copyright 2024, the author, published by the American Chemical Society under a Creative Commons Attribution 4.0 International License.

Fig. 18 highlights the relationship between electrochemical performance, protective packaging, interfacial stability, and controlled degradation in transient energy-storage systems. Biomedical compatibility depends on the complete device architecture rather than individual material properties, because encapsulation, chemical residues, mechanical wear, and degradation products collectively influence operational safety [2, 13, 111, 193-195]. Consequently, device-level evaluation

should integrate electrochemical reliability with biological exposure and environmental degradation assessments [2, 13, 111, 193, 194]. The influence of electrolyte loading and encapsulation on alkaline battery lifetime was investigated using wax-protected cells incorporating platinum-loaded carbon cathodes. Fig. 18(a) presents discharge profiles at 30 μA under air and soil conditions. Increasing gel mass from 36 to approximately 250 mg extended operational lifetime by up to 3.6 times, whereas additional electrolyte beyond 300 mg provided negligible improvement. The corresponding current–voltage characteristics in Fig. 18(b) indicate that oxygen availability within the encapsulated structure strongly influences maximum current output, although electrolyte thickness and internal impedance may also contribute. These results emphasize the need to balance electrolyte quantity, oxygen accessibility, and packaging geometry [198].

Encapsulation durability was further investigated following prolonged discharge. Fig. 18(c) illustrates cracking along wax-package edges after approximately two weeks, primarily associated with zinc-anode expansion. Such defects expose internal components and weaken electrode–electrolyte contact. Additional molten-wax dip-coating improved package integrity. Furthermore, premature oxidation of zinc connection tabs caused electrical failure before complete consumption of the active anode. Increasing tab width approximately threefold, wrapping the connection with copper wire, applying silver paste, and adding wax protection improved electrical continuity and operational reliability [198]. Device miniaturization introduced additional transport limitations. Fig. 18(d) compares batteries with footprints of 0.785 cm^2 and 0.42 cm^2. Their power-density characteristics, presented in Fig. 18(e), were comparable when oxygen transport remained sufficient. However, encapsulating smaller cells reduced peak power density by approximately 50%, while corn-sized packaging further decreased it to approximately 7 mW cm^{-2}, as illustrated in Fig. 18(f). Reduced enclosed oxygen and slower replenishment limited cathodic reactions, demonstrating that packaging dimensions substantially influence miniature battery performance [198]. Separator engineering additionally governs zinc-deposition stability. Scanning electron microscopy images in Fig. 18(g) reveal comparatively uniform zinc deposition with V-NFC-CS separators, whereas NFC-CS produced pronounced surface protrusions and increased dendrite-related short-circuit risks. Atomic force microscopy in Fig. 18(h) further indicates lower zinc-surface roughness with V-NFC-CS. The improved morphology was attributed to separator mechanical reinforcement, which restricts vertical zinc growth and potentially suppresses zinc sulfate hydroxide formation. The same panel also documents separator appearance after soil burial, while Fig. 18(i) presents distribution-of-relaxation-times profiles for Zn||Zn cells during electrochemical cycling, providing complementary characterization of interfacial processes [199].

Biodegradable zinc-ion batteries provide another approach to temporary biomedical power delivery. Fig. 18(j) illustrates a biodegradable zinc-ion battery (BZIB) incorporating a GG–Zn biomacromolecular hydrogel electrolyte, zinc and MnO_2 electrodes, an ultrathin gold current collector, and silk-fibroin substrate and encapsulation layers. The device was investigated for biodegradable energy storage and potential support of neuronal differentiation. Its approximately 850 μm thickness provided flexibility, as shown in Fig. 18(k), while an open-circuit voltage of approximately 1.59 V enabled electronic-watch operation. Degradation experiments in phosphate-buffered saline (PBS, pH 7.4) at 37 °C demonstrated initial silk-fibroin swelling after 10 days, pronounced structural disintegration by day 20, and complete degradation within 60 days, as illustrated in Fig. 18(l). These findings support the feasibility of transient hydrogel-based biomedical batteries [200]. Transient magnesium-based batteries were also investigated to evaluate electrochemical performance and controlled material dissolution. Fig. 18(m) compares Mg–$Mo_2C_{1.0}$g–IG and Mg–Mo configurations at 0.1 mA cm^{-2} . The Mg–$Mo_2C_{1.0}$g–IG 1 cell initially delivered approximately 1.8 V, maintaining a discharge plateau between 1.0 and 1.5 V, approximately 50% higher than the corresponding Mg–Mo–IG 1 configuration. Unlike the Mg–$Mo_2C_{1.0}$g–IL cell, whose voltage decreased to 1.0 V at 0.05 mAh, the ionogel-based cell maintained above 1.0 V until 0.5 mAh.

This improvement was attributed to the EPPOMaC spacer, which facilitated oxygen transport, and the ionogel architecture, which maintained selective contact with the Mo_2C layer. Connecting two cells in series produced an initial voltage of approximately 3.3 V, remaining at 2.8 V after 0.1 mAh discharge. At higher current densities, competing hydrogen evolution increasingly influenced cell polarization. Maximum power densities reached 0.92 and 1.7 mW cm^{-2} for single- and two-cell configurations, respectively. Nevertheless, ionogel dehydration restricted prolonged discharge by reducing electrode contact, highlighting the importance of improved moisture retention and encapsulation [201]. The degradation characteristics of the constituent materials were evaluated through immersion in 100 mL of 10 mM PBS at 37 °C. Fig. 18(n) and (o) illustrate the time-dependent degradation of magnesium and Mo_2C components, respectively, while Fig. 18(p) presents corresponding mass changes and voltage–current characteristics. Degradation was quantified using the normalized remaining-mass ratio:

$$M_c = \frac{M}{M_0} \tag{19}$$

where (M) represents the remaining mass after immersion and M_0 denotes the initial mass. This measurement enables quantitative comparison of dissolution behavior and supports assessment of component suitability for transient biomedical electronics [201]. Collectively, these studies demonstrate that biomedical energy-storage performance depends on the coordinated design of hydrated interfaces, protective encapsulation, electrochemical stability, and predictable degradation. Integrating these requirements is essential for developing safe, conformable, and temporally controlled hydrogel-based power sources for next-generation biomedical and bioelectronic systems.

The degradation fraction can be quantified as:

$$D_{\mathrm{deg}} = \frac{m_0 - m_t}{m_0} \times 100\% \tag{20}$$

where m_o represents the initial dry or wet mass under specified measurement conditions, and m_t denotes the remaining mass after degradation time (t).

These wearable demonstrations establish functional requirements but do not establish complete biodegradability. Implantable systems encounter saline, proteins, enzymes, cells, and motion, which can alter hydrogel swelling, ionic exchange, MXene oxidation, protein adsorption, and encapsulation integrity. Accordingly, testing should include simulated body fluids, protein-rich media, and mechanical cycling [13, 14]. Transient biomedical power sources must maintain predictable output over hours, days, or weeks before controlled degradation. Potential applications include wound sensors, temporary electrophysiology patches, drug-delivery controllers, and postoperative monitors. Wireless charging and energy harvesting can reduce stored-energy requirements, while microsupercapacitors buffer intermittent input. Degradation must avoid sharp residues, toxic products, and uncontrolled electrical discharge [15, 137]. A programmed lifetime also requires coordination between packaging permeability and the degradation of electrodes, current collectors, and electrolyte components. Early encapsulation failure could interrupt therapy, whereas persistent conductive remnants would undermine claims of complete resorption. Monitoring residual mass and identifying dissolved or fragmented products are therefore necessary to distinguish polymer dissolution from safe device-level biodegradation. These assessments should follow the intended exposure environment rather than relying solely on visually observed disappearance.

Matching material disappearance to therapeutic progress remains an important objective. Wound devices may operate until tissue conditions stabilize, whereas neural interfaces may require a defined regeneration period. Stimuli-responsive hydrogels sensitive to pH, enzymes, or ionic composition offer possible control over degradation timing, although integration with MXene storage remains preliminary [13, 46, 142]. Thus, Fig. 17 should be interpreted as evidence of flexible, self-powered monitoring rather than an illustration of implantable degradation pathways. Biomedical translation additionally requires coordinated evaluation of sterilization, swelling, ion exchange, heating, leakage, packaging fatigue, and degradation products alongside electrochemical performance and service-life predictability [13-15, 137].

7.3 Biocompatibility and Degradation Considerations for 2D Fillers

Polymer dissolution, filler transformation, and biological clearance are different endpoints. MXene/PVA/bacterial-cellulose degradation [138], functionalized MoS_2 wound-repair studies [202], and multisensing MoS_2 hydrogels [77] provide formulation-specific evidence. Mg–Mo_2C transient batteries [201], bioresorbable zinc-ion cells [200], and biodegradable separators [199] address different device components. Assessment should identify released species, particulate residue, exposure dose, duration, and tissue response after operation and sterilization.

8. Comparative Analysis and Future Perspectives

Comparative assessment of flexible energy-storage systems remains challenging because reported electrochemical performance depends on measurement configuration, normalization methods, and testing conditions. Three-electrode measurements frequently differ from full-device results, while normalization to active materials rather than complete device mass may overestimate practical performance. Inconsistent mechanical testing further complicates comparison [5, 16, 143]. Gou et al. [203] highlighted the role of high-concentration gel electrolytes and polymer-network engineering in enhancing ionic transport, interfacial stability, and electrochemical performance. Their findings provide useful design strategies for developing efficient and stable hydrogel-based energy-storage systems [203]. Standardized reporting should specify electrode configuration, gravimetric, areal, or volumetric normalization, mechanical testing conditions, electrolyte composition, hydrogel water content, MXene loading, flake size, encapsulation, and storage history [5, 16, 115]. Application-specific evaluation must integrate electrochemical, mechanical, environmental, and safety requirements, as summarized in Tables 4 and 5.

Table 4. *Comparative overview of representative flexible supercapacitor materials, highlighting their energy and power densities, mechanical durability under repeated bending cycles, and electrochemical performance for flexible and wearable energy-storage applications, summarized from Refs. [22, 78, 123, 204-208], and [115].*

Flexible material type	Energy density	Power density	Mechanical test/cycles	Ref.
Carbon-based networks	11.3 Wh kg^{-1}	184 W kg^{-1}	4000 bending cycles	[22]
Conductive polymer composite hydrogels	12.8 Wh kg^{-1}	125 W kg^{-1}	200 bending cycles	[204]
3D microporous graphene foam	37.4 Wh kg^{-1}	2248 W kg^{-1}	1000 bending cycles	[205]
GO/MnO2/CoNi-LDH composite	82.12 Wh kg^{-1}	949.97 W kg^{-1}	1000 bending cycles	[78]
MOF@MXene structures	51 Wh kg^{-1}	1590 W kg^{-1}	1500 bending cycles	[123]
GO/CNT/PSE composite	30 Wh kg^{-1}	430 W kg^{-1}	10000 bending cycles	[206]
NiO@nanofibrillated cellulose	22 Wh kg^{-1}	491.9 W kg^{-1}	10000 bending cycles	[207]
Hexagonal $CoGa_2O_4$ nanoplates	84 Wh kg^{-1}	1200 W kg^{-1}	100 bending cycles	[208]
Ionic conductive hydrogel materials	164.13 Wh kg^{-1}	1283.44 W kg^{-1}	100 cycles	[115]

8.1 Comparative Energy-Storage Performance Across Material Families

Hydrogel-only systems provide mechanical softness and ionic transport but generally require electroactive components. MXene films offer high conductivity and capacitance but suffer restacking, oxidation, brittleness, and limited stretchability. Hydrogel–MXene composites integrate complementary properties when conductive nanosheets remain dispersed and electrochemically accessible [4, 5, 30]. Meaningful benchmarking requires both component-level and full-device measurements, including encapsulation and electrical interconnections [16, 20, 39]. Compared with carbon aerogels, conductive polymers, metal oxides, and layered double hydroxides, hydrogel–MXene composites offer coupled ionic–electronic transport but remain vulnerable to dehydration and oxidation [31, 41, 115].

Application-dependent performance may be expressed conceptually as:

$$S = w_1\eta_{\mathrm{ret}} + w_2E_A + w_3P_A + w_4\mathrm{Bio} \qquad (21)$$

where w_i are explicitly chosen application-specific weights and ηret, E_a, P_a, and Bio must first be normalized to dimensionless scales using declared reference values. Equation (21) is a proposed conceptual score, not a validated universal measure of safety or device merit.

8.2 Materials, Safety, Sustainability, Scalability, and Manufacturing Challenges

MXene oxidation reduces conductivity, electrochemical accessibility, and reproducibility. Polymer wrapping, antioxidants, and oxygen-barrier encapsulation may improve stability, provided ionic transport remains sufficient [30, 41, 45]. Hydrogel dehydration and freezing also impair mechanical compliance and ionic conductivity. Organohydrogels and ionogels offer alternatives, although solvent safety and cost require evaluation [31, 48, 115]. Manufacturing challenges include hazardous fluoride-based etching, inconsistent nanosheet quality, and unstable printable formulations. Alternative synthesis routes and standardized control of ink viscosity, oxidation, adhesion, and flake size are therefore essential [20, 39, 58, 196, 197, 209]. Sustainability additionally requires evaluating processing impacts, repairability, operational lifetime, and complete-device biodegradation rather than polymer degradation alone [14, 137, 142].

Table 5. *Summary of major challenges in nanomaterial–hydrogel-based flexible energy-storage systems, including material oxidation, drying and freezing, nanosheet restacking, mechanical fatigue, biocompatibility, and manufacturing scalability, together with potential design strategies and supporting literature., summarized from Refs.* [14, 15, 20, 25, 30, 31, 39-42, 44, 45, 108, 111, 134], *and* [115].

Challenge	Why it matters	Design strategy	Relevant refs.
MXene oxidation and termination instability	Loss of conductivity, capacitance, and reproducibility	Antioxidant design, polymer wrapping, controlled storage, oxygen-barrier encapsulation	[30, 41, 45]
Hydrogel drying/freezing	Reduced ionic conductivity and mechanical compliance	Organohydrogels, ionic liquids, hygroscopic additives, encapsulation	[31, 115]
Restacking of MXene sheets	Restricted ion transport and lower accessible surface area	3D porous scaffolds, GO/rGO support, spacers, freeze-casting	[42-44]
Mechanical fatigue and delamination	Failure during bending, twisting, and stretching	Dynamic crosslinks, adhesive interfaces, gradient structures, soft interconnects	[25, 108, 134]
Biocompatibility and degradation uncertainty	Limits biomedical translation	Cytocompatible polymers, controlled MXene loading, degradation-product studies	[14, 15, 111]
Scalable manufacturing	Needed for large-area wearable and textile devices	Water-based inks, printing, laser patterning, roll-to-roll compatible processing	[20, 39, 40]

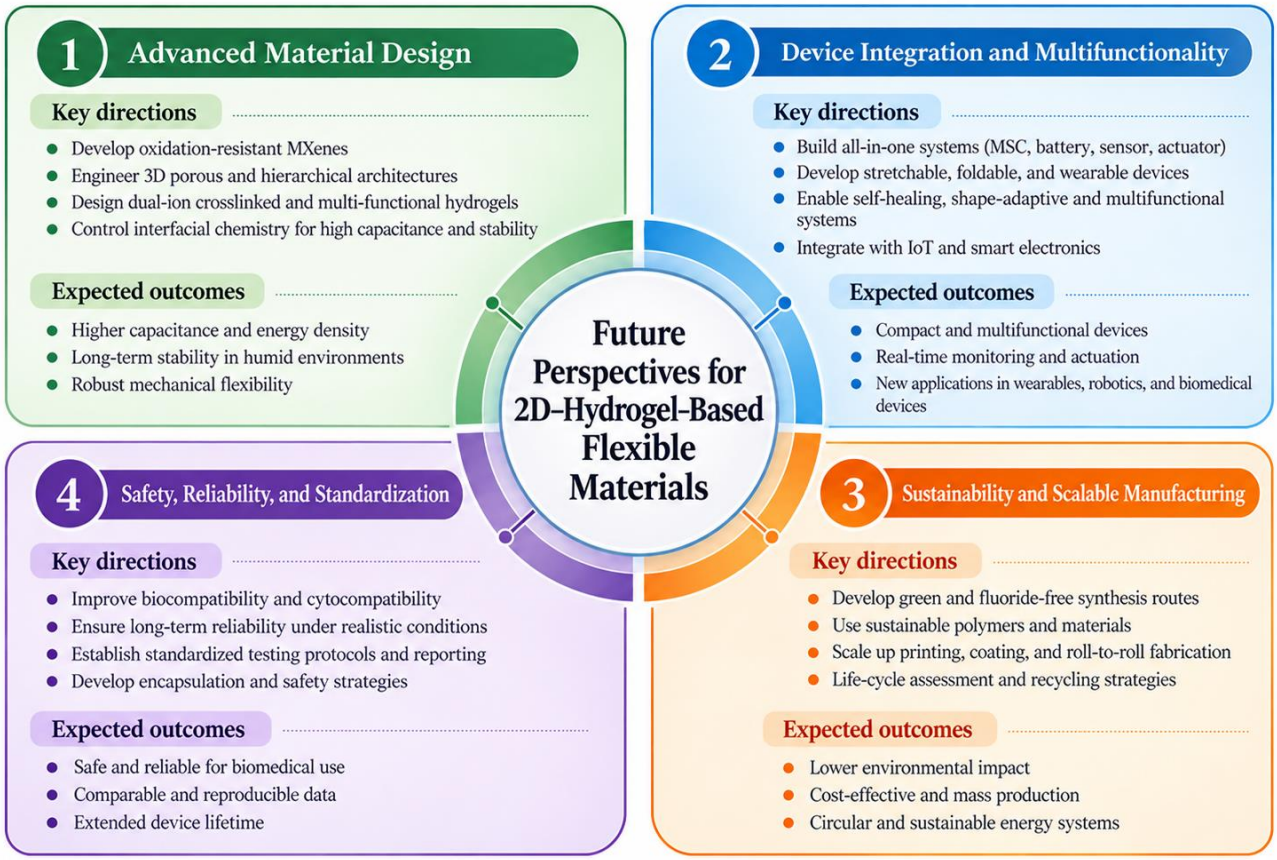


Fig. 19. Future-challenge and opportunity roadmap for hydrogel-MXene flexible energy-storage systems in soft mechatronics. The roadmap summarizes deformation tolerance, long-term durability, energy efficiency, miniaturization, user-specific customization, artificial-intelligence-assisted material/device design, safer MXene synthesis, scalable printing, sustainable processing, and wearable solid-state batteries. Reproduced with permission from Ref. [12]. Copyright 2024, Royal Society of Chemistry. Reproduced with permission from Ref. [58]. Copyright 2023, Elsevier. Reproduced with permission from Ref. [196]. Copyright 2023, Elsevier. Reproduced with permission from Ref. [197]. Copyright 2023, AAAS. Reproduced with permission from Ref. [209]. Copyright 2023, AAAS. Reproduced with permission from Ref. [137]. Copyright 2016, Royal Society of Chemistry. Reproduced with permission from Ref. [14]. Copyright 2021, Wiley-VCH. Reproduced with permission from Ref. [141]. Copyright 2022, Cell Press. Reproduced with permission from Ref. [142]. Copyright 2023, Elsevier. Reproduced with permission from Ref. [39]. Copyright 2020, Wiley-VCH. Reproduced with permission from Ref. [20]. Copyright 2022, Springer Nature. Reproduced with permission from Ref. [40]. Copyright 2023, Wiley-VCH. Reproduced with permission from Ref. [115]. Copyright 2021, Wiley-VCH.

Fig. 19 summarizes interconnected challenges involving MXene oxidation, hydrogel stability, encapsulation, scalable

processing, and sustainable manufacturing. Progress requires coordinated optimization rather than isolated improvements [12, 14, 20, 39, 58, 115, 137, 141, 142, 196, 197, 209].

8.3 Research Gaps and Opportunities for MBenes and Other Emerging 2D Materials

Emerging-material assessment should connect verified nanosheet identity to gel integration and packaged operation. MoO_x/MoB anodes [37], MBene colloidal electrolytes [38], $MoAl_{1-x}B$ electrodes [105], and boride/gel microsupercapacitors [64] establish different stages. MoS_2 network remodeling [66] and carbon-nitride/MXene cycling [67] motivate operando interface studies. Matched polymer, electrolyte, geometry, aging, and deformation controls are proposed benchmarks.

8.4 Future Directions and Conclusions

The evidence examined throughout this review suggests that the principal research priority is to shift optimization from individual electrodes and hydrogel formulations toward complete, mechanically compliant power systems. High MXene conductivity and pseudocapacitive activity are beneficial only when ionic pathways, electrode–electrolyte contact, current collection, and encapsulation remain functional during use. Equally, hydrogel softness, adhesion, and self-healing do not ensure device reliability if dehydration or interfacial resistance limits usable energy. Important trade-offs remain: highly porous networks improve ion accessibility but may reduce volumetric packing density, while protective barriers can improve moisture retention at the expense of compliance or controlled degradation. Future materials selection should therefore be guided by the intended duty cycle and coordinated electrochemical, mechanical, and environmental criteria rather than a single peak performance value [4, 5, 16, 31].

A first requirement is more reproducible and application-relevant benchmarking. Studies should clearly distinguish three-electrode material measurements from two-electrode and fully packaged device performance and report the normalization basis for capacitance, capacity, energy, and power, including active-material mass, device area, and device volume as applicable. MXene loading, flake dimensions, electrolyte composition, hydrogel water content, electrode thickness, encapsulation, and storage history should accompany these metrics. Mechanical protocols should specify bending radius or applied strain, loading mode and frequency, cycle number, recovery interval, and test environment; capacitance or capacity retention, impedance, leakage, and self-discharge should be measured before, during, and after deformation when feasible. Such reporting would distinguish apparent electrochemical durability from actual operational stability and make comparisons among hydrogel-only, MXene-rich, and hybrid devices more defensible [5, 16, 17, 143].

Long-term environmental stability requires particular attention because hydration and chemical degradation are interdependent. Water facilitates ion transport and conformal interfaces but also contributes to evaporation-related failure and can influence MXene oxidation. Oxidation, in turn, alters conductive pathways and active surfaces, while polymer aging or interfacial separation can produce similar electrochemical losses. Polymer wrapping, antioxidant additives, surface-chemistry control, and oxygen-barrier packaging are promising approaches, but excessive isolation of MXene sheets may diminish ion accessibility. Organohydrogels, ionogels, and hygroscopic formulations can improve resistance to drying or freezing, although solvent selection must be assessed against electrolyte conductivity, operating voltage, cost, and biological compatibility. Comparative aging studies should track water content, structural and chemical changes, impedance, and performance under defined temperature, humidity, and oxygen exposure rather than infer stability from cycling retention alone [30, 31, 41, 45, 115].

Interfacial and hierarchical architecture design offers a complementary route to resolving the conductivity–compliance trade-off. MXene/graphene oxide or reduced-graphene-oxide frameworks, porous three-dimensional scaffolds, and controlled nanosheet spacing can mitigate restacking and preserve ion-accessible surfaces, but their effects on packing density and mechanical integrity require concurrent evaluation. Within polymer-rich composites, filler concentration, orientation, surface terminations, crosslink density, and reversible hydrogen-bonding or coordination interactions jointly determine electrical percolation, swelling, adhesion, and fatigue resistance. Gradient structures, mechanically compliant interconnects, and electrode placement near the neutral bending plane may reduce strain localization at assembled interfaces. Claims of self-healing should be supported by recovery of electrochemical function and contact resistance after repeated damage and repair, not solely by restored macroscopic shape or tensile properties [25, 42-44, 108, 134].

Mechanistic studies will also be needed to identify the rate-controlling processes responsible for failure. Coupling electrochemical impedance spectroscopy and charge–discharge analysis with microscopy, structural characterization, rheology, and measurements under imposed deformation could help separate resistance increases arising from nanosheet oxidation, water redistribution, pore collapse, interfacial delamination, or fractured conductors. Where feasible, in situ or operando measurements should connect these evolving structural features to electrochemical output under realistic cycling and environmental conditions. Such datasets would provide a sounder basis for predictive modeling and artificial-intelligence-assisted optimization of polymer chemistry, MXene surface characteristics, filler loading, and processing. The value of data-driven approaches will depend on consistent metadata, comparable test protocols, representation of failed as well as successful formulations, and validation across independently prepared batches rather than predictions based only on small, heterogeneous datasets [12, 25, 32, 45].

Manufacturing research must address the reproducibility and environmental burden of the complete fabrication route. Hazardous fluoride-containing etching chemistries, variations in nanosheet dimensions and surface terminations, dispersion oxidation, and unstable ink rheology complicate translation from laboratory batches to patterned electrodes and larger-area devices. Alternative or safer etching strategies should be

assessed alongside yield, conductivity, residual reagent removal, and batch consistency, rather than by synthesis conditions alone. Water-based ink formulation, screen and inkjet printing, extrusion printing, laser patterning, and potentially roll-to-roll-compatible processing offer relevant routes, provided that adhesion, feature uniformity, conductivity after drying, encapsulation quality, and shelf life remain controlled. Quality assurance should extend from precursor and MXene characterization to the electrochemical performance of completed devices [20, 39, 40, 58, 196, 197, 209].

At the system level, design targets should be derived from real electrical loads and patterns of mechanical use. Microsupercapacitors are well suited to rapid charge delivery and buffering intermittent harvested energy, whereas hydrogel-based batteries can support longer operation; hybrid systems may combine these functions with photovoltaic, triboelectric, or other harvesting modules. Nevertheless, successful integration demands more than short-duration activation of a sensor or light-emitting diode. Demonstrations should quantify energy input, conversion and rectification losses, charging time, leakage and self-discharge, voltage regulation, series-cell balancing, and the power consumed by sensing, communication, and actuation over representative duty cycles. For soft robots, assessment should also consider the influence of storage placement and packaging stiffness on actuation; for wearable devices, it should include repeated attachment, perspiration, body motion, and practical operating duration [17-20, 160].

Biomedical and transient applications warrant a distinct evidence threshold. Skin-conformal demonstrations do not establish implantability, and hydrogel biocompatibility does not establish the safety of MXene-containing electrodes, residual synthesis reagents, current collectors, interconnects, or degradation products. Future work should examine material exposure state, nanosheet release, oxidation products, electrical leakage, tissue irritation, and relevant biological responses under intended durations and conditions of contact. For transient devices, the duration of stable power output must be matched to the required monitoring or therapeutic interval, while encapsulation must prevent premature failure yet permit the intended end-of-life process. Testing in physiologically relevant fluids, with appropriate mechanical and sterilization challenges, should quantify residual components and dissolved or fragmented products. Mass loss or visible disappearance alone is insufficient to demonstrate safe, complete-device biodegradation [13-15, 111, 137, 142, 193].

Sustainability should similarly be evaluated at the device rather than constituent-material level. Processing chemicals, solvent use, fabrication yield, replacement frequency, repairability, and end-of-life management may alter the value of a material initially described as environmentally favorable. The preferred strategy will depend on the application: long-lived soft robotic modules may benefit from repairable or replaceable components, whereas disposable wearables require practical disposal pathways and temporary biomedical systems require predictable degradation and biological clearance. Life-cycle considerations should be incorporated at the design stage without assuming that a degradable polymer network renders the entire device benign [14, 58, 137, 142, 196, 197].

In conclusion, hydrogel–MXene composites offer a versatile materials platform because electronically conductive and electrochemically active nanosheets can be coupled with hydrated, deformable, and multifunctional polymer networks. The reviewed work demonstrates promising routes toward flexible supercapacitors, microsupercapacitors, batteries, and self-powered sensing interfaces, while revealing that conductivity, capacitance, and deformation tolerance do not automatically translate into dependable device operation. Progress will be determined by the ability to preserve coupled ionic–electronic transport and interface integrity through realistic use, establish reproducible full-device metrics, verify biological and environmental outcomes, and manufacture consistent architectures at relevant scales. This materials-to-systems perspective provides a basis for assessing when hydrogel–MXene energy storage can meet the distinct requirements of soft robotics, wearable electronics, smart textiles, and biomedical technologies.

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Supplementary Materials

The original source figures, references, reusable content, permissions for adaptation and reuse, editable figure files, revised versions, and additional raw data files are available through the following GitHub repository link: https://github.com/mawaisashraf3/ZWU-Energy-Environmental-Science

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